EXAM PACK

EEE421

Solid State Devices



Exam Pack

Session 2018–19 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, June 2022, with Class Test 01 and Term Test 2*

IN THIS PACK

1 PAST PAPER SOLUTIONS
Question Paper Solutions 2018–19 — pages 3--95 — 93 pages

2 FORMULA SHEET
Formula Sheet — pages 96--112 — 17 pages

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Prepared by Nur Mohammad

November 2026

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1 PAST PAPER SOLUTIONS
Question Paper Solutions 2018–19

pages 3--95

Question Paper 5
1(a) Temperature Dependence of Carrier Concentration 12
1(b) Electron Concentration in Boron-Doped Germanium 15
1(c) Maximum Electron Energy at Absolute Zero 17
1(d) The Hall Effect and Carrier Concentration 19
2(a) Storage Delay Time and Reverse Recovery 23
2(b) Fermi Energy and the Value of kT 25
2(c) The Einstein Relation 27
2(d) Depletion Capacitance of a PN Junction 30
3(a) MOS Energy Band Diagrams at Different Gate Voltages 34
3(b) Parameters That Affect the Threshold Voltage 37
3(c) The Body Effect of a MOSFET 40
3(d) Amplification Action of a Transistor 42
4(a) Electron and Hole Current Components in a Forward-Biased PN Junction 45
4(b) Factors Causing Base Current in a Transistor 47
4(c) Ohmic and Rectifying Contacts 49
4(d) Base Width Modulation 50
4(e) The Tunnel Diode 53
5(a) Diffusion and the Diffusion Current Density 57
5(b) Ideal Low-Frequency C-V Curve of a p-Type MOS Capacitor 60
5(c) Diffusion Length as the Mean Distance Before Recombination 62
5(d) Metallurgical Channel Thickness of an n-Channel JFET 65
6(a) Cutoff Frequency of an n-Channel Silicon JFET 68
6(b) Emitter-to-Base Doping Ratio for a Given Injection Efficiency 70
6(c) Current Crowding 72
6(d) The Schottky Contact 73
7(a) Density of States and Carrier Concentration 78
7(b) Why the Intrinsic Carrier Concentration Is Not Tiny 79
7(c) Electron Effective Mass and the Heavier Band in Figure 1 81
7(d) Formation of Energy Bands in a Solid 83
8(a) Space Charge Width of a Silicon PN Junction 87
8(b) Fermi Level Position in p-Type GaAs 90
8(c) Mobility, Conductivity and Resistivity of Compensated Silicon 93

2 FORMULA SHEET
Formula Sheet

pages 96--112

1 Bands, effective mass and the Fermi function . . . 99
2 Carriers in equilibrium 99
3 Transport 101
4 Excess carriers 102
5 PN junction 102
6 Metal-semiconductor contacts 103
7 MOS structure 104
8 Bipolar transistor 105
9 JFET 106
10 Key formulas 107
11 Symbol index 109

PAST PAPER SOLUTIONS

EEE421

Solid State Devices

Question Paper Solutions

Session 2018–19

4th Year 1st Semester Final Examination, June 2022, with Class Test 01 and Term Test 2

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8
11 marks	10 marks

EACH ANSWER



Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

100	4 of 6
Total marks	Questions to answer

Q	Sub-part	Marks	Time	Pick
Part – A · (Answer any TWO sets of questions)				
Q1	(a) Temperature Dependence of Carrier Concentration	7		☐
	(b) Electron Concentration in Boron-Doped Germanium	6		
	(c) Maximum Electron Energy at Absolute Zero	5		
	(d) The Hall Effect and Carrier Concentration	7		
Q2	(a) Storage Delay Time and Reverse Recovery	7		☐
	(b) Fermi Energy and the Value of kT	6		
	(c) The Einstein Relation	6		
	(d) Depletion Capacitance of a PN Junction	6		
Q3	(a) MOS Energy Band Diagrams at Different Gate Voltages	8		☐
	(b) Parameters That Affect the Threshold Voltage	6		
	(c) The Body Effect of a MOSFET	5		
	(d) Amplification Action of a Transistor	6		
Part – B · (Answer any TWO sets of questions)				
Q4	(a) Electron and Hole Current Components in a Forward-Biased PN Junction	7		☐
	(b) Factors Causing Base Current in a Transistor	3		
	(c) Ohmic and Rectifying Contacts	4		
	(d) Base Width Modulation	5		
	(e) The Tunnel Diode	6		
Q5	(a) Diffusion and the Diffusion Current Density	7		☐
	(b) Ideal Low-Frequency C-V Curve of a p-Type MOS Capacitor	5		
	(c) Diffusion Length as the Mean Distance Before Recombination	6		
	(d) Metallurgical Channel Thickness of an n-Channel JFET	7		
Q6	(a) Cutoff Frequency of an n-Channel Silicon JFET	6		☐
	(b) Emitter-to-Base Doping Ratio for a Given Injection Efficiency	6		
	(c) Current Crowding	7		
	(d) The Schottky Contact	6		
Class Test 01 · 10 marks · 30 min				
Q7	(a) Density of States and Carrier Concentration	2	6 min	☐
	(b) Why the Intrinsic Carrier Concentration Is Not Tiny	3	9 min	
	(c) Electron Effective Mass and the Heavier Band in Figure 1	3	9 min	
	(d) Formation of Energy Bands in a Solid	3	9 min	
Term Test 2 · 10 marks · 30 min				
Q8	(a) Space Charge Width of a Silicon PN Junction	3	9 min	☐
	(b) Fermi Level Position in p-Type GaAs	4	12 min	
	(c) Mobility, Conductivity and Resistivity of Compensated Silicon	3	9 min	

Formulas, symbols and constants follow Neamen, Semiconductor Physics and Devices (4th ed.).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination-June 2022
Course Code: EEE 421, Session: 2018-19
Course Title: Solid State Devices

Full Marks: 100 **Credits: 3.0** **Exam Durations: 3 Hours**

Part – A

*[(Answer any **TWO** sets of questions)]*

1. (a) Explain the temperature dependance [*sic*: should be "dependence"] of carrier concentration of an extrinsic semiconductor with the help of graph. 7
   (b) A germanium sample is doped with 10^{15} boron atoms per cm^3. Find the electron concentration. Intrinsic carrier concentration of germanium is $2.5 \times 10^{13}/cm^3$ at $300k$. 6
   (c) At absolute zero temperature, what is the maximum energy of an electron in a semiconductor material? 5
   (d) What is the Hall effect? Derive the expression for finding the carrier concentration of a semiconductor from Hall voltage. 7

2. (a) What is storage delay time? What is the impact of reverse recovery time of a junction in switching application? 7
   (b) What is Fermi energy? kT is equal to how many eV at $300k$? 6
   (c) What is Einstein Relation? Derive the expression. 6
   (d) Wat [*sic*: should be "What"] is the depletion capacitance of a PN junction? Explain its variation with reverse bias voltage. 6

3. (a) With brief description sketch the energy band diagram of a MOS structure at different bias voltage. (Use no bias condition, positive bias voltage and negative bias voltage). 8
   (b) What are the parameters that affect the magnitude of threshold voltage in a MOSFET? 6
   (c) What is body effect of MOSFET? 5
   (d) Explain the amplification action of a transistor. 6

Part – B

*[(Answer any **TWO** sets of questions)]*

4. (a) Sketch the ideal electron and hole current components through a pn junction under forward bias. 7
   (b) What are the factors which cause base current in a transistor? 3
   (c) Differentiate between ohmic and rectifying contacts. 4
   (d) What is base width modulation? 5
   (e) Explain the working of Tunnel diode. Draw its characteristics curve. 6

5. (a) Describe diffusion process. Derive the expression for diffusion current density. 7
   (b) Draw the ideal low-frequency capacitance versus gate voltage curve of a MOS capacitor with p-type substrate and indicate different regions. 5
   (c) Show that diffusion length is the average length a carrier diffuses before recombination. 6

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(d) A silicon n-channel JFET at $T = 300K$ has a gate doping concentration of $N_a = 10^{18} cm^{-3}$ and a channel doping concentration of $N_d = 2 \times 10^{16} cm^{-3}$. Determine the metallurgical channel thickness, a, such that the pinchoff voltage is $V_p = -2.50\ V$. 7

6. (a) The parameters of an n-channel silicon JFET are $\mu_n = 1000\ cm^2/V-s$, $N_d = 5 \times 10^{15} cm^{-3}$, $a = 0.50\ \mu m$, and $L = 2\ \mu m$. Determine the cutoff frequency. 6

(b) Design the ratio of emitter doping to base doping in order to achieve an emitter injection efficiency factor of $\gamma = 0.9967$. Assume a *npn* bipolar transistor, and also consider $D_E = D_B$, $L_E = L_B$ and $x_E = X_B$. 6

(c) What is current crowding? What is the impact of current crowding in high power devices? 7

(d) With suitable energy band diagrams, explain Schottky contact. 6

Class Test 01 (its questions 1 to 4 are 7(a) to 7(d))

[Class test-01. Solid State Device. Time-30min. Marks: 10]

7. (a) Define the density of states and carrier concentration. 2

(b) Explain with the help of equations and graphs- why intrinsic carrier concentration (n_i) for a typical semiconductor is high (~10^6-10^9/cc) at 300K even though the Fermi Dirac distribution is low at that temperature. 3

(c) What is electron-effective mass? Two possible conduction bands in the E versus k diagram are given in Figure 1. State which band will result in the heavier electron effective mass;state why. 3

(d) Using the necessary theorem explains how band structure is formed of a particular solid-state material. 3

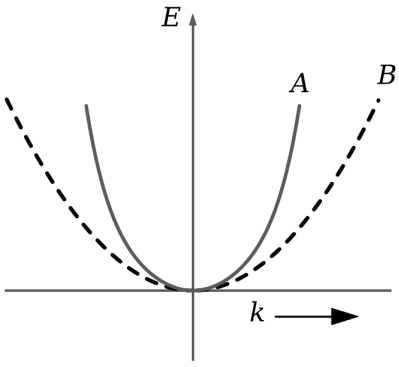


Figure 1: Two possible conduction bands A and B in the E versus k diagram (redrawn from the paper)

Term Test 2 (its questions 1 to 3 are 8(a) to 8(c))

[Course Code: EEE-421, Course Title: Solid State Devices, Credit: 3.0, Session: 2018-19, Total Marks: 10, Total Time: 30 minutes]

8. (a) A silicon pn junction at T = 300 K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{20}$ [*sic*: should be 5×10^{15}] cm^{-3}. Determine x_n, x_p and W. 3

(b) Determine the position of the Fermi level with respect to the valence band energy in p type GaAs at T = 300 K. The doping concentrations are [unclear: N_a] $= 5 \times 10^{19}$ [*sic*: should be 5×10^{16}] cm^{-3} and $N_d = 10^{15}$ [*sic*: should be 4×10^{15}] cm^{-3}. 4

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(c) Silicon at T = 300 K is doped with impurity concentrations of Nd = 5 × 1016 [*sic*: should be 10^{16}] /cm3 and Na = 2 × 1016 [*sic*: should be 10^{16}] /cm3. (a) What are the electron and hole mobilities? (b) Determine the conductivity and resistivity of the [unclear: material]. 3

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Contents

1(a) Temperature Dependence of Carrier Concentration **10**
1 Theory . . . 10
2 Answer . . . 12
3 Keyword Explanation . . . 13
4 Notes and Remarks . . . 13

1(b) Electron Concentration in Boron-Doped Germanium **13**
1 Theory . . . 13
2 Answer . . . 14
3 Keyword Explanation . . . 15
4 Notes and Remarks . . . 15

1(c) Maximum Electron Energy at Absolute Zero **15**
1 Theory . . . 15
2 Answer . . . 16
3 Keyword Explanation . . . 16
4 Notes and Remarks . . . 17

1(d) The Hall Effect and Carrier Concentration **17**
1 Theory . . . 17
2 Answer . . . 18
3 Keyword Explanation . . . 20
4 Notes and Remarks . . . 20
5 Final Concise Answer . . . 20

2(a) Storage Delay Time and Reverse Recovery **21**
1 Theory . . . 21
2 Answer . . . 22
3 Keyword Explanation . . . 23
4 Notes and Remarks . . . 23

2(b) Fermi Energy and the Value of kT **23**
1 Theory . . . 24
2 Answer . . . 24
3 Keyword Explanation . . . 25
4 Notes and Remarks . . . 25

2(c) The Einstein Relation **25**
1 Theory . . . 25
2 Answer . . . 26
3 Keyword Explanation . . . 28
4 Notes and Remarks . . . 28

2(d) Depletion Capacitance of a PN Junction **28**
1 Theory . . . 28
2 Answer . . . 29
3 Keyword Explanation . . . 30
4 Notes and Remarks . . . 31
5 Final Concise Answer . . . 31

3(a) MOS Energy Band Diagrams at Different Gate Voltages **32**

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1 Theory . . . 32
2 Answer . . . 33
3 Keyword Explanation . . . 34
4 Notes and Remarks . . . 35

3(b) Parameters That Affect the Threshold Voltage 35
1 Theory . . . 35
2 Answer . . . 35
3 Keyword Explanation . . . 37
4 Notes and Remarks . . . 37

3(c) The Body Effect of a MOSFET 38
1 Theory . . . 38
2 Answer . . . 38
3 Keyword Explanation . . . 39
4 Notes and Remarks . . . 39

3(d) Amplification Action of a Transistor 40
1 Theory . . . 40
2 Answer . . . 40
3 Keyword Explanation . . . 41
4 Notes and Remarks . . . 41
5 Final Concise Answer . . . 42

4(a) Electron and Hole Current Components in a Forward-Biased PN Junction 43
1 Theory . . . 43
2 Answer . . . 44
3 Keyword Explanation . . . 45
4 Notes and Remarks . . . 45

4(b) Factors Causing Base Current in a Transistor 45
1 Theory . . . 45
2 Answer . . . 46
3 Keyword Explanation . . . 46
4 Notes and Remarks . . . 47

4(c) Ohmic and Rectifying Contacts 47
1 Theory . . . 47
2 Answer . . . 47
3 Keyword Explanation . . . 48
4 Notes and Remarks . . . 48

4(d) Base Width Modulation 48
1 Theory . . . 48
2 Answer . . . 49
3 Keyword Explanation . . . 50
4 Notes and Remarks . . . 50

4(e) The Tunnel Diode 51
1 Theory . . . 51
2 Answer . . . 52
3 Keyword Explanation . . . 53

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4 Notes and Remarks . . . 53
5 Final Concise Answer . . . 53

5(a) Diffusion and the Diffusion Current Density 55
1 Theory . . . 55
2 Answer . . . 56
3 Keyword Explanation . . . 58
4 Notes and Remarks . . . 58

5(b) Ideal Low-Frequency C-V Curve of a p-Type MOS Capacitor 58
1 Theory . . . 58
2 Answer . . . 59
3 Keyword Explanation . . . 60
4 Notes and Remarks . . . 60

5(c) Diffusion Length as the Mean Distance Before Recombination 60
1 Theory . . . 60
2 Answer . . . 61
3 Keyword Explanation . . . 62
4 Notes and Remarks . . . 62

5(d) Metallurgical Channel Thickness of an n-Channel JFET 63
1 Theory . . . 63
2 Answer . . . 64
3 Keyword Explanation . . . 64
4 Notes and Remarks . . . 64
5 Final Concise Answer . . . 65

6(a) Cutoff Frequency of an n-Channel Silicon JFET 66
1 Theory . . . 66
2 Answer . . . 66
3 Keyword Explanation . . . 67
4 Notes and Remarks . . . 67

6(b) Emitter-to-Base Doping Ratio for a Given Injection Efficiency 68
1 Theory . . . 68
2 Answer . . . 68
3 Keyword Explanation . . . 69
4 Notes and Remarks . . . 69

6(c) Current Crowding 70
1 Theory . . . 70
2 Answer . . . 70
3 Keyword Explanation . . . 71
4 Notes and Remarks . . . 71

6(d) The Schottky Contact 71
1 Theory . . . 71
2 Answer . . . 72
3 Keyword Explanation . . . 73
4 Notes and Remarks . . . 73
5 Final Concise Answer . . . 73

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7(a) Density of States and Carrier Concentration **76**
1 Theory . . . 76
2 Answer . . . 76
3 Keyword Explanation . . . 77
4 Notes and Remarks . . . 77

7(b) Why the Intrinsic Carrier Concentration Is Not Tiny **77**
1 Theory . . . 77
2 Answer . . . 77
3 Keyword Explanation . . . 79
4 Notes and Remarks . . . 79

7(c) Electron Effective Mass and the Heavier Band in Figure 1 **79**
1 Theory . . . 79
2 Answer . . . 80
3 Keyword Explanation . . . 81
4 Notes and Remarks . . . 81

7(d) Formation of Energy Bands in a Solid **81**
1 Theory . . . 81
2 Answer . . . 82
3 Keyword Explanation . . . 83
4 Notes and Remarks . . . 83
5 Final Concise Answer . . . 84

8(a) Space Charge Width of a Silicon PN Junction **85**
1 Theory . . . 85
2 Answer . . . 86
3 Keyword Explanation . . . 88
4 Notes and Remarks . . . 88

8(b) Fermi Level Position in p-Type GaAs **88**
1 Theory . . . 88
2 Answer . . . 89
3 Keyword Explanation . . . 90
4 Notes and Remarks . . . 90

8(c) Mobility, Conductivity and Resistivity of Compensated Silicon **91**
1 Theory . . . 91
2 Answer . . . 91
3 Keyword Explanation . . . 93
4 Notes and Remarks . . . 93
5 Final Concise Answer . . . 93

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1. (a) Explain the temperature dependance [*sic*: should be "dependence"] of carrier concentration of an extrinsic semiconductor with the help of graph. Marks: 7
   (b) A germanium sample is doped with 10^{15} boron atoms per cm^3. Find the electron concentration. Intrinsic carrier concentration of germanium is $2.5 \times 10^{13}/cm^3$ at $300k$. Marks: 6
   (c) At absolute zero temperature, what is the maximum energy of an electron in a semiconductor material? Marks: 5
   (d) What is the Hall effect? Derive the expression for finding the carrier concentration of a semiconductor from Hall voltage. Marks: 7

1.(a) Explain the temperature dependance [*sic*: should be "dependence"] of carrier concentration of an extrinsic semiconductor with the help of graph. Marks: 7

1. Theory

1. Carriers in a semiconductor (Core Idea)

In a crystal the allowed electron energies form **bands**. Two of them matter in a semiconductor:

- the **valence band**, with top edge E_v, full of the electrons that make the covalent bonds;
- the **conduction band**, with bottom edge E_c, almost empty.

Between them lies the **band gap** $E_g = E_c - E_v$ (1.12 eV for Si). An electron that gains at least E_g jumps from the valence band into the conduction band. It leaves an empty bond behind, the **hole**, which moves like a positive carrier. The densities of free electrons and holes are written n_0 and p_0 (cm^{-3}, the 0 meaning thermal equilibrium).

In a pure (**intrinsic**) crystal, electrons and holes are made only in pairs, so $n_0 = p_0 = n_i$. The **intrinsic carrier concentration** is (Neamen, Eq. 4.23)

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$$

where N_c, N_v are the effective densities of states of the two bands ($\propto T^{3/2}$), $k = 8.617 \times 10^{-5}$ eV/K and T is the absolute temperature. The exponential makes n_i grow extremely fast with T: for Si it is 1.5×10^{10} cm^{-3} at 300 K, but 5.2×10^{12} at 400 K and 2.1×10^{15} at 600 K.

2. Extrinsic semiconductor: donors

An **extrinsic** semiconductor is doped on purpose. A **donor** atom (P or As in Si) has five valence electrons; four make bonds and the fifth is held only weakly, at the **donor level** E_d just below E_c. The **ionization energy** $E_c - E_d$ is small, about 0.045 eV for P in Si. When the fifth electron leaves, it becomes a free conduction electron and the donor becomes a fixed positive ion.

Two sources of free electrons therefore compete:

- **ionization of donors:** needs only $E_c - E_d \approx 0.045$ eV per electron, but there are only N_d donors;
- **band-to-band generation:** needs the whole $E_g = 1.12$ eV, but there are about 5×10^{22} bonds per cm^3 to break.

Which one wins depends on the temperature. This is why n_0 follows three distinct regions.

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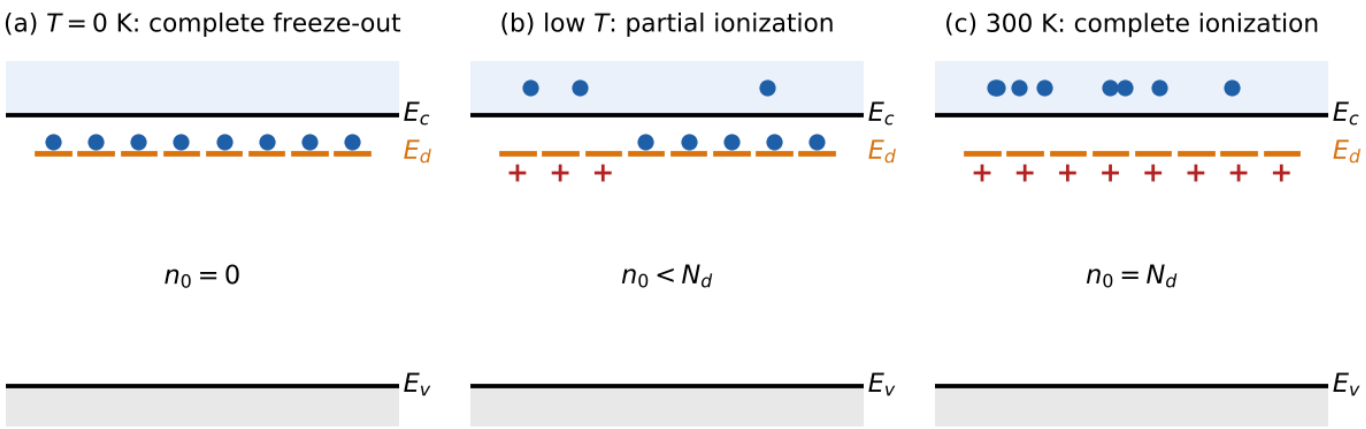


Figure 1: Donor states at 0 K, at a low temperature and at 300 K

Reading the figure. The short orange bars are the donor levels E_d, 0.045 eV below E_c (the gap is drawn larger for clarity). A filled blue dot is an electron; a red plus sign is an ionized donor. In panel (a), at 0 K, all electrons sit on the donors: complete freeze-out, $n_0 = 0$. In panel (b), at a low temperature, some donors are ionized and their electrons are in the conduction band. In panel (c), at 300 K, all donors are ionized and all their electrons are free: complete ionization, $n_0 = N_d$.

3. The charge-neutrality equation

In equilibrium the crystal is electrically neutral: negative charges equal positive charges. For an n-type sample with donors only,

$$n_0 = N_d^+ + p_0$$

where N_d^+ is the number of ionized donors. With the mass-action law $p_0 = n_i^2/n_0$ and complete ionization ($N_d^+ = N_d$) this is a quadratic in n_0, whose positive root is (Neamen, Eq. 4.60)

$$n_0 = \frac{N_d}{2} + \sqrt{\left(\frac{N_d}{2}\right)^2 + n_i^2}$$

- If $n_i \ll N_d$: $n_0 \approx N_d$.
- If $n_i \gg N_d$: $n_0 \approx n_i$.

At low temperature not all donors are ionized. The fraction still holding its electron is (Neamen, Eq. 4.55)

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \dfrac{N_c}{2N_d}\exp\left[-\dfrac{E_c - E_d}{kT}\right]}$$

which tends to 1 as $T \to 0$, because then the exponential vanishes.

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2. Answer

The electron concentration n_0 of an n-type semiconductor changes with temperature in **three regions**, because two sources of electrons (donor ionization and band-to-band generation) dominate at different temperatures.

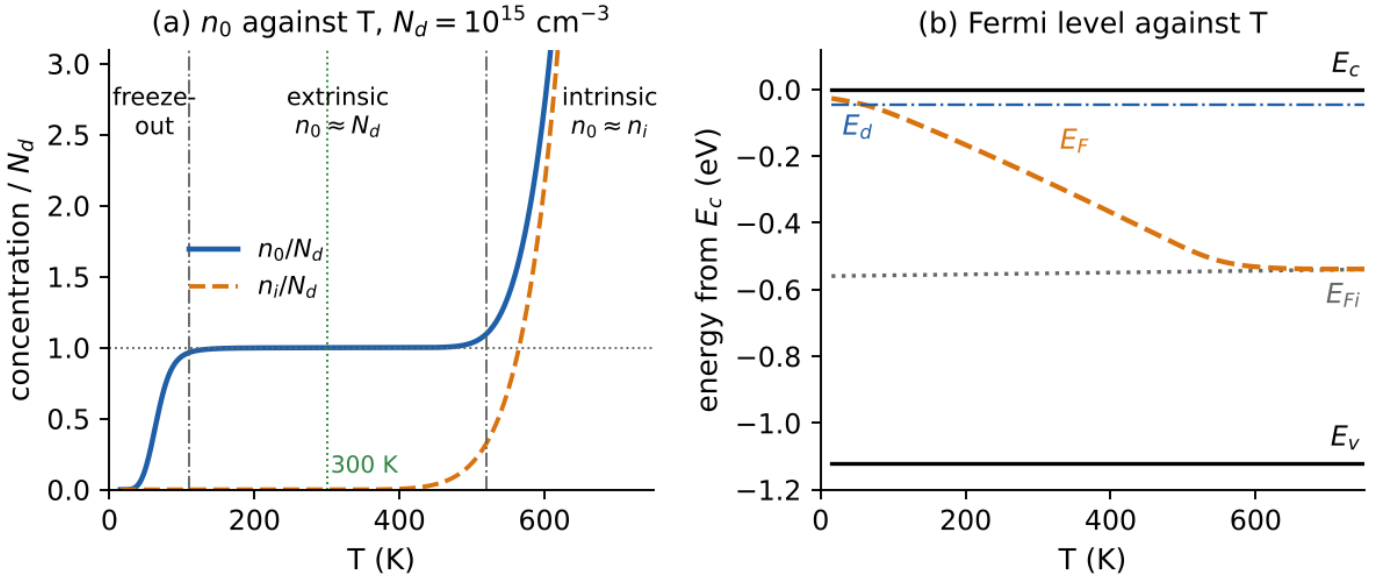


Figure 2: Electron concentration and Fermi level of Si with $N_d = 10^{15}$ cm^{-3} against temperature

Reading the figure. Panel (a): the solid blue curve is n_0/N_d and the dashed orange curve n_i/N_d for Si with $N_d = 10^{15}$ cm^{-3} of phosphorus (computed from the neutrality equation with partial ionization). The dash-dot grey lines divide the three regions; the dotted green line marks 300 K. Below about 100 K n_0 falls towards zero (freeze-out); from about 100 K to 500 K it stays flat at N_d (extrinsic region); above about 500 K it climbs with n_i (intrinsic region). Panel (b): the dashed orange Fermi level starts between E_c and the dash-dot blue donor level E_d at low T, falls through the gap as T rises, and approaches the dotted grey intrinsic level E_{Fi} at high T.

1. Freeze-out region (low temperature, below about 100 K for Si).

- At $T = 0$ K every donor electron stays bound to its donor: $n_0 = 0$ (complete freeze-out).
- As T rises, the small energy $E_c - E_d$ is supplied by heat and more donors ionize. n_0 rises steeply, roughly as $\sqrt{N_c N_d/2}\,\exp[-(E_c - E_d)/2kT]$.
- Example: Si, $N_d = 10^{15}$ cm^{-3}: $n_0 = 1.53 \times 10^{14}$ cm^{-3} (15 % of N_d) at 50 K and 9.39×10^{14} cm^{-3} (94 %) at 100 K.

2. Extrinsic (saturation) region (about 100 K to 500 K for this doping).

- All donors are ionized: $N_d^+ = N_d$.
- Band-to-band generation is still negligible: $n_i \ll N_d$.
- So $n_0 \approx N_d$, **independent of temperature**: the flat part of the curve. At 300 K, $n_0 = 0.9996\, N_d$.
- This is the useful operating range of devices: the carrier density is fixed by the doping, not by the temperature.

3. Intrinsic region (high temperature, above about 500 K here).

- n_i grows as $T^{3/2}\exp(-E_g/2kT)$ and finally exceeds N_d ($n_i = N_d$ at about 565 K for $N_d = 10^{15}$ cm^{-3}).
- Then $n_0 \approx n_i$ and $p_0 \approx n_i$: the material behaves as **intrinsic**, the doping no longer matters.
- Example: $n_0 = 2.70 \times 10^{15}$ cm^{-3} at 600 K and 1.32×10^{16} cm^{-3} at 700 K.

Fermi level. At low T the Fermi level lies between E_d and E_c; as T rises it moves down towards midgap, and in the intrinsic region it reaches E_{Fi} (panel (b) of the figure).

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For a p-type semiconductor the same three regions appear for the hole concentration p_0: acceptor freeze-out, $p_0 \approx N_a$, then $p_0 \approx n_i$.
Summary: freeze-out ($n_0 < N_d$, rising) → extrinsic ($n_0 = N_d$, flat) → intrinsic ($n_0 \approx n_i$, rising exponentially).

3. Keyword Explanation

1. Extrinsic semiconductor

A semiconductor whose carrier concentration is set by added impurity atoms (donors or acceptors) rather than by thermal generation.

2. Freeze-out

The low-temperature state in which the donor electrons stay on their donors (or acceptors stay empty), so few carriers are free.

3. Complete ionization

Every donor has given its electron to the conduction band, $N_d^+ = N_d$.

4. Intrinsic temperature

The temperature at which n_i becomes comparable with the doping; above it the semiconductor behaves as intrinsic.

4. Notes and Remarks

- **Heavier doping widens the extrinsic region:** a larger N_d needs a higher temperature before n_i catches up, and its freeze-out also starts at a higher temperature.
- **Wider band gap, higher limit:** GaAs ($E_g = 1.42$ eV) stays extrinsic to higher temperatures than Si; Ge ($E_g = 0.66$ eV) becomes intrinsic much sooner, which is why Ge devices fail at lower temperatures.
- **Numbers:** the values above are computed (calc script) with $n_i(T) \propto T^{3/2}\exp(-E_g/2kT)$ scaled from 1.5×10^{10} cm^{-3} at 300 K, E_g held at 1.12 eV, and the neutrality equation with partial ionization; Neamen's Fig. 4.16 shows the same curve.
- The same question is 2(c) of the 2019–20 paper (2019–20 2(c)), extra reading only.

1.(b) A germanium sample is doped with 10^{15} boron atoms per cm^3. Find the electron concentration. Intrinsic carrier concentration of germanium is $2.5 \times 10^{13}/cm^3$ at $300k$. Marks: 6

1. Theory

1. Acceptors make a p-type semiconductor (Core Idea)

Germanium, like silicon, has four valence electrons per atom. **Boron** has only three. A boron atom on a Ge site completes three bonds and leaves the fourth bond one electron short. An electron from a neighbouring bond easily moves in to fill it, which needs only a small energy (the **acceptor level** E_a just above E_v). Then:

- the boron atom becomes a fixed **negative ion** (it has accepted an electron);
- the bond the electron came from is now a free **hole**.

So each boron atom is an **acceptor** and gives one hole. A sample doped with acceptors is **p-type**: holes are the majority carriers and electrons the minority carriers.

2. Two equations fix the two concentrations

Charge neutrality. The sample as a whole is neutral. With N_a ionized acceptors (negative) and no donors,

$$p_0 = n_0 + N_a$$

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Mass-action law. In thermal equilibrium the product of electron and hole concentrations is fixed by the material and temperature (Neamen, Eq. 4.43):

$$n_0 p_0 = n_i^2$$

Adding acceptors raises p_0, so it must **lower** n_0: the extra holes recombine with electrons until the product is back at n_i^2.

3. Solving them together

Put $n_0 = n_i^2/p_0$ in the neutrality equation and multiply by p_0:

$$p_0^2 - N_a p_0 - n_i^2 = 0 \quad \Rightarrow \quad p_0 = \frac{N_a}{2} + \sqrt{\left(\frac{N_a}{2}\right)^2 + n_i^2}$$

(the positive root; Neamen, Eq. 4.62 with $N_d = 0$). When $N_a \gg n_i$, $p_0 \approx N_a$ and $n_0 \approx n_i^2/N_a$.

4. Why germanium needs care

Ge has a small band gap (0.66 eV), so its n_i is large: 2.5×10^{13} cm^{-3} here, against 1.5×10^{10} cm^{-3} for Si. A doping of 10^{15} cm^{-3} is only 40 times n_i, so it is worth checking that the simple form $p_0 = N_a$ is good enough.

2. Answer

Given:

- Ge at $T = 300$ K, boron (acceptor) doping $N_a = 10^{15}$ cm^{-3}, no donors
- $n_i = 2.5 \times 10^{13}$ cm^{-3}
- complete ionization at 300 K

Required: the electron concentration n_0.

Step 1: Hole concentration from neutrality and mass action

$$p_0 = \frac{N_a}{2} + \sqrt{\left(\frac{N_a}{2}\right)^2 + n_i^2}$$

$$\left(\frac{N_a}{2}\right)^2 = (5 \times 10^{14})^2 = 2.5 \times 10^{29}, \qquad n_i^2 = (2.5 \times 10^{13})^2 = 6.25 \times 10^{26}$$

$$\sqrt{2.5 \times 10^{29} + 6.25 \times 10^{26}} = 5.00625 \times 10^{14}$$

$$p_0 = 5 \times 10^{14} + 5.00625 \times 10^{14} = 1.000625 \times 10^{15} \text{ cm}^{-3}$$

$$\boxed{p_0 = 1.0006 \times 10^{15} \text{ cm}^{-3}}$$

So p_0 is only 0.06 % above N_a: the simple $p_0 \approx N_a$ is fine.

Step 2: Electron concentration from the mass-action law

$$n_0 = \frac{n_i^2}{p_0} = \frac{6.25 \times 10^{26}}{1.000625 \times 10^{15}} = 6.246 \times 10^{11} \text{ cm}^{-3}$$

$$\boxed{n_0 = 6.25 \times 10^{11} \text{ cm}^{-3}}$$

(The simple form $n_0 = n_i^2/N_a = 6.25 \times 10^{11}$ cm^{-3} gives the same three figures.)

Step 3: Meaning

The electrons are the minority carriers: $p_0/n_0 = 1.6 \times 10^3$, so there are 1600 holes for every electron. The Fermi level lies $E_{Fi} - E_F = kT \ln(p_0/n_i) = 0.0259 \ln(40.0) = 0.0956$ eV below the intrinsic level.

Final Answer: $n_0 = 6.25 \times 10^{11}$ cm^{-3} (with $p_0 = 1.0 \times 10^{15}$ cm^{-3}).

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3. Keyword Explanation

1. Acceptor

An impurity atom with one valence electron fewer than the host; it takes an electron from the valence band and leaves a hole.

2. Majority and minority carriers

In p-type material holes are the majority carriers ($p_0 \approx N_a$) and electrons the minority carriers ($n_0 = n_i^2/N_a$).

3. Mass-action law

$n_0 p_0 = n_i^2$ for a non-degenerate semiconductor in thermal equilibrium, whatever the doping.

4. Notes and Remarks

- **Neamen's n_i for Ge** is 2.4×10^{13} cm^{-3} (Table 4.2); with it $n_0 = 5.76 \times 10^{11}$ cm^{-3}. The paper's 2.5×10^{13} is used above.
- **Higher temperature:** n_i of Ge grows fast; near 400 K it reaches 10^{15} cm^{-3} and this sample would no longer be clearly p-type (1(a)).
- **Units:** the paper writes "300k"; it means 300 K.

1.(c) At absolute zero temperature, what is the maximum energy of an electron in a semiconductor material? Marks: 5

1. Theory

1. Allowed energies and Pauli's principle (Core Idea)

Electrons in a crystal can only take the energies of the allowed bands. Each allowed state can hold at most **one electron** (the Pauli exclusion principle; with spin counted as part of the state). So electrons cannot all crowd into the lowest state: they fill the states one by one from the bottom up, like water filling a vessel.

2. The Fermi-Dirac function at 0 K

The probability that a state at energy E is occupied in thermal equilibrium is (Neamen, Eq. 3.79)

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

At $T = 0$ K the exponent is $-\infty$ for $E < E_F$ and $+\infty$ for $E > E_F$, so

$$f_F(E) = 1 \quad (E < E_F), \qquad f_F(E) = 0 \quad (E > E_F)$$

Every state below E_F is full and every state above it is empty: a sharp step. With no thermal energy, no electron can be lifted above the lowest arrangement.

3. Where the states are in a semiconductor

In a metal the band that holds the valence electrons is only partly full, so at 0 K the electrons fill it up to E_F, and E_F itself is the highest occupied energy. A semiconductor is different:

- the valence band holds exactly as many states as there are valence electrons, so the electrons fill it **completely**;
- the next allowed states, at E_c and above, are separated by the gap E_g in which **no states exist**;
- E_F lies inside the gap, where there are no states to fill.

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2. Answer

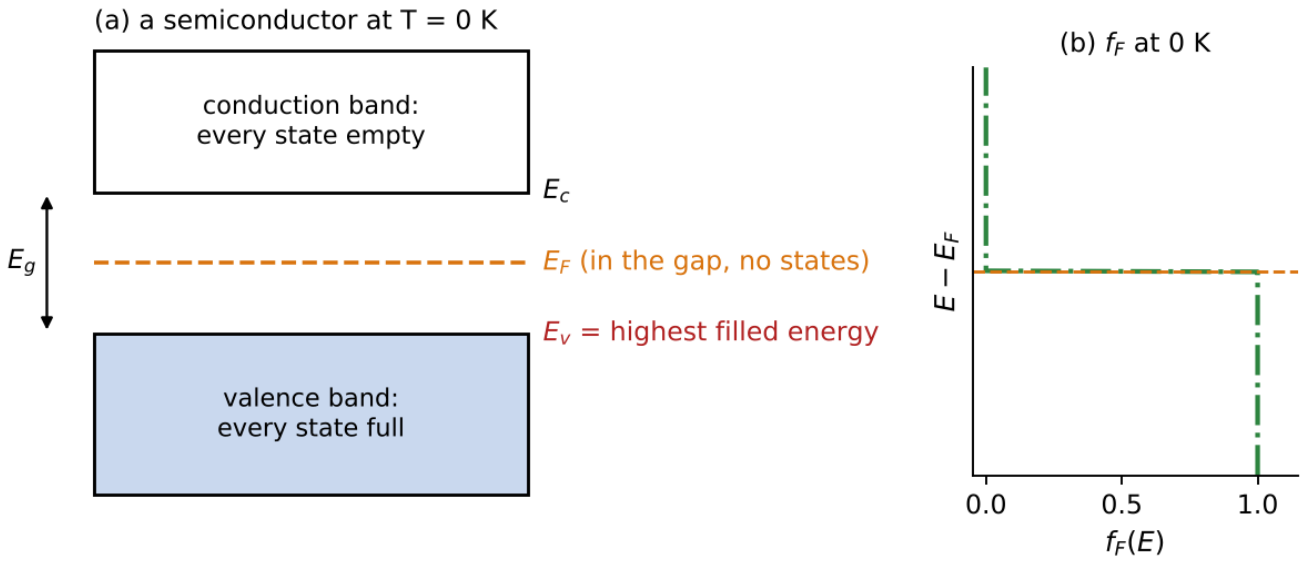


Figure 3: A semiconductor at 0 K: band picture and the Fermi-Dirac step

Reading the figure. Panel (a): the shaded valence band is completely full and the conduction band (empty box) completely empty; the dashed orange Fermi level lies in the gap, where there are no states. The red label marks E_v, the highest filled energy. Panel (b): the dash-dot green Fermi-Dirac function at 0 K is a step: 1 below E_F and 0 above it.

At $T = 0$ K:

- There is no thermal energy, so the electrons take the lowest possible arrangement. By the Pauli principle they fill the allowed states one per state, from the bottom upwards.
- In a semiconductor the valence band is then **completely full** and the conduction band **completely empty** ($f_F = 1$ below E_F, $f_F = 0$ above, and E_F lies in the gap).
- No electron can sit in the band gap, because there are no states there.

Therefore the maximum energy of an electron at 0 K is the top of the valence band, E_v.

$$\boxed{E_{max}(T = 0) = E_v}$$

Every electron has an energy $E \leq E_v$; there are no free electrons in the conduction band, so a pure semiconductor at 0 K is a perfect **insulator**. Measured from the conduction band edge, the highest electron is E_g below it (1.12 eV in Si).

In terms of the Fermi energy: the Fermi level is defined as the energy below which all states are filled at 0 K. In a metal it is itself the maximum electron energy. In a semiconductor E_F is the dividing line between full and empty states and lies in the gap, so the highest *occupied* level is E_v, just below E_F.

3. Keyword Explanation

1. Pauli exclusion principle

No two electrons can occupy the same quantum state.

2. Fermi energy

The energy at which the Fermi-Dirac probability is 1/2; at 0 K, the boundary between the filled and empty states.

3. Valence band

The highest band that is completely filled at 0 K; its top edge is E_v.

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4. Notes and Remarks

- **Doped semiconductor at 0 K:** donor electrons are frozen onto their donor levels (1(a)), so the highest filled energy is then E_d, just below E_c, and E_F lies between E_d and E_c.
- **Metal:** the answer "the Fermi energy" is correct for a metal, where the band is half full; for silver E_F is about 5.5 eV above the band bottom.
- The quantity E_F at 0 K is also defined for a semiconductor, and some texts say "at 0 K all states up to E_F are filled"; that is the same statement, since no states lie between E_v and E_F.

1.(d) What is the Hall effect? Derive the expression for finding the carrier concentration of a semiconductor from Hall voltage. Marks: 7

1. Theory

1. Force on a moving charge in a magnetic field (Core Idea)

A charge q moving with velocity $\mathbf{v}$ through a magnetic field $\mathbf{B}$ feels the **Lorentz force** (Neamen, Eq. 5.48)

$$\mathbf{F} = q\,\mathbf{v} \times \mathbf{B}$$

It is perpendicular to both the velocity and the field. So a current flowing along x in a field along z is pushed sideways, along y.

2. What happens in a semiconductor bar

Take a bar carrying a current I_x along x, placed in a field B_z along z:

- **Holes** move along $+x$: $\mathbf{v} \times \mathbf{B} = v_x\hat{x} \times B_z\hat{z} = -v_xB_z\hat{y}$, so with $q = +e$ the force is along $-y$.
- **Electrons** move along $-x$ (the current direction is still $+x$): $q = -e$ and $\mathbf{v} = -v_x\hat{x}$, so the force is $(-e)(-v_x)(\hat{x} \times \hat{z})B_z = -ev_xB_z\hat{y}$: also along $-y$.

So both kinds of carrier are pushed to the same face, $y = 0$. In a p-type bar positive charge piles up there; in an n-type bar negative charge. The piled-up charge creates an electric field across the bar, the **Hall field** E_H, which grows until it exactly cancels the magnetic force. The voltage across the bar is the **Hall voltage** V_H.

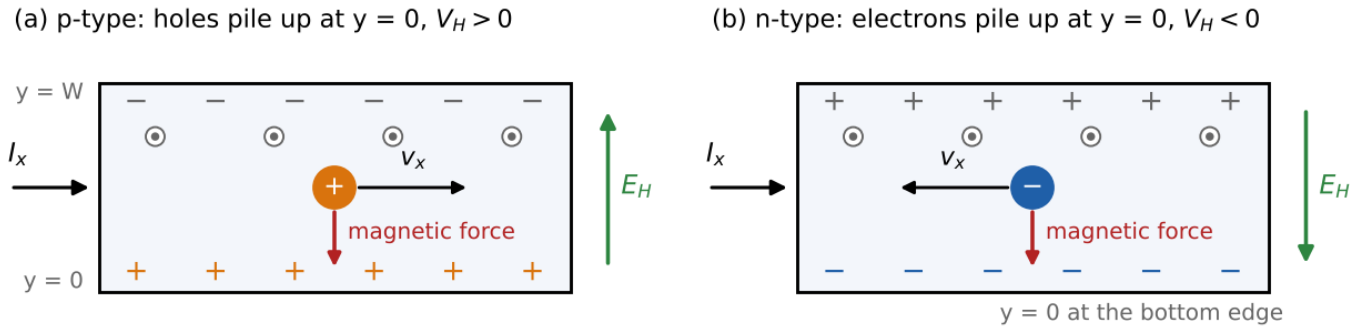


Figure 4: Top view of a p-type and an n-type bar in the Hall effect: carriers pile up on the y = 0 face

Reading the figure. Both bars are seen from above, with B_z coming out of the page (dots) and I_x flowing to the right. In panel (a) a hole (filled circle, +) moves right; the red arrow, the magnetic force, pushes it down to $y = 0$, where plus signs pile up. The green arrow E_H then points up $(+y)$, and $V_H > 0$. In panel (b) an electron (filled circle, −) moves left, is pushed down to $y = 0$ too, and minus signs pile up there. E_H points down $(-y)$, and $V_H < 0$.

3. Why it is useful

- The **sign** of V_H tells whether the majority carriers are holes (p-type) or electrons (n-type).
- Its **size** gives the majority carrier concentration, as derived below.
- With the concentration known, a measurement of the voltage along the bar gives the **mobility**.
- Hall sensors also measure magnetic fields (position and current sensors).

2. Answer

The Hall effect: when a current-carrying semiconductor is placed in a magnetic field perpendicular to the current, the magnetic force pushes the carriers to one side of the sample; the charge that builds up creates a transverse electric field (the Hall field) and a voltage across the sample (the Hall voltage) that is perpendicular to both the current and the field.
The derivation uses the geometry of Neamen's Fig. 5.13, and every step is written out below.

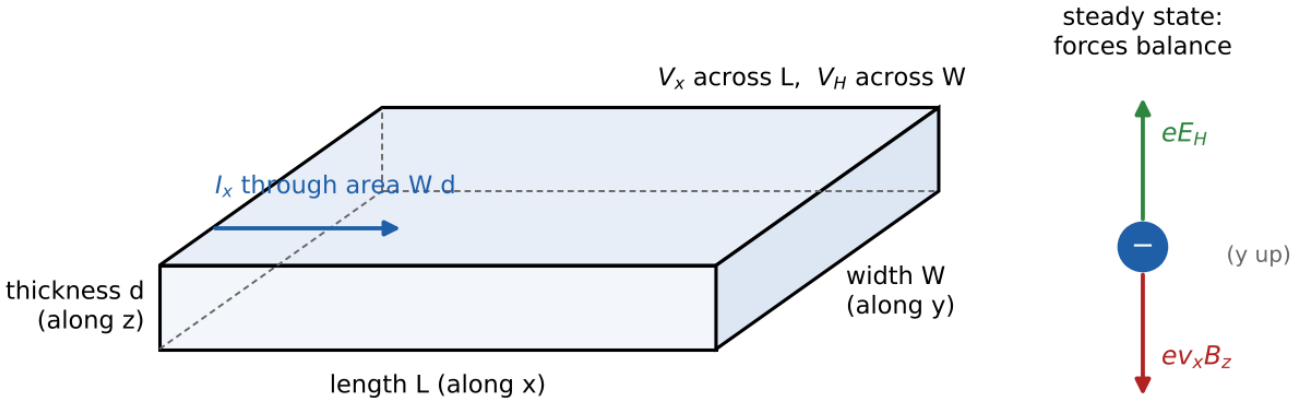


Figure 5: The bar for the Hall effect derivation: dimensions and the force balance on one electron

Reading the figure. The bar on the left has length L along x, width W along y (the depth of the drawing) and thickness d along z. The current I_x (blue arrow) crosses the area Wd. On the right is one electron in steady state, with y upwards: the magnetic force ev_xB_z (red, downwards, towards $y = 0$) is balanced by the force of the Hall field eE_H (green, upwards).

Step 1: The set-up

A bar of length L (along x), width W (along y) and thickness d (along z) carries a current I_x along $+x$ through the cross-section Wd. A magnetic field B_z points along $+z$. The Hall voltage V_H is measured across the width W, between the faces $y = 0$ and $y = W$.

Step 2: Force balance in steady state (p-type first)

Holes drifting with velocity v_x feel the magnetic force $-ev_xB_z$ along y and pile up at $y = 0$. The Hall field E_H they create points along $+y$ and pushes them back. In steady state the net sideways force is zero (Neamen, Eq. 5.49a):

$$\mathbf{F} = e\,[\mathbf{E} + \mathbf{v} \times \mathbf{B}] = 0 \quad \Rightarrow \quad eE_y = ev_xB_z$$

(Neamen, Eq. 5.49b). So $E_H = v_xB_z$.

Step 3: Hall voltage from the Hall field

The Hall field is uniform across the width, so the voltage across the bar is (Neamen, Eq. 5.50)

$$V_H = E_HW$$

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Putting in Step 2 (Neamen, Eq. 5.51):

$$V_H = v_x W B_z$$

Step 4: The drift velocity from the current

The current density is $J_x = ep\,v_x$ (charge per unit volume times velocity), and $J_x = I_x/(Wd)$ because the current crosses the area Wd. So (Neamen, Eq. 5.52)

$$v_x = \frac{J_x}{ep} = \frac{I_x}{ep\,Wd}$$

Step 5: Substitute and solve for the hole concentration

Put Step 4 into Step 3; the width W cancels:

$$V_H = \frac{I_x}{epWd}\,WB_z = \frac{I_x B_z}{epd} \qquad \text{(Neamen, Eq. 5.53)}$$

Solving for p:

$$\boxed{p = \frac{I_x B_z}{e\,d\,V_H}} \qquad \text{(Neamen, Eq. 5.54)}$$

For p-type material V_H is positive.

Step 6: The same for an n-type sample

Electrons move along $-x$ with speed v_x; the current density is $J_x = -en(-v_x) = env_x$, so $v_x = I_x/(enWd)$. The magnetic force again pushes them to $y = 0$, but they are negative, so the Hall field points along $-y$: $E_H = -v_x B_z$. Then

$$V_H = E_H W = -v_x W B_z = -\frac{I_x B_z}{ned} \qquad \text{(Neamen, Eq. 5.55)}$$

$$\boxed{n = -\frac{I_x B_z}{e\,d\,V_H}} \qquad \text{(Neamen, Eq. 5.56)}$$

V_H is negative for n-type material, so n comes out positive.

Step 7: Mobility (the use of the result)

Once p (or n) is known, the drift equation $J_x = ep\mu_p E_x$ with $J_x = I_x/(Wd)$ and $E_x = V_x/L$ gives (Neamen, Eqs. 5.59, 5.60)

$$\mu_p = \frac{I_x L}{epV_x Wd}, \qquad \mu_n = \frac{I_x L}{enV_x Wd}$$

Steps to remember

Lorentz force pushes carriers to $y = 0 \to$ steady state $eE_H = ev_x B_z \to V_H = E_H W = v_x W B_z \to v_x = I_x/(epWd) \to V_H = I_x B_z/(epd) \to p = I_x B_z/(edV_H)$; for electrons $n = -I_x B_z/(edV_H)$.

Example (Neamen, Example 5.8)

$I_x = 1.0$ mA, $B_z = 5 \times 10^{-2}$ T, $d = 10^{-5}$ m, $V_H = -6.25$ mV. The negative sign means n-type, and in SI units

$$n = -\frac{(10^{-3})(5 \times 10^{-2})}{(1.6 \times 10^{-19})(10^{-5})(-6.25 \times 10^{-3})} = 5 \times 10^{21}\ \text{m}^{-3} = 5 \times 10^{15}\ \text{cm}^{-3}$$

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3. Keyword Explanation

1. Lorentz force

The force $q\mathbf{v} \times \mathbf{B}$ on a charge moving in a magnetic field.

2. Hall field

The transverse electric field E_H set up by the piled-up carriers; it balances the magnetic force.

3. Hall voltage

$V_H = E_H W$, the voltage across the width of the sample; positive for p-type, negative for n-type in this geometry.

4. Hall coefficient

$R_H = 1/(ep)$ for p-type and $-1/(en)$ for n-type, so that $V_H = R_H I_x B_z / d$.

4. Notes and Remarks

- **Units:** use SI units throughout (m, T, A, V), then convert n from m^{-3} to cm^{-3} by dividing by 10^6; mixing cm with tesla gives wrong answers (Neamen's comment on Example 5.8).
- **Assumption:** one type of carrier dominates (an extrinsic sample). If both carry comparable current, the Hall voltage depends on both concentrations and mobilities.
- **Thickness, not width:** the concentration formula has the thickness d (along B), not the width W across which V_H is measured; W cancels in Step 5.
- The same derivation with numbers given is the numerical 5(d) of the 2020–21 paper (2020–21 5(d)), extra reading only.

5. Final Concise Answer

(a) n_0 of an n-type semiconductor against T has three regions: **freeze-out** at low T (donors not ionized, n_0 rises from 0; Si with $N_d = 10^{15}$ cm^{-3}: 15 % of N_d at 50 K), the flat **extrinsic** region where all donors are ionized and $n_0 = N_d$ (about 100 to 500 K), and the **intrinsic** region at high T where n_i exceeds N_d (565 K here) and $n_0 \approx n_i \propto T^{3/2} e^{-E_g/2kT}$; E_F moves from near E_c down to E_{Fi}.

(b) $p_0 = N_a/2 + \sqrt{(N_a/2)^2 + n_i^2} = 1.0006 \times 10^{15}$ cm^{-3}, so $n_0 = n_i^2/p_0 = 6.25 \times 10^{11}$ cm^{-3}.

(c) At 0 K the electrons fill the states from the bottom up (Pauli), the valence band is completely full and the conduction band empty, and E_F lies in the gap where there are no states; so the maximum electron energy is the top of the valence band, E_v.

(d) A current in a magnetic field perpendicular to it is pushed sideways; the piled-up charge makes a Hall field and a Hall voltage across the sample. Balance $eE_H = ev_x B_z$, $V_H = E_H W$ and $v_x = I_x/(epWd)$ give $V_H = I_x B_z/(epd)$, so $p = I_x B_z/(edV_H)$ and, for n-type, $n = -I_x B_z/(edV_H)$; the sign of V_H gives the carrier type.

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2. (a) What is storage delay time? What is the impact of reverse recovery time of a junction in switching application? Marks: 7
   (b) What is Fermi energy? kT is equal to how many eV at $300k$? Marks: 6
   (c) What is Einstein Relation? Derive the expression. Marks: 6
   (d) Wat [*sic*: should be "What"] is the depletion capacitance of a PN junction? Explain its variation with reverse bias voltage. Marks: 6

2.(a) What is storage delay time? What is the impact of reverse recovery time of a junction in switching application? Marks: 7

1. Theory

1. Stored minority charge in a forward-biased junction (Core Idea)

Take a p^+n diode (the p side much more heavily doped). Under forward bias the barrier is lowered and holes are **injected** from the p side into the n side. There they are minority carriers. They diffuse away from the junction and recombine over about one diffusion length L_p. In steady state the excess hole concentration on the n side is (Neamen, Eq. 8.14)

$$\delta p_n(x) = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right] e^{-(x-x_n)/L_p}$$

The area under this curve, times e and the junction area, is a **stored charge** Q_p. Charge balance says it is kept up by the forward current: holes recombine at the rate Q_p/τ_{p0}, so

$$Q_p = I_F\,\tau_{p0}$$

where τ_{p0} is the minority carrier (hole) lifetime.

2. Why a diode cannot switch off at once

The junction voltage is tied to the minority concentration at the edge of the depletion region by the **law of the junction**: $p_n(x_n) = p_{n0}\exp(eV_a/kT)$. The junction stays forward biased as long as $p_n(x_n) > p_{n0}$, that is, as long as there is excess charge stored at the edge. To switch the diode off, this stored charge must first be removed, by a reverse current and by recombination. That takes time.

3. The switching circuit

Neamen (Sec. 8.4) switches the diode from a forward current I_F to a reverse voltage at $t = 0$ through a resistor:

- for $t < 0$: forward current $I_F \approx V_F/R_F$ (Eq. 8.107);
- at $t = 0$ the source jumps to $-V_R$. Because the junction is still forward biased (a few tenths of a volt), the current reverses at once to $-I_R \approx -V_R/R_R$ (Eq. 8.108), limited only by the resistor;
- the stored holes flow back across the junction (this is the reverse current) and recombine.

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2. Answer

Storage delay time (storage time) t_s: when a forward-biased diode is suddenly switched to reverse bias, the current reverses at once and stays at a nearly constant value $-I_R$ (set by the external circuit) while the minority carriers stored near the junction are being removed. The time this takes, from the instant of switching until the excess minority concentration at the edge of the space charge region has fallen to zero ($p_n(x_n) = p_{n0}$), is the **storage delay time** t_s.

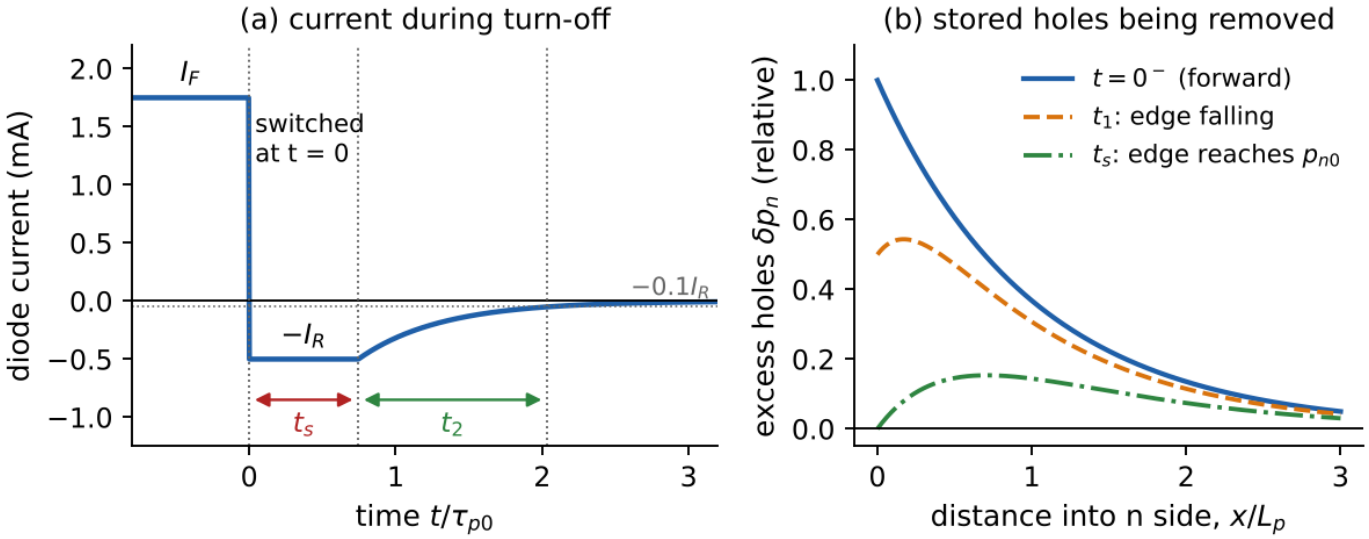


Figure 6: Turn-off of a pn junction diode: the current and the stored holes in the n region

Reading the figure. Panel (a): the solid blue current is $I_F = 1.75$ mA before $t = 0$; at $t = 0$ it jumps to $-I_R = -0.5$ mA and stays there for the storage time t_s (red arrow, $0.745\,\tau_{p0}$), then decays towards zero during the decay time t_2 (green arrow) until it reaches $-0.1I_R$ (dotted grey line). Panel (b): the stored excess holes in the n region; the solid blue curve is the forward profile, the dashed orange curve a moment later (the edge value falling, its slope fixed by the constant I_R), and the dash-dot green curve the end of the storage phase, when the edge value has reached zero ($p_n = p_{n0}$).

What happens during turn-off (three phases):

1. **Storage phase ($0 < t < t_s$):** the stored excess holes keep the junction forward biased, so the junction voltage is small and the reverse current is fixed by the circuit, $I_R \approx V_R/R_R$. The holes are pulled back across the junction (and recombine) until $p_n(x_n)$ falls to p_{n0}. Neamen's result (Eq. 8.109) is

$$\operatorname{erf}\sqrt{\frac{t_s}{\tau_{p0}}} = \frac{I_F}{I_F + I_R}$$

2. **Decay phase ($t_s < t < t_s + t_2$):** the junction becomes reverse biased, the space charge region widens, and the current decays to the small reverse saturation value. The **decay time** t_2 is found from (Eq. 8.111)

$$\operatorname{erf}\sqrt{\frac{t_2}{\tau_{p0}}} + \frac{\exp(-t_2/\tau_{p0})}{\sqrt{\pi t_2/\tau_{p0}}} = 1 + 0.1\left(\frac{I_R}{I_F}\right)$$

3. **Reverse recovery (turn-off) time:** $t_{rr} = t_s + t_2$, the time until the diode really blocks.

Example (Neamen, TYU 8.7)

a one-sided p^+n Si diode with $I_F = 1.75$ mA is switched to $V_R = 2$ V through $R_R = 4$ kΩ; $\tau_{p0} = 10^{-7}$ s.

- $I_R = V_R/R_R = 2/4000 = 0.5$ mA, so $I_F/(I_F + I_R) = 1.75/2.25 = 0.7778$.

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- $\mathrm{erf}(u) = 0.7778$ gives $u = 0.8631$, so $t_s/\tau_{p0} = u^2 = 0.745$:

$$\boxed{t_s = 0.745 \times 10^{-7}\ \mathrm{s} = 74.5\ \mathrm{ns}}$$

- The decay equation with $1 + 0.1(0.5/1.75) = 1.0286$ gives $t_2/\tau_{p0} = 1.29$, $t_2 = 1.29 \times 10^{-7}$ s, so the turn-off time is $t_s + t_2 = 2.03 \times 10^{-7} \approx 0.2\ \mu$s.

Impact of the reverse recovery time in switching applications:

- **Limits the switching speed:** the diode cannot turn off faster than $t_s + t_2$; for a switching period comparable to this time it does not rectify properly. This sets the highest frequency at which a pn diode can be used as a switch or rectifier.
- **Reverse current spike:** during t_s a large reverse current I_R flows through a device that should be blocking. It adds **switching loss** ($P \approx V_R I_R$ during t_{rr}, times the switching frequency), heats the diode and the transistor that switches it, and causes electromagnetic noise (ringing) in power converters.
- **Distortion and delay in logic:** in digital circuits the stored charge delays the turn-off of diodes and saturated bipolar transistors (the same storage effect), which limits logic speed (the reason for Schottky-clamped TTL).
- **Design remedies:** a short minority lifetime (gold doping or electron irradiation adds recombination centres), a small forward current, a large reverse current (t_s falls as I_R/I_F grows), or a majority carrier device such as the **Schottky diode**, which stores almost no minority charge (6(d)).

3. Keyword Explanation

1. Minority carrier lifetime

τ_{p0}, the mean time an excess hole survives in the n region before recombining.

2. Stored charge

The excess minority carriers held near a forward-biased junction, $Q_p = I_F \tau_{p0}$.

3. Decay time

t_2, the time after t_s for the reverse current to fall to 10 % of I_R.

4. Reverse recovery time

$t_{rr} = t_s + t_2$, the total time a diode needs to switch from conducting to blocking.

4. Notes and Remarks

- **The approximate formula:** Neamen also gives $t_s \approx \tau_{p0} \ln(1 + I_F/I_R)$ (Eq. 8.110), from a simpler charge-control model. Here it gives 1.50×10^{-7} s, about twice the erf result; the erf result (Eq. 8.109) is the one the book uses for its answer.
- **Book check:** Neamen's answers to TYU 8.7 are (a) 0.746×10^{-7} s, (b) 1.25×10^{-7} s, (c) $\approx 2 \times 10^{-7}$ s. The storage time agrees; the decay time solved here numerically is 1.29×10^{-7} s (the book reads its value from a plot), and the total agrees.
- **Trend:** t_s/τ_{p0} is 0.093, 0.228 and 0.468 for $I_F/I_R = 0.5$, 1 and 2; a large reverse current empties the store quickly.

2.(b) What is Fermi energy? kT is equal to how many eV at $300k$? Marks: 6

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1. Theory

1. Filling the allowed states (Core Idea)

An electron in a solid can only take certain energies (the allowed states), and each state holds at most one electron (Pauli). In thermal equilibrium the probability that a state at energy E is occupied is the **Fermi-Dirac distribution function** (Neamen, Eq. 3.79)

$$f_F(E) = \frac{1}{1 + \exp\left(\dfrac{E - E_F}{kT}\right)}$$

where

- E_F is the **Fermi energy** (Fermi level), in eV
- $k = 8.617 \times 10^{-5}$ eV/K $= 1.38 \times 10^{-23}$ J/K is Boltzmann's constant
- T is the absolute temperature in K

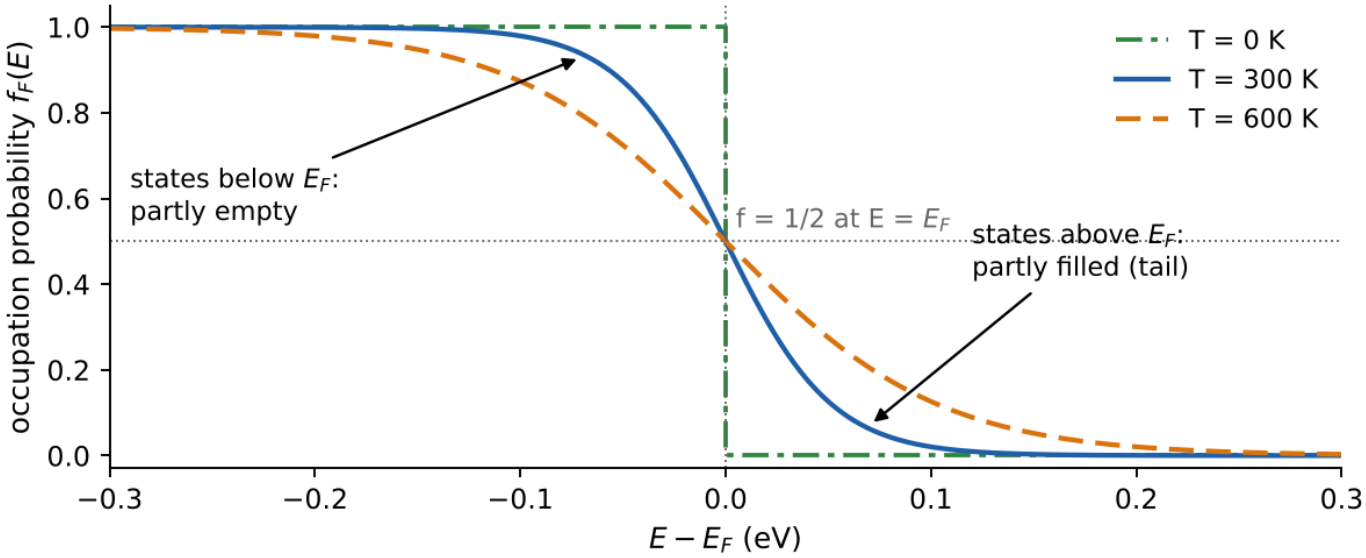


Figure 7: The Fermi-Dirac function at 0 K, 300 K and 600 K

Reading the figure. The dash-dot green line is the sharp step at 0 K. The solid blue curve (300 K) and the dashed orange curve (600 K) all pass through $f_F = 1/2$ at $E = E_F$. The hotter the crystal, the wider the region where the probability changes from 1 to 0. Already $3kT$ (0.078 eV) above E_F, f_F is only 0.047 at 300 K.

2. What kT means

kT is the **thermal energy scale**: the typical energy that random thermal motion gives to one particle. It sets how wide the "blurred" part of the Fermi function is (a few kT), and it appears in every carrier formula, $n_0 = N_c \exp[-(E_c - E_F)/kT]$. A level a few kT above E_F is almost empty; a few kT below, almost full.

2. Answer

Fermi energy: the Fermi energy E_F is the energy level at which the probability of a state being occupied by an electron is exactly one half, $f_F(E_F) = 1/2$. At $T = 0$ K it is the boundary between filled and empty states: every state below E_F is full and every state above it is empty. At $T > 0$ the occupancy changes from nearly 1 to nearly 0 over a few kT around E_F.

Key points:

- In thermal equilibrium E_F is the **same everywhere** in a system (a flat Fermi level), which is the basis of every junction band diagram.

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- Its position tells the carrier densities: the nearer E_F is to E_c, the more electrons; the nearer to E_v, the more holes. In an intrinsic semiconductor it is near midgap (E_{Fi}); donors raise it, acceptors lower it.
- It is the **electrochemical potential** of the electrons: the average energy needed to add one electron to the system.

Value of kT at 300 K:

$$kT = (1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K}) = 4.14 \times 10^{-21}\ \text{J}$$

Divide by $e = 1.6 \times 10^{-19}$ C to convert joules to electron-volts:

$$kT = \frac{4.14 \times 10^{-21}}{1.6 \times 10^{-19}} = 0.02588\ \text{eV}$$

(Or directly: $(8.617 \times 10^{-5}\ \text{eV/K})(300\ \text{K}) = 0.02585$ eV.)

$$\boxed{kT \approx 0.0259\ \text{eV} \approx \frac{1}{40}\ \text{eV at 300 K}}$$

The matching thermal voltage is $V_t = kT/e = 0.0259$ V ≈ 26 mV.

3. Keyword Explanation

1. Fermi-Dirac distribution

The probability $f_F(E)$ that an allowed state at energy E is occupied by an electron.

2. Electron-volt

The energy an electron gains through 1 V: 1 eV $= 1.6 \times 10^{-19}$ J.

3. Thermal voltage

$V_t = kT/e$, 0.0259 V at 300 K.

4. Notes and Remarks

- **Rounded value:** Neamen uses $kT = 0.0259$ eV in all his examples; the exact product $k \times 300$ is 0.02585 eV.
- **Boltzmann limit:** at $E - E_F = 3kT$ the exact f_F is 0.0474 and the Boltzmann form $e^{-3} = 0.0498$, an error of about 5 %; this is why $3kT$ is the usual limit of the approximation.

2.(c) What is Einstein Relation? Derive the expression. Marks: 6

1. Theory

1. Two ways a carrier can move (Core Idea)

Carriers make a current in two ways (Neamen, Eq. 5.35):

- **Drift:** an electric field E pushes them. $J_{n,drift} = en\mu_n E$, where the **mobility** μ_n (cm^2/V s) is the drift velocity per unit field.
- **Diffusion:** they spread from where there are many to where there are few, by random thermal motion alone. $J_{n,diff} = eD_n\, dn/dx$ (Neamen, Eq. 5.33), where the **diffusion coefficient** D_n (cm^2/s) measures how fast they spread.

$$J_n = en\mu_n E_x + eD_n \frac{dn}{dx}$$

2. Why they should be related

Both motions come from the **same random walk** of the carrier: free flight between collisions, a new random direction at each collision. Drift is that random walk with a small push; diffusion is the same

walk with more carriers on one side. Both depend on the mean time between collisions, so μ and D cannot be independent.

3. A graded doping makes a built-in field

To find the link, Neamen (Sec. 5.3) uses a sample in which drift and diffusion must cancel: an n-type semiconductor whose donor concentration $N_d(x)$ falls with x, in thermal equilibrium.

- More donors on the left means more electrons there; electrons **diffuse** to the right.
- The donor ions they leave behind are uncovered, which creates an **induced electric field** E_x pointing to the right.
- The field pushes electrons back to the left. In equilibrium this **drift** exactly cancels the diffusion.

In equilibrium the Fermi level is flat, so E_c rises where N_d is small: a sloping band means a field.

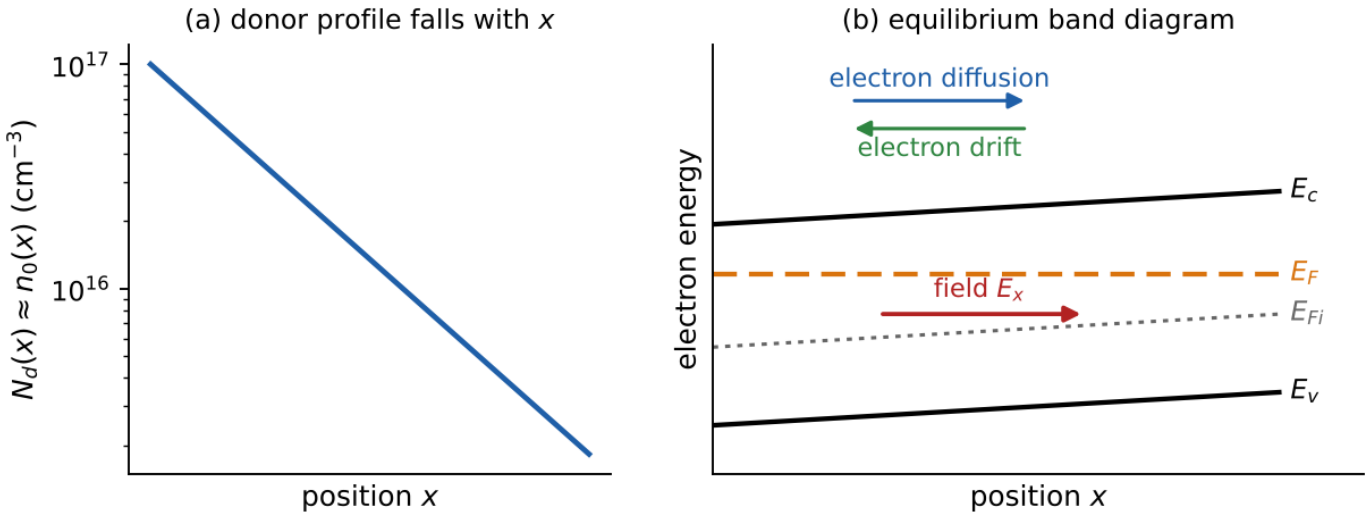


Figure 8: Graded n-type doping: donor profile and the equilibrium band diagram with the induced field

Reading the figure. Panel (a): the solid blue line is the donor concentration, falling with x (log scale). Panel (b): E_F (dashed orange) is flat; E_c and E_v (solid black) and E_{Fi} (dotted grey) all rise to the right, because there are fewer electrons there. The blue arrow shows electrons diffusing to the right (down their concentration gradient); the green arrow shows them drifting to the left, pushed by the induced field (red arrow, pointing to $+x$).

4. Field from a band slope

The electron energy is $E_c = \text{constant} - e\phi$, where ϕ is the electric potential, and $E_x = -d\phi/dx$. So a band edge that rises to the right means a field to the right:

$$E_x = \frac{1}{e}\frac{dE_c}{dx}$$

2. Answer

Einstein relation: in a non-degenerate semiconductor, the diffusion coefficient and the mobility of a carrier are related by

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$$

so the ratio is the thermal voltage, 0.0259 V at 300 K. It holds because drift and diffusion are two results of the same random thermal motion, limited by the same scattering.

Derivation (Neamen, Sec. 5.3). Set-up: a non-uniformly doped n-type semiconductor, $N_d(x)$ decreasing with x, in thermal equilibrium (band diagram in the Theory). Step 3 is a basic calculus step that can be skipped by a student who knows it.

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Step 1: Zero electron current in equilibrium

In thermal equilibrium no current flows, so the drift and diffusion parts of the electron current cancel (Neamen, Eq. 5.43):

$$J_n = en\mu_n E_x + eD_n \frac{dn}{dx} = 0$$

Step 2: The electron concentration in terms of the band

In equilibrium E_F is constant in x, and

$$n = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Only E_c changes with x (it follows the doping).

Step 3 (Basic): Derivative of an exponential

For $y = e^{u(x)}$ the chain rule gives $dy/dx = e^u\, du/dx = y\, du/dx$. Here $u = -(E_c - E_F)/kT$, so $du/dx = -\dfrac{1}{kT}\dfrac{dE_c}{dx}$ (E_F and kT are constants), and

$$\frac{dn}{dx} = -\frac{n}{kT}\frac{dE_c}{dx}$$

Step 4: The field from the band slope

$E_c = \text{const} - e\phi$ and $E_x = -d\phi/dx$ give $dE_c/dx = -e\, d\phi/dx = eE_x$ (Theory 4). Step 3 becomes

$$\frac{dn}{dx} = -\frac{e\,n}{kT}\,E_x$$

Step 5: Substitute into the zero-current condition

Put Step 4 into Step 1:

$$0 = en\mu_n E_x - eD_n \frac{e\,n}{kT}\,E_x = enE_x\left(\mu_n - \frac{eD_n}{kT}\right)$$

The graded doping makes $E_x \neq 0$, and $n \neq 0$, so the bracket must be zero:

$$\boxed{\frac{D_n}{\mu_n} = \frac{kT}{e}}$$

(Neamen, Eq. 5.46a).

Step 6: The same derivation for holes

Take a graded p-type sample, $N_a(x)$ decreasing with x, in equilibrium. The hole current is zero:

$$J_p = ep\mu_p E_x - eD_p \frac{dp}{dx} = 0$$

With $p = N_v \exp[-(E_F - E_v)/kT]$ and only E_v depending on x, the chain rule (Step 3) gives $dp/dx = (p/kT)\, dE_v/dx$. E_v also equals a constant minus $e\phi$, so $dE_v/dx = eE_x$ and $dp/dx = (ep/kT)E_x$. Then

$$0 = ep\mu_p E_x - eD_p \frac{e\,p}{kT}E_x = epE_x\left(\mu_p - \frac{eD_p}{kT}\right) \quad \Rightarrow \quad \boxed{\frac{D_p}{\mu_p} = \frac{kT}{e}}$$

(Neamen, Eq. 5.46b). Together (Neamen, Eq. 5.47):

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e} = V_t = 0.0259 \text{ V at } 300 \text{ K}$$

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Steps to remember

Equilibrium: $J_n = 0 \to n = N_c e^{-(E_c - E_F)/kT}$ with E_F flat $\to dn/dx = -(n/kT)\, dE_c/dx \to dE_c/dx = eE_x \to$ substitute: $\mu_n = eD_n/kT \to$ same for holes.

Use: for Si at 300 K, $D_n = (0.0259)(1350) = 35.0$ cm^2/s and $D_p = (0.0259)(480) = 12.4$ cm^2/s. The mobility is easy to measure (Hall effect, 1(d)); the relation then gives the diffusion coefficient.

3. Keyword Explanation

1. Diffusion coefficient

The constant D (cm^2/s) in the diffusion current $J = \pm eD\, dn/dx$.

2. Mobility

The drift velocity per unit field, $\mu = v_d/E$ (cm^2/V s).

3. Induced electric field

The built-in field of a non-uniformly doped semiconductor, created as majority carriers diffuse away from their ions.

4. Thermal equilibrium

No applied voltage, light or current; the Fermi level is flat and every current is zero.

4. Notes and Remarks

- **Neamen's form:** Neamen writes $n = n_i \exp[(E_F - E_{Fi})/kT]$ and $E_x = (1/e)\, dE_{Fi}/dx$ (Eq. 5.38); the derivation is the same, since E_c and E_{Fi} are parallel.
- **Validity:** non-degenerate statistics (Boltzmann form). In degenerately doped material D/μ is larger than kT/e.
- **Units:** (cm^2/s)/(cm^2/V s) = V, the unit of kT/e ✔.
- The same derivation is 2(b) of the 2020–21 paper (2020–21 2(b)), extra reading only.

2.(d) Wat [*sic*: should be "What"] **is the depletion capacitance of a PN junction? Explain its variation with reverse bias voltage. Marks: 6**

1. Theory

1. The space charge region (Core Idea)

When p-type and n-type silicon meet, electrons diffuse from the n side into the p side and holes the other way. They leave behind uncovered **fixed ions**: positive donor ions on the n side and negative acceptor ions on the p side. This **space charge region** (depletion region) of width $W = x_n + x_p$ has almost no free carriers, and its charge creates the built-in field and the **built-in potential** (Neamen, Eq. 7.10)

$$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

The charge on the two sides is equal and opposite (Neamen, Eq. 7.17): $N_a x_p = N_d x_n$.

2. A reverse bias widens the region

A reverse voltage V_R adds to the barrier: the total potential across the junction becomes $V_{bi} + V_R$ (Eq. 7.33). Solving Poisson's equation for the abrupt junction gives (Neamen, Eq. 7.34)

$$W = \left\{\frac{2\epsilon_s(V_{bi} + V_R)}{e}\left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}$$

So the width, and with it the uncovered charge, grows with V_R.

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3. Capacitance means charge that changes with voltage

A capacitor stores charge that changes with the voltage across it. The junction does the same: when V_R increases by dV_R, the region widens by dx_n on the n side and dx_p on the p side, uncovering an extra charge $dQ' = eN_d\,dx_n = eN_a\,dx_p$ per unit area (Eq. 7.39). The charge sits at the two edges, like the charge on two plates.

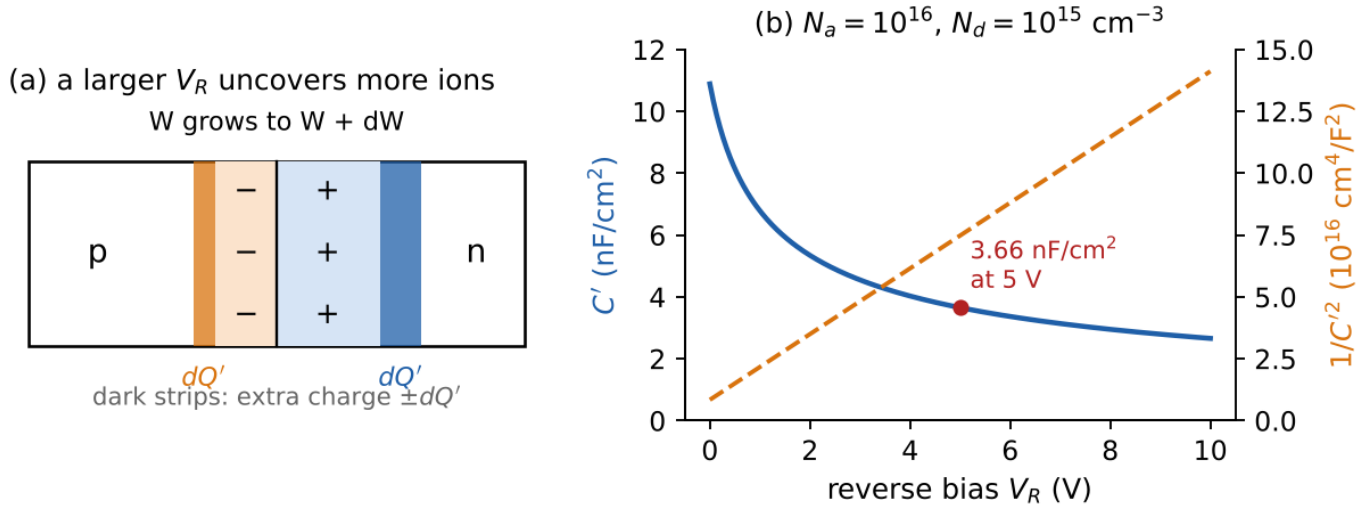


Figure 9: Depletion capacitance: a larger reverse bias uncovers more ions; the capacitance falls with reverse bias

Reading the figure. Panel (a): the light orange negative and light blue positive regions are the space charge at a reverse bias V_R; the dark orange and dark blue strips at the edges are the extra charge $\pm dQ'$ uncovered when V_R increases, so the "plates" move apart. Panel (b): for $N_a = 10^{16}$, $N_d = 10^{15}$ cm^{-3} the solid blue curve (left axis) is C', falling from 10.9 nF/cm^2 at 0 V to 3.66 nF/cm^2 at 5 V (red dot); the dashed orange line (right axis) is $1/C'^2$, a straight line in V_R.

2. Answer

Depletion (junction) capacitance: the capacitance of a reverse-biased pn junction that comes from the change of the fixed space charge in its depletion region with the applied voltage. Per unit area it is defined as (Neamen, Eq. 7.38)

$$C' = \frac{dQ'}{dV_R}$$

where dQ' is the extra charge uncovered on each side when the reverse bias increases by dV_R.

Step 1: The incremental charge

The extra charge per unit area on the n side is $dQ' = eN_d\,dx_n$ (Eq. 7.39), so

$$C' = eN_d\frac{dx_n}{dV_R} \qquad \text{(Eq. 7.41)}$$

Step 2: The edge position under reverse bias

From Poisson's equation (the same solution as at zero bias, with V_{bi} replaced by $V_{bi} + V_R$; Eq. 7.40):

$$x_n = \left\{\frac{2\epsilon_s(V_{bi}+V_R)}{e}\left[\frac{N_a}{N_d}\right]\left[\frac{1}{N_a+N_d}\right]\right\}^{1/2}$$

Step 3: Differentiate

x_n has the form $A(V_{bi}+V_R)^{1/2}$, with A constant, so $dx_n/dV_R = \frac{1}{2}A(V_{bi}+V_R)^{-1/2} = x_n/[2(V_{bi}+V_R)]$. Then

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$$C' = eN_d \frac{x_n}{2(V_{bi}+V_R)} = \left\{ \frac{e\epsilon_s N_a N_d}{2(V_{bi}+V_R)(N_a+N_d)} \right\}^{1/2}$$

(Neamen, Eq. 7.42). Comparing with W gives the simple form (Eq. 7.43)

$$\boxed{C' = \frac{\epsilon_s}{W}}$$

the same as a parallel-plate capacitor with plate spacing W.

Variation with reverse bias voltage:

- As V_R increases, $W \propto (V_{bi}+V_R)^{1/2}$ **increases**, so $C' = \epsilon_s/W$ **decreases**:

$$C' \propto \frac{1}{\sqrt{V_{bi}+V_R}}$$

- For a one-sided p^+n junction ($N_a \gg N_d$; Eq. 7.48)

$$\left(\frac{1}{C'}\right)^2 = \frac{2(V_{bi}+V_R)}{e\epsilon_s N_d}$$

a **straight line** against V_R, whose slope gives the doping N_d of the lightly doped side and whose intercept on the voltage axis is $-V_{bi}$ (the usual C-V measurement).

- **Forward bias** narrows W and raises C', but there the diffusion capacitance of the stored minority charge soon dominates.

Example (Neamen, Example 7.5)

Si, $N_a = 10^{16}$, $N_d = 10^{15}$ cm^{-3}, $T = 300$ K. $V_{bi} = 0.0259 \ln[(10^{16})(10^{15})/(1.5\times10^{10})^2] = 0.635$ V. At $V_R = 5$ V:

$$C' = \left\{ \frac{(1.6\times10^{-19})(11.7)(8.85\times10^{-14})(10^{16})(10^{15})}{2(0.635+5)(10^{16}+10^{15})} \right\}^{1/2} = 3.66\times10^{-9} \text{ F/cm}^2$$

For an area of 10^{-4} cm^2, $C = 0.366$ pF. At 0, 1, 5 and 10 V, C' is 10.9, 6.79, 3.66 and 2.66 nF/cm^2 and W is 0.95, 1.53, 2.83 and 3.89 μm: the capacitance falls as the width grows.

Use: a reverse-biased junction is a voltage-controlled capacitor (the **varactor** diode) for tuning circuits.

3. Keyword Explanation

1. Space charge region

The region around the junction emptied of free carriers, holding the fixed ions.

2. Built-in potential

$V_{bi} = V_t \ln(N_a N_d/n_i^2)$, the potential barrier of the junction at zero bias.

3. One-sided junction

A junction with one side much more heavily doped; the depletion region lies almost entirely in the lightly doped side.

4. Varactor

A diode used as a capacitor whose value is set by the reverse bias.

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4. Notes and Remarks

- **Why reverse bias:** under reverse bias almost no current flows, so the junction behaves as a nearly ideal capacitor; under forward bias the large current and the diffusion capacitance hide it.
- **Graded junction:** for a linearly graded junction $C' \propto (V_{bi} + V_R)^{-1/3}$ (Neamen, Sec. 7.5); the exponent tells the doping profile.
- **Book check:** Neamen's Example 7.5 gives $C' = 3.66 \times 10^{-9}$ F/cm^2 and $C = 0.366$ pF, as above.

5. Final Concise Answer

(a) Storage delay time t_s: after a forward-biased diode is switched to reverse bias, the time during which the reverse current stays at $-I_R \approx -V_R/R_R$ while the stored minority charge ($Q_p = I_F \tau_{p0}$) is removed, until $p_n(x_n) = p_{n0}$; $\text{erf}\sqrt{t_s/\tau_{p0}} = I_F/(I_F + I_R)$ (TYU 8.7: $t_s = 74.5$ ns, $t_2 = 129$ ns, $t_{rr} \approx 0.2$ μs). The reverse recovery time $t_s + t_2$ limits the switching speed and frequency, causes a reverse current spike, switching loss and noise; it is reduced by a short lifetime, a large I_R/I_F, or a Schottky diode.

(b) E_F is the energy at which a state is occupied with probability 1/2; at 0 K all states below it are full and all above it empty; it is flat through a system in equilibrium. $kT = (1.38 \times 10^{-23})(300)/(1.6 \times 10^{-19}) = 0.0259$ eV $\approx 1/40$ eV at 300 K.

(c) $D_n/\mu_n = D_p/\mu_p = kT/e$ (0.0259 V at 300 K). Derived from a graded n-type sample in equilibrium: $J_n = en\mu_n E_x + eD_n\, dn/dx = 0$, with $n = N_c e^{-(E_c - E_F)/kT}$, $dn/dx = -(n/kT)\, dE_c/dx = -(en/kT)E_x$, which gives $\mu_n = eD_n/kT$.

(d) $C' = dQ'/dV_R = \{e\epsilon_s N_a N_d/[2(V_{bi} + V_R)(N_a + N_d)]\}^{1/2} = \epsilon_s/W$: the capacitance of the fixed charge uncovered at the edges of the space charge region. It falls as $(V_{bi} + V_R)^{-1/2}$ with reverse bias because W grows; $1/C'^2$ is linear in V_R (Example 7.5: 3.66 nF/cm^2 at 5 V).

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3. (a) With brief description sketch the energy band diagram of a MOS structure at different bias voltage. (Use no bias condition, positive bias voltage and negative bias voltage). Marks: 8
   (b) What are the parameters that affect the magnitude of threshold voltage in a MOSFET? Marks: 6
   (c) What is body effect of MOSFET? Marks: 5
   (d) Explain the amplification action of a transistor. Marks: 6

3.(a) With brief description sketch the energy band diagram of a MOS structure at different bias voltage. (Use no bias condition, positive bias voltage and negative bias voltage). Marks: 8

1. Theory

1. The MOS structure (Core Idea)

A **MOS** (metal-oxide-semiconductor) structure is a sandwich:

- a **metal** gate (aluminium, or heavily doped polysilicon) on top;
- a thin **oxide** (SiO_2, a few nm to tens of nm thick), a very good insulator with a band gap of about 9 eV;
- the **semiconductor** body, here p-type silicon, with a contact at its back.

No current flows through the oxide, so the structure is a capacitor. The gate voltage V_G puts charge Q'_m on the metal; an equal and opposite charge must appear in the semiconductor just under the oxide. **What kind** of charge appears there (holes, ionized acceptors, or electrons) depends on the sign and size of V_G, and the band diagram shows it.

2. Rules for drawing the band diagram

- **Equilibrium (no current):** the Fermi level is flat inside the semiconductor, because no current flows into it (the oxide blocks any current).
- **Gate voltage moves the metal Fermi level:** a voltage V_G between gate and body separates the two Fermi levels by eV_G. A **positive** V_G **lowers** the metal Fermi level E_{Fm} by eV_G relative to E_F of the semiconductor; a negative V_G raises it.
- **Band bending:** where the potential in the semiconductor changes near the surface, all its bands (E_c, E_{Fi}, E_v) bend together by the **surface potential** $e\phi_s$. A positive surface potential bends them *down*.
- **Carrier densities follow the distance to E_F:** $p = n_i \exp[(E_{Fi} - E_F)/kT]$ and $n = n_i \exp[(E_F - E_{Fi})/kT]$. Where E_v comes closer to E_F there are more holes; where E_{Fi} drops below E_F there are more electrons than holes.

3. The ideal structure

To keep the picture simple, take the **ideal** case: the metal and semiconductor work functions are equal ($\phi_{ms} = 0$) and there is no charge in the oxide. Then at $V_G = 0$ nothing happens: the bands are flat (the **flat-band** condition). In a real device a gate voltage $V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$ (Neamen, Eq. 10.25) is needed to make them flat; every statement below then holds with V_G measured from V_{FB}.

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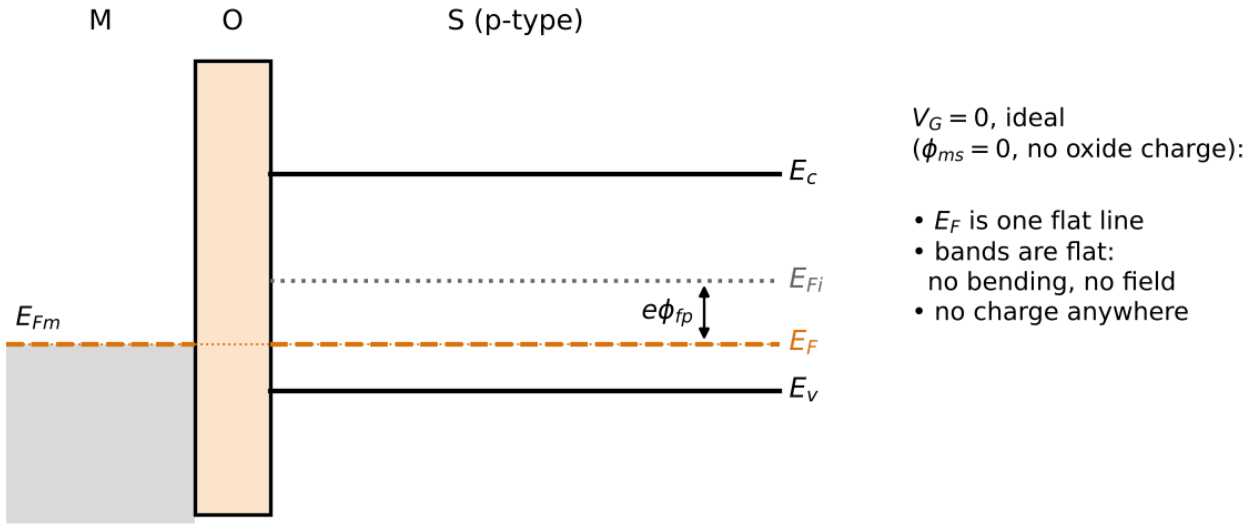


Figure 10: Ideal MOS structure on p-type silicon at zero gate voltage

Reading the figure. From left to right: the metal (grey, filled up to E_{Fm}), the oxide (tall orange box) and the p-type silicon. The dashed orange Fermi level is one straight line through the whole structure. E_c and E_v (solid black) and E_{Fi} (dotted grey) are flat; E_F lies $e\phi_{fp}$ below E_{Fi}, as in any p-type material. No band bends, so there is no field and no charge anywhere.

4. The bulk Fermi potential

The distance of E_F below E_{Fi} in the p-type body is written $e\phi_{fp}$ (Neamen, Eq. 10.4):

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right)$$

For $N_a = 10^{16}$ cm^{-3}, $\phi_{fp} = 0.347$ V. The surface is **inverted** when the bands bend down so far that E_{Fi} at the surface falls below E_F; strong inversion (threshold) is reached when $\phi_s = 2\phi_{fp}$ (Neamen, Sec. 10.1.2).

2. Answer

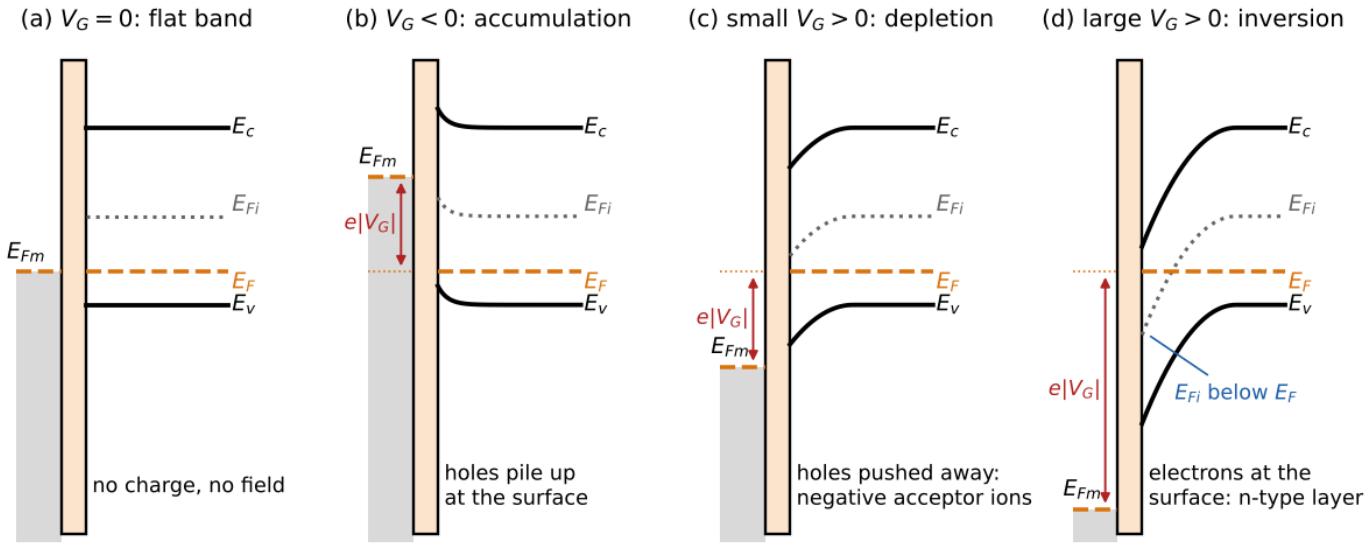


Figure 11: Energy band diagrams of an ideal MOS capacitor on p-type silicon: zero, negative and positive gate voltages

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Reading the figure. Each panel shows the metal (grey, its Fermi level E_{Fm} dashed orange), the oxide (orange box) and the p-type silicon with E_c, E_v (solid black), E_{Fi} (dotted grey) and E_F (dashed orange). The red arrow $e|V_G|$ is the offset of E_{Fm} from E_F. (a) $V_G = 0$: all bands flat. (b) $V_G < 0$: E_{Fm} is raised, and the semiconductor bands bend *up* at the surface, bringing E_v close to E_F. (c) small $V_G > 0$: E_{Fm} is lowered and the bands bend *down*. (d) large $V_G > 0$: the bands bend down so far that E_{Fi} (blue label) crosses below E_F at the surface.

The structure is metal / SiO_2 / p-type Si, ideal ($\phi_{ms} = 0$, no oxide charge).

1. No bias, $V_G = 0$ (flat band), panel (a).

- The metal and semiconductor Fermi levels line up; there is no voltage across the oxide.
- No charge on the gate, so no charge in the semiconductor: the bands are **flat** right up to the oxide.
- The hole concentration at the surface equals that in the bulk, $p_s = N_a$.

2. Negative bias, $V_G < 0$ (accumulation), panel (b).

- A negative gate puts negative charge on the metal; its Fermi level moves *up* by $e|V_G|$.
- The negative gate charge **attracts holes** (the majority carriers) to the oxide-silicon interface, where they form a thin **accumulation layer** of positive charge.
- The bands **bend upwards** near the surface: E_v comes closer to E_F, and $p = n_i \exp[(E_{Fi} - E_F)/kT]$ shows the larger hole density there.
- Because holes respond at once and sit right at the surface, the capacitance is that of the oxide, $C' = C_{ox}$.

3. Small positive bias, $0 < V_G < V_T$ (depletion), panel (c).

- A positive gate puts positive charge on the metal; its Fermi level moves *down* by eV_G.
- The positive gate **repels holes** from the surface. They leave behind the uncovered negatively charged acceptor ions over a **depletion layer** of width $x_d = (2\epsilon_s\phi_s/eN_a)^{1/2}$ (Neamen, Eq. 10.5); its charge $-eN_ax_d$ balances the gate charge.
- The bands **bend downwards** by $e\phi_s$; E_v moves away from E_F, so the surface holds fewer holes.

4. Large positive bias, $V_G \geq V_T$ (inversion), panel (d).

- The bands bend down so far that E_{Fi} at the surface falls **below** E_F. There E_F is closer to E_c than to E_v, so the surface behaves as **n-type**: an **inversion layer** of electrons (minority carriers) forms under the oxide.
- **Threshold (strong) inversion** is reached when $\phi_s = 2\phi_{fp}$: the electron density at the surface equals the hole density in the bulk. Beyond this the depletion width stops growing at $x_{dT} = (4\epsilon_s\phi_{fp}/eN_a)^{1/2}$ (Eq. 10.6), and extra gate charge is balanced by more electrons in the inversion layer.
- This electron layer is the **channel** of an n-channel MOSFET.

Summary: negative bias → bands up, holes accumulate; small positive bias → bands down, holes depleted, ions exposed; large positive bias → E_{Fi} crosses E_F, electrons form an inversion layer. (For an n-type body every sign is reversed.)

3. Keyword Explanation

1. Flat band

The condition with no band bending in the semiconductor; $V_G = V_{FB}$ (zero in the ideal structure).

2. Accumulation

Majority carriers gathered at the semiconductor surface by a gate voltage that attracts them.

3. Depletion

Majority carriers pushed away from the surface, leaving fixed ionized dopants.

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4. Inversion

The surface turned to the opposite type, with a layer of minority carriers, once E_{Fi} crosses E_F.

5. Surface potential

ϕ_s, the total band bending (in volts) from the bulk to the surface.

4. Notes and Remarks

- **Real gates:** with an aluminium gate on p-type Si, $\phi_{ms} \approx -0.9$ V, so the bands are already bent down at $V_G = 0$ and the flat-band voltage is negative (Neamen, Fig. 10.14); the three cases then sit around V_{FB} instead of 0.
- **Oxide drop:** part of V_G drops across the oxide (V_{ox}) and the rest bends the silicon bands (ϕ_s): $V_G = V_{ox} + \phi_s + \phi_{ms}$ (Neamen, Eq. 10.21). The panels show E_{Fm} shifted by the whole eV_G.
- **Numbers:** for $N_a = 10^{16}$ cm^{-3}, $\phi_{fp} = 0.347$ V and threshold inversion needs $\phi_s = 0.695$ V, with $x_{dT} = 0.30$ μm.
- The same diagrams appear in 4(b) of the 2020–21 paper (2020–21 4(b)), extra reading only.

3.(b) What are the parameters that affect the magnitude of threshold voltage in a MOSFET? Marks: 6

1. Theory

1. What the threshold voltage is (Core Idea)

The **threshold voltage** V_T of an n-channel MOSFET is the gate voltage at which an electron inversion layer (the channel) just forms under the gate: the surface potential reaches $\phi_s = 2\phi_{fp}$ (Neamen, Sec. 10.1.2). Below V_T the transistor is off; above it, the channel conducts.

2. Where the gate voltage goes

At threshold the gate voltage must do three jobs:

- **Bend the bands** by $2\phi_{fp}$ to invert the surface.
- **Support the charge** in the semiconductor: the depletion charge $|Q'_{SD}(\max)| = eN_a x_{dT}$ (Eq. 10.27) is balanced by gate charge, which needs a voltage across the oxide capacitance $C_{ox} = \epsilon_{ox}/t_{ox}$. Any fixed positive oxide charge Q'_{ss} helps, because it already supplies part of the balancing charge.
- **Make up the work function difference** ϕ_{ms} between gate and semiconductor, which is present even at $V_G = 0$.

Adding the three (Neamen, Eqs. 10.28 and 10.30):

$$V_{TN} = (|Q'_{SD}(\max)| - Q'_{ss})\left(\frac{t_{ox}}{\epsilon_{ox}}\right) + \phi_{ms} + 2\phi_{fp}$$

with

$$\phi_{fp} = V_t \ln\left(\frac{N_a}{n_i}\right), \qquad x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}, \qquad |Q'_{SD}(\max)| = eN_a x_{dT}$$

The flat-band voltage collects the last two charge-free terms: $V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$, so $V_{TN} = |Q'_{SD}(\max)|/C_{ox} + V_{FB} + 2\phi_{fp}$.

2. Answer

The threshold voltage of an n-channel MOSFET is (Neamen, Eq. 10.30)

$$V_{TN} = \frac{|Q'_{SD}(\max)| - Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp}$$

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Every symbol in it is a parameter that affects V_T:

1. **Substrate (body) doping N_a:** it sets $\phi_{fp} = V_t \ln(N_a/n_i)$ and the depletion charge $|Q'_{SD}(\max)| = \sqrt{4e\epsilon_s N_a \phi_{fp}}$. Higher doping → more charge to support and more band bending → **higher** V_{TN}. This is the main design knob, adjusted by ion implantation (threshold adjust implant).
2. **Oxide thickness t_{ox} (oxide capacitance $C_{ox} = \epsilon_{ox}/t_{ox}$):** the charge term is divided by C_{ox}. A thicker oxide (smaller C_{ox}) **magnifies** the charge term, so when the depletion charge exceeds the oxide charge (the usual case) V_{TN} **rises**. A thin oxide shrinks the charge term, and V_{TN} comes close to $\phi_{ms} + 2\phi_{fp}$.
3. **Oxide permittivity ϵ_{ox}:** a high-κ dielectric raises C_{ox} at the same thickness, with the same effect as a thinner oxide.
4. **Fixed oxide (interface) charge Q'_{ss}:** positive charge trapped in the oxide or at the interface supplies part of the charge that balances the depletion charge, so it **lowers** V_{TN} by Q'_{ss}/C_{ox}. A clean oxide process keeps it small (about 10^{10} to $10^{11}\, e/\text{cm}^2$).
5. **Metal-semiconductor work function difference ϕ_{ms} (gate material):** set by the gate material and the body doping. An Al gate on p-type Si gives $\phi_{ms} \approx -0.9$ V; an n^+ polysilicon gate about -1.1 V; a p^+ polysilicon gate about $+0.2$ to $+0.3$ V. Changing the gate material shifts V_T directly by the change in ϕ_{ms}.
6. **Temperature:** through n_i and ϕ_{fp}; V_T falls by a few mV per K as the temperature rises.
7. **Source-to-body voltage V_{SB} (body effect):** a reverse bias between source and body widens the depletion layer and **raises** V_T (3(c)).
8. **Channel length and width** (short- and narrow-channel effects): in short channels the source and drain share part of the depletion charge, so V_T falls; in narrow channels the edge depletion charge raises it.

Example (Neamen, Example 10.4)

p-type Si, $N_a = 10^{15}$ cm^{-3}, $Q'_{ss} = 10^{10}\, e/\text{cm}^2$, $t_{ox} = 12$ nm, Al gate ($\phi_{ms} = -0.88$ V):

- $\phi_{fp} = (0.0259)\ln(10^{15}/1.5 \times 10^{10}) = 0.2877$ V
- $x_{dT} = [4(11.7)(8.85 \times 10^{-14})(0.2877)/(1.6 \times 10^{-19})(10^{15})]^{1/2} = 8.63 \times 10^{-5}$ cm
- $|Q'_{SD}(\max)| = (1.6 \times 10^{-19})(10^{15})(8.63 \times 10^{-5}) = 1.381 \times 10^{-8}$ C/cm^2
- $C_{ox} = (3.9)(8.85 \times 10^{-14})/(12 \times 10^{-7}) = 2.876 \times 10^{-7}$ F/cm^2

$$V_{TN} = \frac{1.381 \times 10^{-8} - 1.6 \times 10^{-9}}{2.876 \times 10^{-7}} - 0.88 + 2(0.2877) = 0.0480 - 0.0056 - 0.88 + 0.5754 = -0.262 \text{ V}$$

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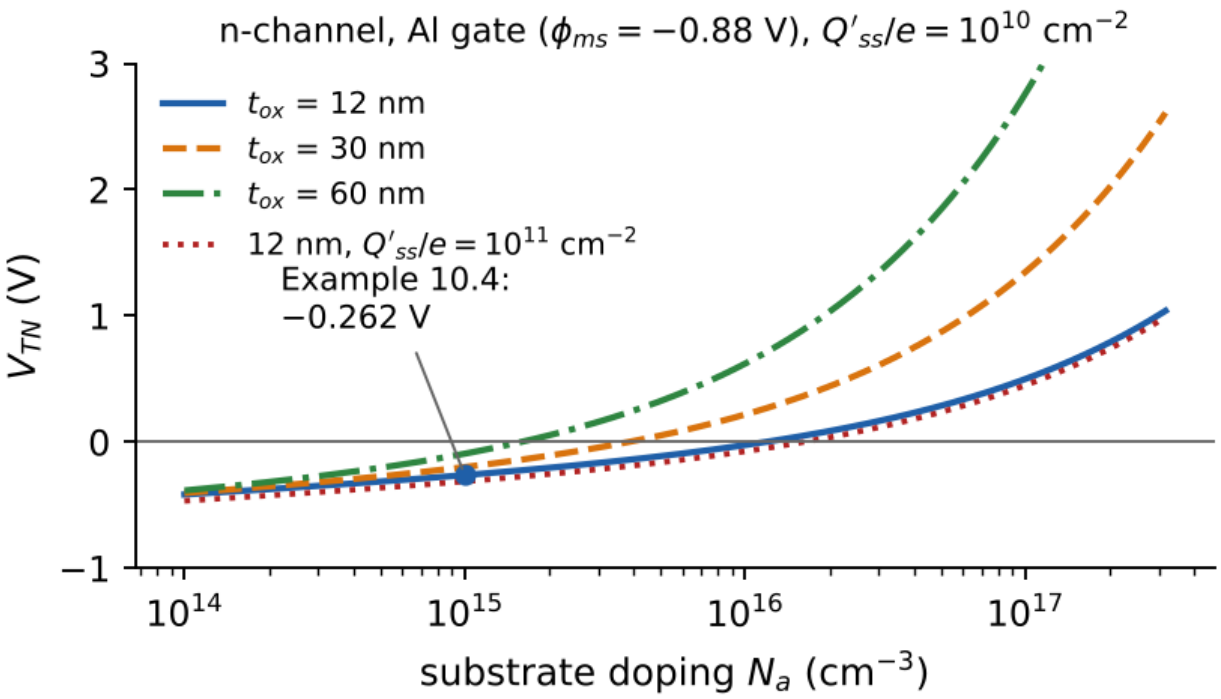


Figure 12: Threshold voltage against substrate doping for three oxide thicknesses and two oxide charges

Reading the figure. Each curve is V_{TN} from the formula above with an Al gate. The solid blue curve is $t_{ox} = 12$ nm with $Q'_{ss}/e = 10^{10}$ cm^{-2}, the dashed orange 30 nm and the dash-dot green 60 nm; the dotted red curve is 12 nm with ten times more oxide charge. V_{TN} rises with doping, and more steeply for a thicker oxide; more oxide charge lowers it. The blue dot is Example 10.4 (-0.262 V at 10^{15} cm^{-3}).

3. Keyword Explanation

1. Threshold inversion

The point $\phi_s = 2\phi_{fp}$ at which the surface electron density equals the bulk hole density.

2. Work function difference

$\phi_{ms} = \phi'_m - (\chi' + E_g/2e + \phi_{fp})$ (Neamen, Eq. 10.15), the built-in potential between gate and body.

3. Oxide charge

Fixed positive charge Q'_{ss} at or near the Si-SiO_2 interface.

4. Oxide capacitance

$C_{ox} = \epsilon_{ox}/t_{ox}$ per unit area.

4. Notes and Remarks

- **Sign:** a negative V_{TN} (as in Example 10.4) means the channel already exists at $V_G = 0$: a **depletion-mode** device. Raising N_a to about 10^{16} cm^{-3} makes it an enhancement-mode device (0 V at 1.2×10^{16} cm^{-3} for 12 nm in the figure).
- **p-channel:** $V_{TP} = (-|Q'_{SD}(\text{max})| - Q'_{ss})(t_{ox}/\epsilon_{ox}) + \phi_{ms} - 2\phi_{fn}$ (Neamen, Eq. 10.32), with the same parameters.
- **Book check:** Neamen's Example 10.4 gives $V_{TN} = -0.262$ V, as above.

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3.(c) What is body effect of MOSFET? Marks: 5

1. Theory

1. The four terminals of a MOSFET (Core Idea)

A MOSFET has four terminals: gate, source, drain and **body** (substrate, B). In an n-channel MOSFET the body is p-type and the source and drain are n^+ regions in it. The source-body and drain-body junctions must never be forward biased, so the body is tied to the most negative voltage in the circuit. When the body is tied to the source, $V_{SB} = 0$. In integrated circuits many transistors share one body, and the sources of some of them sit above it, so $V_{SB} > 0$ (a reverse bias on the source-body junction).

2. Threshold needs a certain depletion charge

At threshold the gate must support the depletion charge under the channel, $|Q'_{SD}(\max)| = eN_a x_{dT}$ (Eq. 10.27). With $V_{SB} = 0$ the band bending at inversion is $2\phi_{fp}$, and

$$|Q'_{SD}(\max)| = \sqrt{2e\epsilon_s N_a (2\phi_{fp})}$$

(Neamen, Eq. 10.78). The question is what V_{SB} does to this charge.

2. Answer

Body effect: the change of the threshold voltage of a MOSFET when a voltage is applied between the source and the body (substrate). For an n-channel MOSFET a reverse bias $V_{SB} > 0$ **increases** the threshold voltage. It is also called the **substrate bias effect** or back-gate effect.

Why it happens (Neamen, Sec. 10.3.5):

- With $V_{SB} > 0$ the inversion layer (connected to the source) is at a higher potential than the body. The induced pn junction between the channel and the body is reverse biased by V_{SB}.
- To form the inversion layer the surface must now be bent by $2\phi_{fp} + V_{SB}$ instead of $2\phi_{fp}$: the electrons of the channel would otherwise flow out to the more positive source.
- The depletion layer under the channel becomes **wider**, so it holds *more* negative charge (Eq. 10.79):

$$|Q'_{SD}| = \sqrt{2e\epsilon_s N_a (2\phi_{fp} + V_{SB})}$$

- The gate must supply this extra charge, so a larger gate voltage is needed to reach inversion.

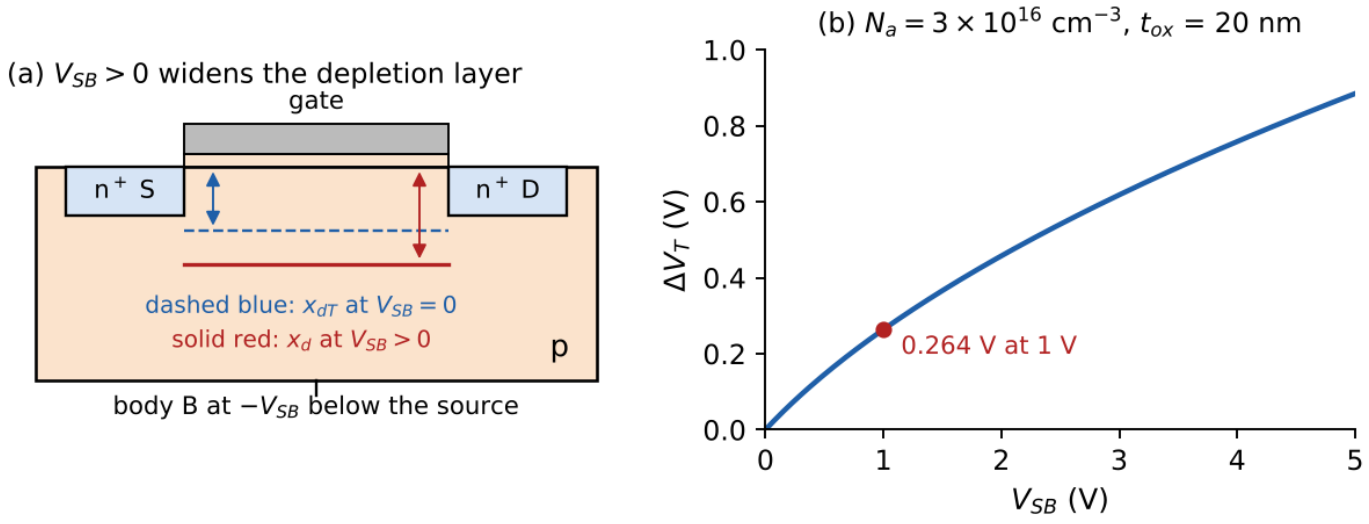


Figure 13: The body effect: a source-to-body bias widens the depletion layer and raises the threshold voltage

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Reading the figure. Panel (a): an n-channel MOSFET; the dashed blue line is the edge of the depletion layer at threshold with $V_{SB} = 0$ (x_{dT}) and the solid red line its edge with $V_{SB} > 0$ (wider, x_d). Panel (b): the solid blue curve is the threshold shift ΔV_T against V_{SB} for $N_a = 3 \times 10^{16}$ cm^{-3} and $t_{ox} = 20$ nm; it grows as a square root, 0.264 V at $V_{SB} = 1$ V (red dot).

Step 1: The change in depletion charge

$$\Delta Q'_{SD} = \sqrt{2e\epsilon_s N_a}\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right]$$

Step 2: The threshold shift

The gate supplies this charge across C_{ox}, so $\Delta V_T = \Delta Q'_{SD}/C_{ox}$ (Neamen, Eq. 10.81):

$$\boxed{\Delta V_T = \gamma\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right], \qquad \gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}}$$

γ is the **body-effect coefficient** (V$^{1/2}$, Eq. 10.82).

Example (Neamen, Example 10.9)

$N_a = 3 \times 10^{16}$ cm^{-3}, $t_{ox} = 20$ nm, $V_{SB} = 1$ V:

- $\phi_{fp} = 0.0259 \ln(3 \times 10^{16}/1.5 \times 10^{10}) = 0.3758$ V, $C_{ox} = 1.726 \times 10^{-7}$ F/cm^2
- $\gamma = [2(1.6 \times 10^{-19})(11.7)(8.85 \times 10^{-14})(3 \times 10^{16})]^{1/2}/1.726 \times 10^{-7} = 0.578$ V$^{1/2}$
- $\Delta V_T = 0.578[\sqrt{0.7516 + 1} - \sqrt{0.7516}] = 0.264$ V

Effects:

- The threshold rises with V_{SB} (here by 0.26 V at 1 V and 0.88 V at 5 V), so the drain current at a given gate voltage falls.
- In stacked transistors (NAND gates, pass transistors, source followers) the upper transistors have $V_{SB} > 0$ and are slower; a source follower's gain falls below 1.
- The body can be used as a **second gate**: varying V_{SB} tunes V_T (adaptive body bias for leakage control).
- A lightly doped body and a thin oxide (small γ) reduce the effect.

3. Keyword Explanation

1. Body (substrate)

The semiconductor region in which the channel forms; the fourth terminal of the MOSFET.

2. Body-effect coefficient

$\gamma = \sqrt{2e\epsilon_s N_a}/C_{ox}$, which sets how strongly V_{SB} shifts V_T.

3. Back gate

The body acting as a second (weaker) gate through the depletion charge.

4. Notes and Remarks

- **Book check:** Neamen's Example 10.9 gives $\gamma = 0.5776$ V$^{1/2}$ and $\Delta V_T = 0.264$ V, as above.
- **p-channel:** with $V_{BS} > 0$ (n-type body above the source) $|V_{TP}|$ grows in the same way.
- The depletion width grows from 0.180 μm at $V_{SB} = 0$ to 0.275 μm at 1 V in this example.

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3.(d) Explain the amplification action of a transistor. Marks: 6

1. Theory

1. The bipolar transistor (Core Idea)

A bipolar junction transistor (BJT) has three regions, here an **npn**: a heavily doped n^+ **emitter**, a thin p-type **base** and an n-type **collector**, forming two junctions. In the **forward-active mode** the base-emitter (B-E) junction is forward biased and the base-collector (B-C) junction reverse biased.

2. How the current flows

- The forward-biased B-E junction **injects** electrons from the emitter into the base, in number $\propto \exp(eV_{BE}/kT)$.
- The base is very thin compared with the electron diffusion length, so almost all these electrons diffuse across it without recombining.
- At the reverse-biased B-C junction the field sweeps them into the collector. This is the **collector current**:

$$i_C = I_S \exp\left(\frac{v_{BE}}{V_t}\right)$$

(Neamen, Sec. 12.1), set by the B-E voltage and almost independent of the collector voltage.

- The base current is small: holes injected from base to emitter plus holes that replace recombined electrons. So $i_C = \beta i_B$ with β about 100.

3. Why this can amplify

The input (the B-E junction) is a forward-biased diode: a small voltage change makes a large current change, because of the exponential. The output (the collector) is a reverse-biased junction: it delivers that current almost regardless of the collector voltage, like a current source, so it can drive a large resistor and develop a large voltage.

2. Answer

Amplification action: a small change of the base-emitter voltage (or base current) produces a large change of the collector current, which, through a load resistor, gives a much larger output voltage. This is possible because the input junction (B-E, forward biased) has a low resistance and the output junction (B-C, reverse biased) a very high one: the same current is transferred from a low-resistance to a high-resistance circuit (**trans-resistor**, hence "transistor").

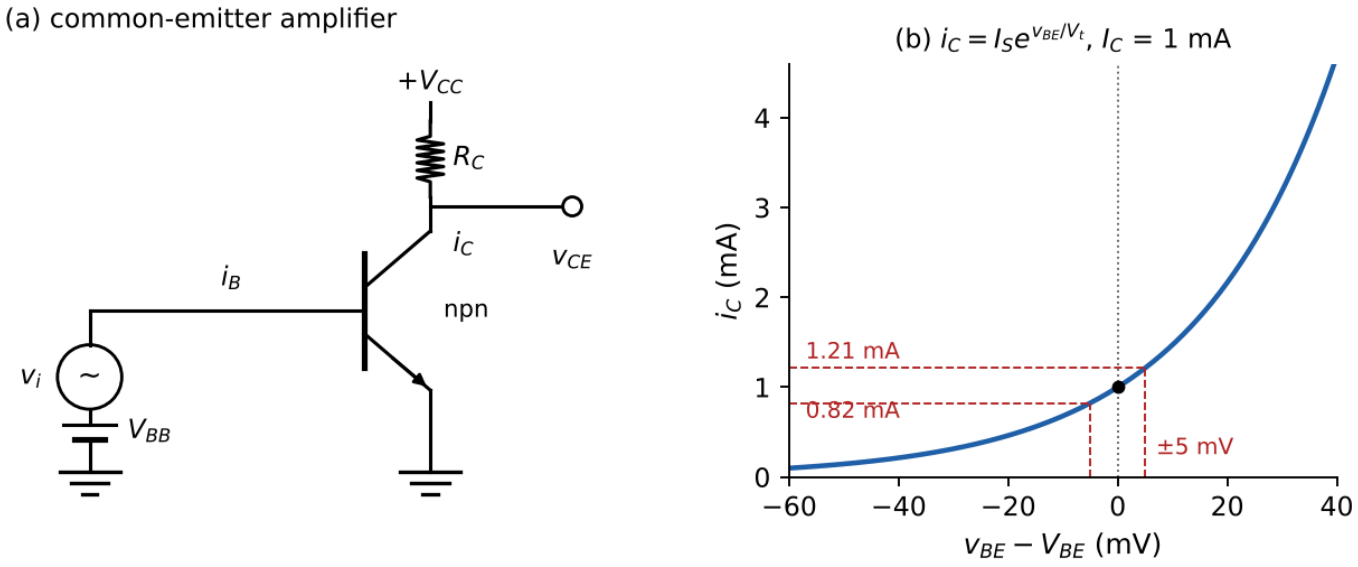


Figure 14: Common-emitter amplifier and the exponential collector characteristic

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Reading the figure. Panel (a): an npn transistor in the common-emitter circuit; the input signal v_i in series with the bias V_{BB} drives the base, and the collector is fed from $+V_{CC}$ through R_C; the output is v_{CE}. Panel (b): the solid blue curve is $i_C = I_S e^{v_{BE}/V_t}$ around the bias point $I_C = 1$ mA (black dot). A change of only ± 5 mV in v_{BE} (red dashed lines) moves i_C between 0.82 and 1.21 mA.

How it works (npn, common emitter):

1. **Bias:** V_{BB} forward biases the B-E junction (about 0.6 to 0.7 V for Si); V_{CC} through R_C keeps the B-C junction reverse biased (forward-active mode).
2. **Input:** the signal v_i adds a small v_{be} to V_{BE}. The injected electron current varies as $\exp(v_{BE}/V_t)$.
3. **Transport:** nearly all injected electrons cross the thin base and are collected; only a small fraction recombines in the base, so $i_c = \beta i_b$.
4. **Output:** the collector current flows through R_C, so $v_{CE} = V_{CC} - i_C R_C$; the signal part is $v_{ce} = -i_c R_C$, much larger than v_{be}.

Gains: for small signals $i_c = g_m v_{be}$ with the **transconductance** $g_m = I_C/V_t$, so the voltage gain is

$$A_v = \frac{v_{ce}}{v_{be}} = -g_m R_C = -\frac{I_C R_C}{V_t}$$

Example

$I_C = 1$ mA, $\beta = 100$, $R_C = 5$ kΩ, $V_t = 0.0259$ V:

- $g_m = 1/0.0259 = 38.6$ mA/V
- $v_{be} = 5$ mV peak $\rightarrow i_c = (38.6)(0.005) = 0.193$ mA peak and $i_b = i_c/\beta = 1.93$ μA
- $v_{ce} = -i_c R_C = -0.965$ V peak, so $A_v = -193$

A 5 mV input gives nearly 1 V at the output (the minus sign: the output is inverted). The current gain is $\beta = 100$ and the power gain $|A_v|\beta \approx 1.9 \times 10^4$; the extra power comes from the supply V_{CC}, which the small input signal controls.

3. Keyword Explanation

1. Forward-active mode

B-E junction forward biased, B-C junction reverse biased; the mode used for amplification.

2. Common-emitter current gain

$\beta = i_C/i_B$, typically 50 to 300.

3. Transconductance

$g_m = di_C/dv_{BE} = I_C/V_t$, the change of collector current per volt of input.

4. Notes and Remarks

- **Small-signal limit:** the linear gain holds for $v_{be} \ll V_t$; at 5 mV the curve is already a little unequal (+0.21 mA, −0.18 mA), as panel (b) shows.
- **Base transport and emitter injection:** a high β needs a thin base and an emitter much more heavily doped than the base (6(b)).
- **MOSFET:** a field-effect transistor amplifies in the same way, with the gate voltage controlling the channel current ($g_m = 2I_D/(V_{GS} - V_T)$ in saturation).

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5. Final Concise Answer

(a) Ideal metal/SiO_2/p-Si: at $V_G = 0$ the Fermi level is flat and the bands are flat (no charge). $V_G < 0$ raises E_{Fm}, attracts holes and bends the bands *up* (accumulation). Small $V_G > 0$ lowers E_{Fm}, repels holes, uncovers acceptor ions over x_d and bends the bands *down* (depletion). Large $V_G > 0$ bends them so far that E_{Fi} falls below E_F at the surface: an electron inversion layer (threshold at $\phi_s = 2\phi_{fp}$).

(b) $V_{TN} = (|Q'_{SD}(\text{max})| - Q'_{ss})/C_{ox} + \phi_{ms} + 2\phi_{fp}$: substrate doping N_a (through ϕ_{fp} and Q'_{SD}), oxide thickness and permittivity (C_{ox}), fixed oxide charge Q'_{ss}, gate material (ϕ_{ms}), temperature, source-body bias (body effect) and channel dimensions. Example 10.4: $V_{TN} = -0.262$ V.

(c) A source-to-body reverse bias V_{SB} widens the depletion layer under the channel, so more charge must be supported and V_T rises: $\Delta V_T = \gamma[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}]$, $\gamma = \sqrt{2e\epsilon_s N_a}/C_{ox}$ (Example 10.9: $\gamma = 0.578$ $V^{1/2}$, $\Delta V_T = 0.264$ V at 1 V).

(d) In forward-active mode the forward-biased B-E junction injects carriers $\propto e^{v_{BE}/V_t}$, nearly all cross the thin base and are collected at the reverse-biased B-C junction; a small input voltage (or base current) controls a large collector current, which develops a large voltage across R_C: $A_v = -g_m R_C = -I_C R_C/V_t$ (-193 for 1 mA and 5 kΩ), $i_c = \beta i_b$.

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4. (a) Sketch the ideal electron and hole current components through a *pn* junction under forward bias. Marks: 7
   (b) What are the factors which cause base current in a transistor? Marks: 3
   (c) Differentiate between ohmic and rectifying contacts. Marks: 4
   (d) What is base width modulation? Marks: 5
   (e) Explain the working of Tunnel diode. Draw its characteristics curve. Marks: 6

4.(a) Sketch the ideal electron and hole current components through a *pn* junction under forward bias. Marks: 7

1. Theory

1. Forward bias injects minority carriers (Core Idea)

In a pn junction at zero bias, the built-in barrier eV_{bi} stops the majority carriers from crossing. A forward voltage V_a (p side positive) lowers the barrier to $e(V_{bi} - V_a)$. Then:

- electrons from the n side cross into the p side, where they are **minority carriers**;
- holes from the p side cross into the n side, also as minority carriers.

This is **minority carrier injection**. At the edges of the space charge region the minority concentrations rise to (the **law of the junction**, Neamen, Eqs. 8.6 and 8.7)

$$n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right), \qquad p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

where $n_{p0} = n_i^2/N_a$ and $p_{n0} = n_i^2/N_d$ are the equilibrium minority concentrations.

2. The injected carriers diffuse and recombine

In the neutral regions there is almost no field, so the injected minority carriers move by **diffusion**, and they recombine with the majority carriers as they go. The steady-state diffusion equation $D_p\, d^2\delta p_n/dx^2 - \delta p_n/\tau_{p0} = 0$ (Neamen, Eq. 8.9) has the decaying solution (Eq. 8.14)

$$\delta p_n(x) = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right] \exp\left(\frac{x_n - x}{L_p}\right), \quad x \geq x_n$$

and similarly for electrons on the p side, with the **diffusion lengths** $L_p = \sqrt{D_p\tau_{p0}}$, $L_n = \sqrt{D_n\tau_{n0}}$.

3. Minority current is diffusion current

The minority hole current in the n region is a diffusion current (Neamen, Eqs. 8.20 to 8.22):

$$J_p(x) = -eD_p\frac{d(\delta p_n)}{dx} = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right] \exp\left(\frac{x_n - x}{L_p}\right)$$

It is largest at x_n and falls exponentially, because the holes recombine. The electron current on the p side behaves the same way, mirrored.

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2. Answer

Assumptions of the ideal diode (Neamen, Sec. 8.1): abrupt junction, low injection, no generation or recombination inside the space charge region, so each current component is **constant across the space charge region**; neutral regions longer than the diffusion lengths.

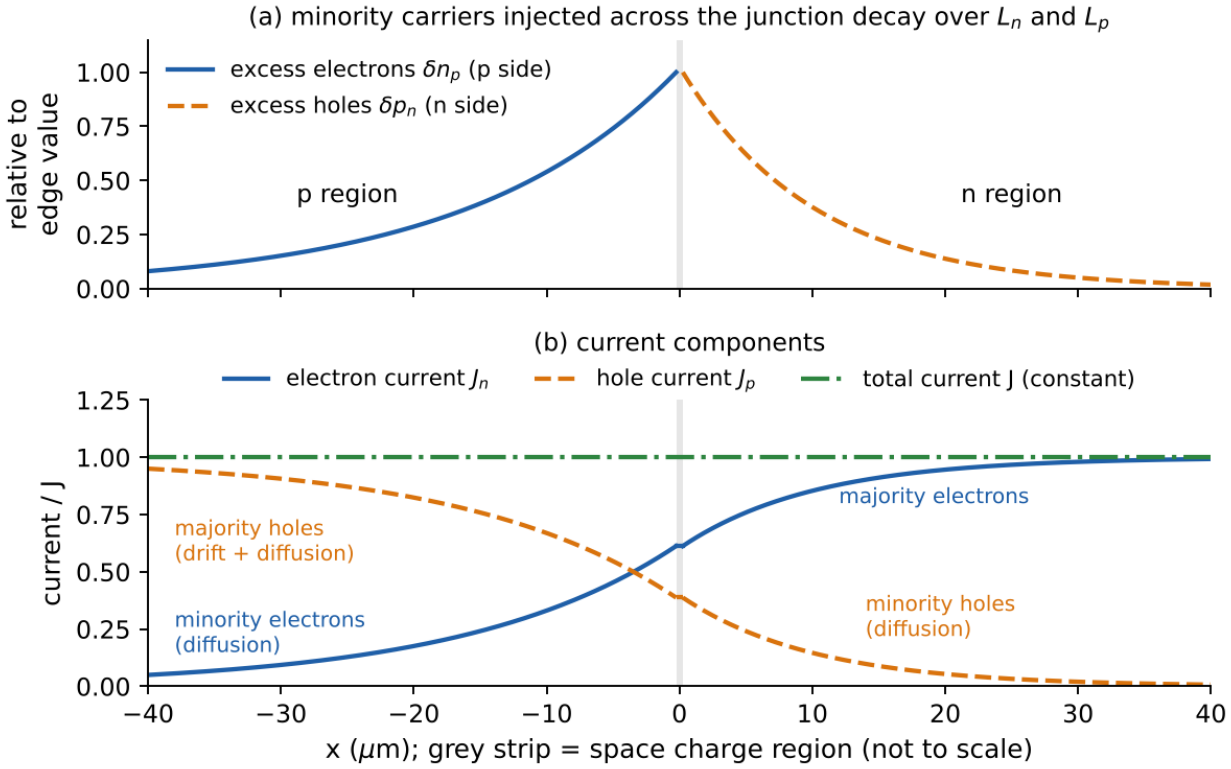


Figure 15: Ideal forward-biased pn junction: excess minority carriers and the electron and hole current components

Reading the figure. Silicon, $N_a = N_d = 10^{16}$ cm^{-3}, $V_a = 0.6$ V, $L_n = 15.8$ μm, $L_p = 10$ μm; the grey strip is the space charge region (not to scale). Panel (a): the solid blue curve is the excess electron concentration on the p side and the dashed orange curve the excess hole concentration on the n side, each relative to its edge value, decaying over L_n and L_p. Panel (b): the dash-dot green line is the total current J, constant everywhere. The solid blue curve is the electron current and the dashed orange curve the hole current. On the p side (left) the electron current is minority diffusion current that falls away from the junction, and the hole current (majority) rises to make up the total; on the n side (right) the roles are swapped.

How the components are drawn:

1. **Minority diffusion currents at the edges** (Neamen, Eqs. 8.22 and 8.24):

$$J_p(x_n) = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right], \qquad J_n(-x_p) = \frac{eD_n n_{p0}}{L_n}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

2. **Through the space charge region** both are constant (no recombination there), so the **total current** is their sum (Eq. 8.25):

$$J = J_p(x_n) + J_n(-x_p) = J_s\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right], \qquad J_s = \frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}$$

3. **In the n region ($x > x_n$):** the minority hole current decays as $e^{-(x-x_n)/L_p}$ as the holes recombine. Each recombining hole takes one electron, so the **majority electron current** rises by the same amount: $J_n(x) = J - J_p(x)$. Far from the junction the whole current is carried by electrons (drift plus diffusion of majority carriers).

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4. **In the p region ($x < -x_p$):** the minority electron current decays as $e^{(x+x_p)/L_n}$, and the **majority hole current** makes up the rest, $J_p(x) = J - J_n(x)$.
5. **The total current J is the same at every x** (a series circuit in steady state).

Numbers for the figure: $n_{p0} = p_{n0} = 2.25 \times 10^4$ cm^{-3}, $D_n = 25$, $D_p = 10$ cm^2/s, $\tau = 10^{-7}$ s:

$$J_n(-x_p) = 0.655 \text{ A/cm}^2, \qquad J_p(x_n) = 0.414 \text{ A/cm}^2, \qquad J = 1.069 \text{ A/cm}^2$$

so electrons carry 61 % and holes 39 % of the current across the junction (electrons diffuse faster, $D_n > D_p$, at equal doping).

3. Keyword Explanation

1. Minority carrier injection

The flow of carriers across a forward-biased junction into the region where they are minorities.

2. Diffusion length

$L = \sqrt{D\tau}$, the mean distance a minority carrier diffuses before recombining (5(c)).

3. Majority carrier current

The current carried by the majority carriers in a neutral region; it supplies the carriers that recombine with the injected minorities.

4. Reverse saturation current density

$J_s = eD_p p_{n0}/L_p + eD_n n_{p0}/L_n$.

4. Notes and Remarks

- **One-sided junction:** in a p$^+$n diode $p_{n0} \gg n_{p0}$, so nearly all current across the junction is hole injection into the n side.
- **Real diodes:** recombination in the space charge region adds a current $\propto \exp(eV_a/2kT)$ (Neamen, Sec. 8.2), which dominates at small forward bias; the ideal components above leave it out.
- **Neamen's figure:** this is the content of Neamen's Fig. 8.10 (ideal electron and hole current components in a forward-biased junction).
- The same currents, with the full derivation of the diode equation, are 5(b) of the 2019–20 paper (2019–20 5(b)), extra reading only.

4.(b) What are the factors which cause base current in a transistor? Marks: 3

1. Theory

1. The forward-active npn transistor (Core Idea)

In an npn transistor in the forward-active mode the B-E junction is forward biased and the B-C junction reverse biased. The emitter injects electrons into the thin p-type base; they diffuse across it and are swept into the collector. The base terminal supplies **holes**, the majority carriers of the base. Every process that uses up holes in the base, or sends them out of it, must be fed by the base current. The current components are named in Neamen's Fig. 12.13.

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2. Answer

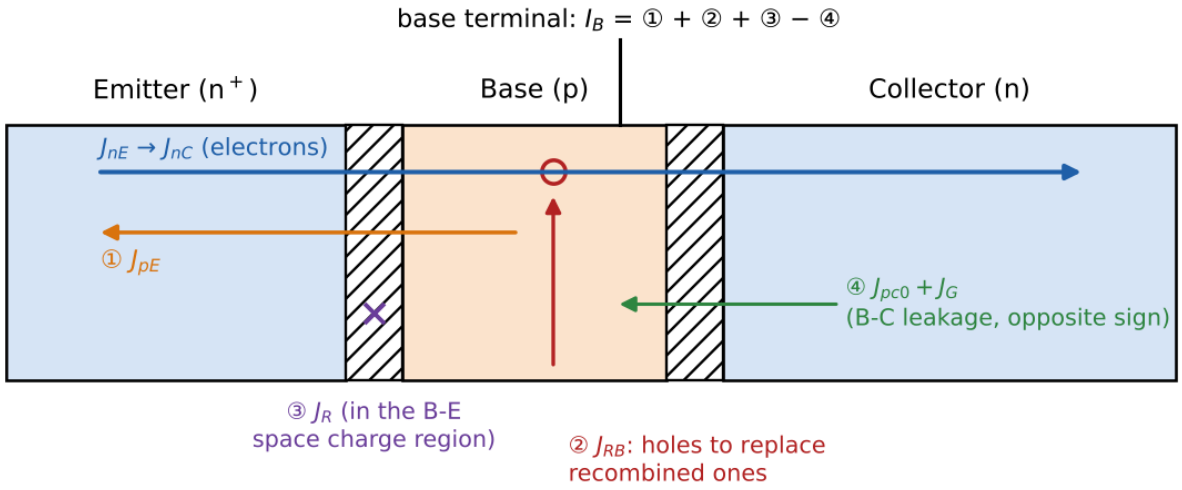


Figure 16: The current components of an npn transistor in the forward-active mode and the parts that make up the base current

Reading the figure. The blue arrow is the electron current injected from the emitter that reaches the collector ($J_{nE} \rightarrow J_{nC}$); the red circle marks recombination in the base. The numbered arrows are the base-current components: ① orange, holes injected from the base into the emitter (J_{pE}); ② red, holes that replace those lost by recombination in the base (J_{RB}); ③ purple cross, recombination in the B-E space charge region (J_R); ④ green, the B-C reverse current ($J_{pc0} + J_G$), which flows the opposite way. The base current is made up of:

1. **Hole injection into the emitter (J_{pE}):** the forward-biased B-E junction also injects holes from the base into the emitter. They come from the base contact. This is the main component in a good transistor, and is reduced by doping the emitter much more heavily than the base (6(b)).
2. **Recombination in the base (J_{RB}):** some of the electrons crossing the base recombine with holes there; each lost hole must be replaced through the base terminal. It is reduced by a thin base ($x_B \ll L_B$).
3. **Recombination in the B-E space charge region (J_R):** electrons and holes recombining inside the forward-biased depletion layer draw holes from the base. It matters most at low currents.
4. **The B-C reverse current ($J_{pc0} + J_G$, opposite sign):** thermally generated carriers and minority holes of the collector swept into the base by the reverse-biased B-C junction supply holes to the base and so **reduce** the external base current (this is I_{CBO}).

$$I_B \approx I_{pE} + I_{RB} + I_R - I_{CBO}$$

3. Keyword Explanation

1. Emitter injection efficiency

$\gamma = J_{nE}/(J_{nE} + J_{pE})$, the fraction of emitter current carried by the useful electrons (6(b)).

2. Base transport factor

$\alpha_T = J_{nC}/J_{nE}$, the fraction of injected electrons that cross the base without recombining.

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4. Notes and Remarks

- **Gain:** $\beta = I_C/I_B$ is high (about 100) only because all four components are small compared with I_C; each factor above corresponds to one of γ, α_T and the recombination factor δ in $\alpha = \gamma\alpha_T\delta$ (Neamen, Sec. 12.3.1).

4.(c) Differentiate between ohmic and rectifying contacts. Marks: 4

1. Theory

1. Work function and electron affinity (Core Idea)

The **work function** $e\phi$ of a material is the energy needed to take an electron from its Fermi level to the vacuum level (just outside the material). For a metal it is $e\phi_m$; for a semiconductor $e\phi_s = e\chi + (E_c - E_F)$, where the **electron affinity** $e\chi$ is the distance from the conduction band edge to the vacuum level (4.01 eV for Si).

2. What happens at contact

When a metal touches a semiconductor, electrons flow from the side with the **higher Fermi level** (smaller work function) to the other until the Fermi levels line up. This leaves a space charge in the semiconductor and bends its bands near the interface:

- For an **n-type** semiconductor with $\phi_m > \phi_s$, electrons leave the semiconductor; a depletion layer of positive donor ions forms, and the bands bend *up*. A barrier $e\phi_{B0} = e(\phi_m - \chi)$ (Neamen, Eq. 9.1) appears between the metal Fermi level and E_c at the interface.
- For an n-type semiconductor with $\phi_m < \phi_s$, electrons flow into the semiconductor and pile up at the surface (accumulation); the bands bend *down* and there is no barrier.

2. Answer

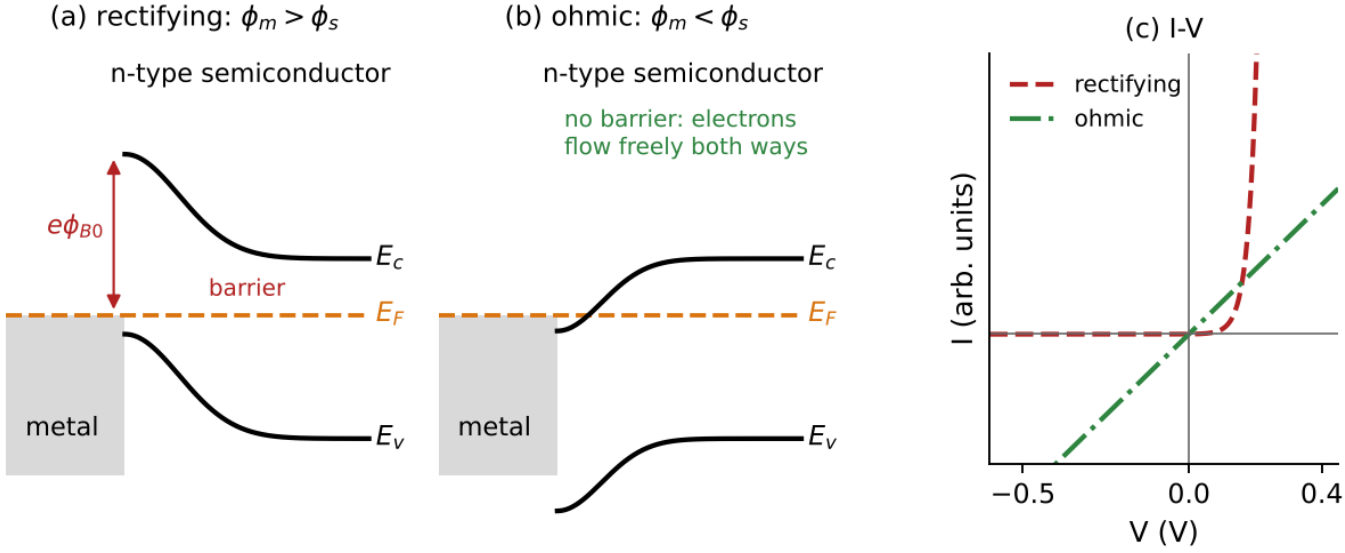


Figure 17: Rectifying and ohmic metal contacts on n-type silicon and their I-V curves

Reading the figure. Panel (a): rectifying contact, $\phi_m > \phi_s$: the bands of the n-type semiconductor (solid black) bend up near the metal, and the red arrow marks the barrier $e\phi_{B0}$ from the dashed orange Fermi level to E_c at the interface. Panel (b): ohmic contact, $\phi_m < \phi_s$: the bands bend down, so electrons meet no barrier. Panel (c): the dashed red I-V curve of the rectifying contact is diode-like; the dash-dot green line of the ohmic contact is straight through the origin.

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Point	Ohmic contact	Rectifying (Schottky) contact
Definition	Metal-semiconductor contact with a negligibly small resistance compared with the bulk, conducting equally in both directions	Metal-semiconductor contact with a potential barrier that passes current easily in one direction only
Condition (n-type)	$\phi_m < \phi_s$ (accumulation), or very heavy doping so that the barrier is thin enough to tunnel through	$\phi_m > \phi_s$: barrier $e\phi_{B0} = e(\phi_m - \chi)$
Band diagram	Bands bend down (or barrier only a few nm thick); no barrier to electron flow	Bands bend up; depletion layer and Schottky barrier
I-V curve	Linear, symmetric: $V = IR_c$	Diode-like: $J = J_{sT}[\exp(eV_a/kT) - 1]$ (Neamen, Eq. 9.25)
Specific contact resistance	Very small ($R_c \leq 10^{-5}\ \Omega\ \text{cm}^2$)	Large in reverse, small in forward
Use	Connecting devices to the outside circuit (every terminal of a transistor or diode)	Schottky diodes, clamps, MESFET gates, fast rectifiers

Table 1: Summary

3. Keyword Explanation

1. Work function

The energy from the Fermi level to the vacuum level.

2. Schottky barrier height

$e\phi_{B0} = e(\phi_m - \chi)$, the barrier electrons in the metal see at an ideal rectifying contact on n-type material.

3. Tunnelling contact

An ohmic contact made by doping the semiconductor so heavily ($\gtrsim 10^{19}\ \text{cm}^{-3}$) that the depletion layer is thin enough for electrons to tunnel through it (Neamen, Sec. 9.2).

4. Notes and Remarks

- **p-type:** the conditions reverse: a rectifying contact needs $\phi_m < \phi_s$ and an ohmic contact $\phi_m > \phi_s$.
- **In practice** surface states pin the barrier height, so most metal contacts on Si are rectifying unless the surface is heavily doped; real ohmic contacts are almost always tunnelling contacts.
- **Example**

  n-Si with $N_d = 10^{16}\ \text{cm}^{-3}$ has $\phi_s = \chi + (E_c - E_F)/e = 4.01 + 0.206 = 4.216$ V; tungsten ($\phi_m = 4.55$ V) on it gives a rectifying contact with $\phi_{B0} = 0.54$ V (6(d)).
- The same comparison is 1(d) of the 2019–20 paper (2019–20 1(d)), extra reading only.

4.(d) What is base width modulation? Marks: 5

1. Theory

1. The neutral base (Core Idea)

The base of a bipolar transistor is the p region between two depletion layers (npn). The part that has no depletion is the **neutral base**, of width x_B. The electrons injected from the emitter diffuse across it. If $x_B \ll L_B$, their concentration falls almost linearly from $n_B(0) = n_{B0}\exp(eV_{BE}/kT)$ at the emitter

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edge to about zero at the collector edge, and the collector current is the diffusion current (Neamen, Example 12.8):

$$|J_C| = eD_B \left|\frac{dn_B}{dx}\right| \approx \frac{eD_B n_{B0}}{x_B} \exp\left(\frac{eV_{BE}}{kT}\right)$$

So J_C is inversely proportional to the neutral base width.

2. The B-C depletion layer depends on V_{CB}

The B-C junction is reverse biased. Its depletion layer reaches into the base by x_{dB}, which grows with V_{CB} like any junction width ($\propto \sqrt{V_{bi} + V_{CB}}$). So the neutral base width $x_B = x_{B0} - x_{dB}$ depends on the collector voltage.

2. Answer

Base width modulation (the Early effect): the change of the neutral base width of a bipolar transistor caused by the change of the B-C space charge width with the collector-base (or collector-emitter) voltage. A larger reverse bias V_{CB} widens the B-C depletion layer into the base, so the **neutral base becomes narrower**.

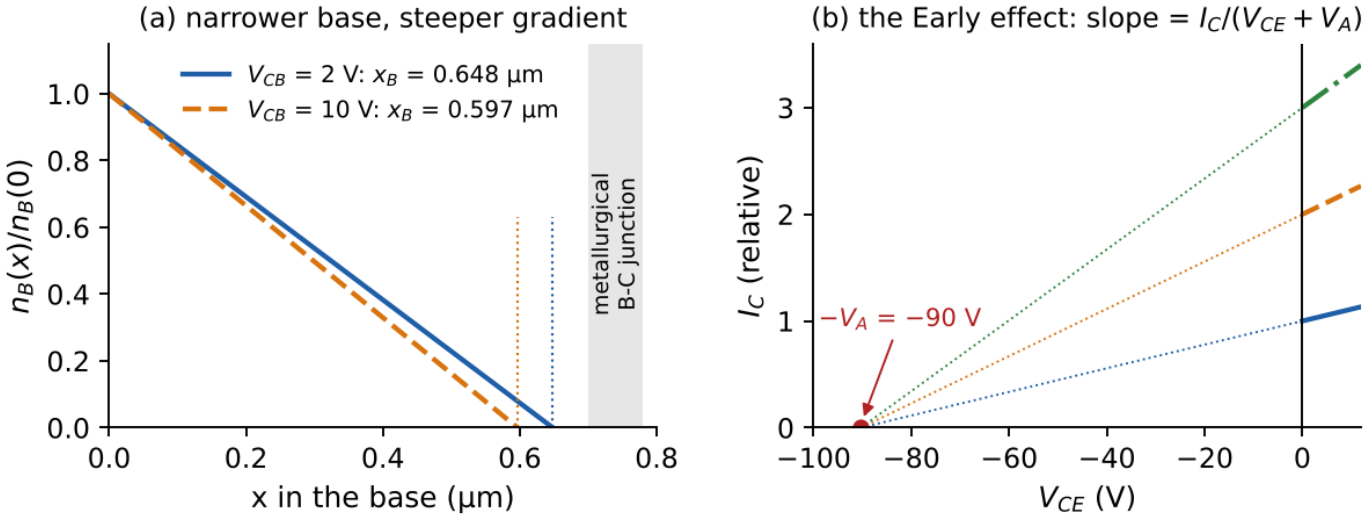


Figure 18: Base width modulation: narrower neutral base and the Early effect on the output characteristics

Reading the figure. Panel (a): the minority electron concentration in the base for Neamen's Example 12.8; the solid blue line (V_{CB} = 2 V, x_B = 0.648 μm) and the dashed orange line (V_{CB} = 10 V, x_B = 0.597 μm) start at the same value, set by V_{BE}, but the second reaches zero sooner, so its slope (the collector current) is larger. The grey strip is the metallurgical B-C junction. Panel (b): output characteristics I_C against V_{CE} for three base drives; the solid lines are the forward-active region, and their dotted extensions all meet the voltage axis at $-V_A = -90$ V (red dot), the Early voltage.

Effects:

1. **Collector current rises with V_{CE}:** a narrower base gives a steeper electron gradient, so I_C increases instead of staying constant: the output characteristics slope upwards. Their extensions meet the V_{CE} axis at $-V_A$, the **Early voltage** (typically 100 to 300 V); the slope is (Neamen, Eq. 12.45a)

$$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE} + V_A}$$

2. **Finite output resistance:** $r_o = V_A/I_C$ instead of infinity; this limits the voltage gain of an amplifier.
3. **Current gain rises with V_{CE}:** less recombination in a narrower base increases α_T and β.

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4. **Punch-through:** at a large enough V_{CB} the B-C depletion layer reaches the B-E one ($x_B = 0$) and the transistor loses control: a breakdown mechanism.

Example (Neamen, Example 12.8)

Si npn, $N_B = 5 \times 10^{16}$, $N_C = 2 \times 10^{15}$ cm^{-3}, $x_{B0} = 0.70$ μm, $D_B = 25$ cm^2/s, $V_{BE} = 0.60$ V. With $n_{B0} = n_i^2/N_B = 4.5 \times 10^3$ cm^{-3}:

- $V_{CB} = 2$ V: $x_{dB} = 0.0518$ μm, $x_B = 0.648$ μm, $J_C = 3.195$ A/cm^2
- $V_{CB} = 10$ V: $x_{dB} = 0.103$ μm, $x_B = 0.597$ μm, $J_C = 3.470$ A/cm^2

An 8 V change raises J_C by 8.6 %; $\Delta J_C/\Delta V = 0.0344 = J_C/(0.6 + 2 + V_A)$ gives $V_A \approx 90$ V.

Reduced by: a more heavily doped base than collector (the depletion layer then spreads mainly into the collector), and a wider base (at the cost of gain).

3. Keyword Explanation

1. Neutral base width

x_B, the width of the base outside the two depletion layers.

2. Early voltage

V_A, the voltage at which the extended output characteristics meet the V_{CE} axis (at $-V_A$).

3. Punch-through

The B-C depletion layer reaching the B-E depletion layer, so the neutral base disappears.

4. Notes and Remarks

- **Book check:** Neamen's Example 12.8 gives $J_C = 3.195$ and 3.469 A/cm^2 and $V_A = 90.7$ V; computed here without rounding the intermediate values, $V_A = 90.3$ V.
- The depletion width into the base is $x_{dB} = \{[2\epsilon_s(V_{bi} + V_{CB})/e][N_C/N_B][1/(N_C + N_B)]\}^{1/2}$, the junction formula of 2(d) for the base side; $V_{bi} = 0.695$ V here.

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4.(e) Explain the working of Tunnel diode. Draw its characteristics curve. Marks: 6

1. Theory

1. Tunnelling (Core Idea)

In classical physics an electron that meets an energy barrier higher than its own energy is always turned back. Quantum mechanically the electron is a wave. Inside the barrier the wave does not stop at once; it **decays** exponentially. If the barrier is thin enough, part of the wave is still there on the other side, so the electron has a real chance of appearing beyond it without ever having the energy to go over it. This is **tunnelling** (Neamen, Sec. 2.3.4). The transmission probability falls roughly as $\exp(-2\kappa a)$ for a barrier of width a, so it is only important when a is a few nanometres.

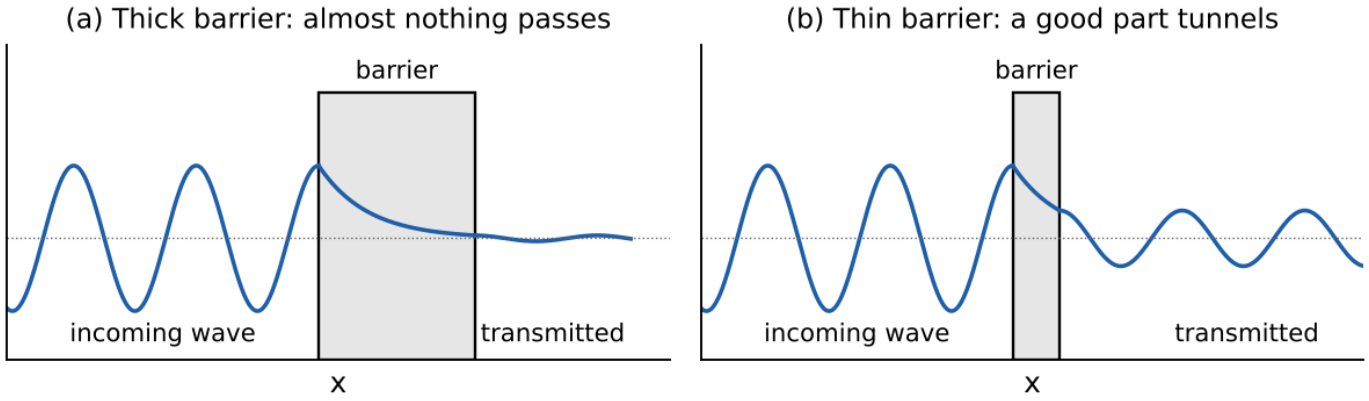


Figure 19: An electron wave meeting a thick and a thin energy barrier

Reading the figure. The solid blue curve is the electron wave. In panel (a) it decays inside the wide grey barrier to almost nothing, so the wave on the right is very small. In panel (b) the barrier is narrow; the wave has not decayed much when it reaches the far side, so a clear wave continues: the electron tunnels.

2. Degenerate doping makes the junction thin

The width of a junction depletion region falls as the doping rises, $W \propto 1/\sqrt{N}$. In a **tunnel diode** both sides are **degenerately** doped ($\gtrsim 10^{19}$ cm^{-3}):

- the Fermi level lies **inside** the conduction band on the n^+ side and inside the valence band on the p^+ side;
- the depletion region is only about 5 to 10 nm wide, thin enough for electrons to tunnel straight through it.

3. Tunnelling needs filled states facing empty states

An electron tunnels horizontally in the band diagram (at constant energy). A tunnelling current flows only where **filled states on one side face empty allowed states at the same energy on the other side**. The bias changes how much the two bands overlap, and that sets the current.

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2. Answer

Tunnel diode: a pn junction diode in which both the p and n regions are degenerately doped, so that the depletion region is very thin and electrons can tunnel through it; its forward I-V characteristic has a region of **negative differential resistance** (Neamen, Sec. 8.5).

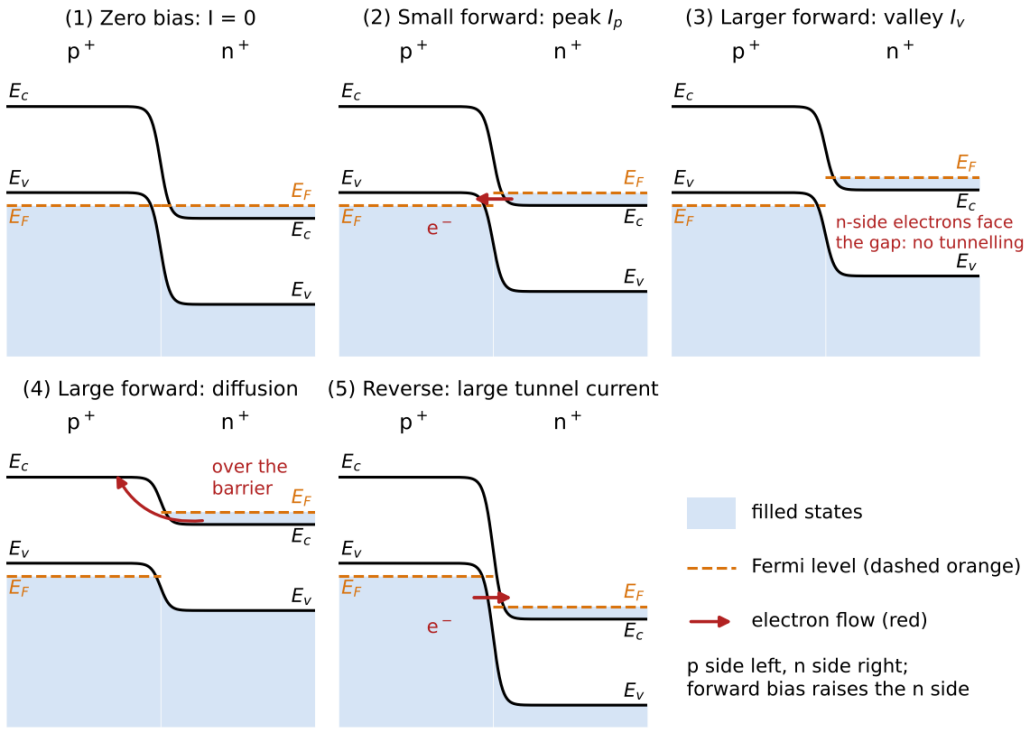


Figure 20: Band diagrams of a tunnel diode at zero bias, small forward bias, larger forward bias, large forward bias and reverse bias

Reading the figure. The p^+ side is on the left and the n^+ side on the right; solid black lines are E_c and E_v, the dashed orange lines the Fermi levels, and the shaded areas the filled states. Red arrows show where electrons go.

Working (the numbers match the band diagrams (1) to (5) and the red numbers on the I-V curve below):

1. **Zero bias:** E_F is flat. Filled states on each side face filled states on the other at the same energies, so equal small tunnelling flows cancel: $I = 0$.
2. **Small forward bias, up to the peak:** the n side is raised. Filled electron states in the n^+ conduction band now face **empty** states (holes) in the p^+ valence band at the same energy, and electrons tunnel from n to p: the current rises quickly. When this overlap of filled n-side states with empty p-side states is greatest, the tunnelling current reaches its **peak** I_p at the peak voltage V_p.
3. **Larger forward bias, down to the valley:** the conduction band of the n side rises above the top of the valence band of the p side; the filled states start to face the **band gap**, where there are no states. Tunnelling decreases, and the current **falls** as the voltage rises: **negative differential resistance**, down to the **valley** current I_v at V_v.
4. **Large forward bias:** tunnelling stops; the ordinary diffusion current of a pn diode (injection over the lowered barrier) takes over and rises exponentially.
5. **Reverse bias:** filled valence-band states of the p side face empty conduction-band states of the n side, and electrons tunnel from p to n freely, so the reverse current is large even at small voltages (the diode conducts well in reverse).

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Characteristic curve:

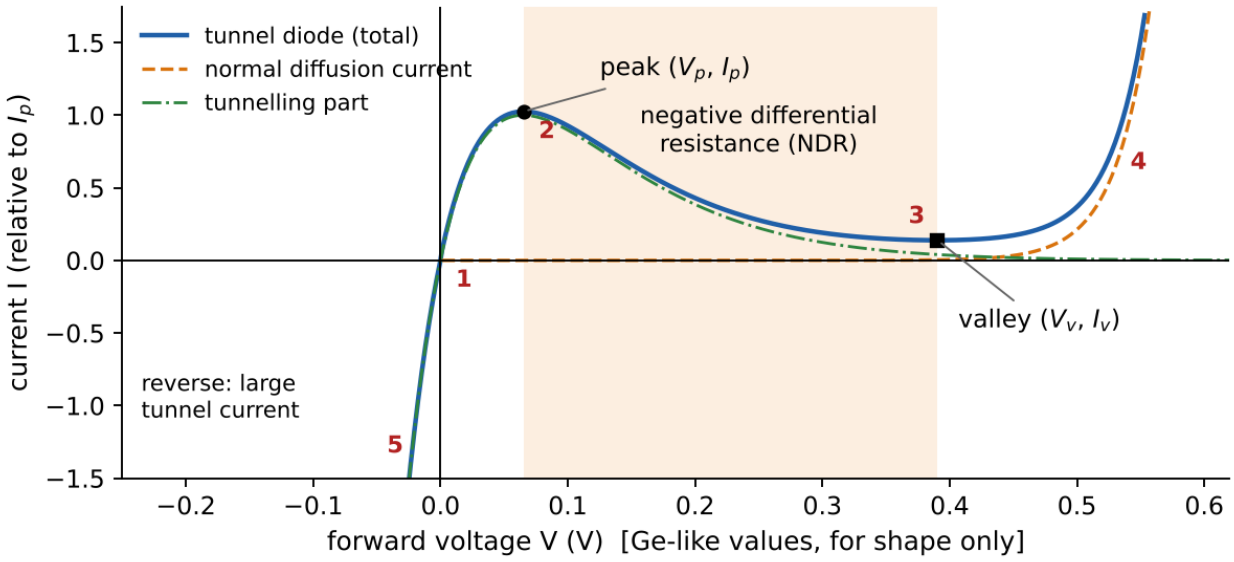


Figure 21: I-V characteristic of a tunnel diode, with the peak, the valley, the negative-resistance region and the diffusion current

Reading the figure. The solid blue curve is the tunnel diode current. It rises fast from 0 to the peak (filled circle, V_p, I_p), falls through the shaded negative-resistance region to the valley (filled square, V_v, I_v), then rises again along the dashed orange diffusion current of an ordinary diode. The dash-dot green curve is the tunnelling part alone. In reverse the current is large. The red numbers 1 to 5 match the five band diagrams. For a Ge tunnel diode, V_p is about 0.05 to 0.1 V, V_v about 0.3 V and $I_p/I_v \approx 8$.

Uses: the negative resistance cancels circuit losses, so tunnel diodes make microwave **oscillators** and amplifiers; tunnelling is very fast (no minority charge storage), which suits high-speed switching.

3. Keyword Explanation

1. Degenerate semiconductor

A semiconductor doped so heavily that E_F lies inside a band.

2. Negative differential resistance

A region where $dI/dV < 0$: the current falls as the voltage rises.

3. Peak-to-valley ratio

I_p/I_v, a figure of merit of a tunnel diode.

4. Notes and Remarks

- **Materials:** Ge, GaAs and GaSb tunnel diodes have higher peak-to-valley ratios than Si.
- **Limits:** low voltage swing (well under 1 V), two-terminal device (input and output not isolated), so it has been replaced by transistors in most uses.
- The same diode is 5(c) of the 2020–21 paper (2020–21 5(c)), extra reading only.

5. Final Concise Answer

(a) Forward bias injects electrons into the p side and holes into the n side; they diffuse and recombine, so the minority currents $J_n(-x_p) = (eD_n n_{p0}/L_n)(e^{eV_a/kT} - 1)$ and $J_p(x_n) = (eD_p p_{n0}/L_p)(e^{eV_a/kT} - 1)$ decay exponentially away from the junction, both are constant through the space charge region, the majority currents rise to make up the difference, and the total $J = J_s(e^{eV_a/kT} - 1)$ is the same everywhere (Si, 10^{16} cm^{-3} each side, 0.6 V: $J_n = 0.655$, $J_p = 0.414$, $J = 1.07$ A/cm^2).

(b) Holes injected from base to emitter (J_{pE}), recombination of injected electrons in the base (J_{RB}),

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recombination in the B-E space charge region (J_R), minus the B-C reverse current (I_{CBO}).

(c) Ohmic: no barrier (n-type with $\phi_m < \phi_s$, or a tunnelling contact on heavy doping), linear symmetric I-V, tiny resistance, used for device terminals. Rectifying: barrier $e\phi_{B0} = e(\phi_m - \chi)$ (n-type with $\phi_m > \phi_s$), diode I-V $J = J_{sT}(e^{eV_a/kT} - 1)$, used as Schottky diodes.

(d) The narrowing of the neutral base as the B-C depletion layer widens with V_{CB} (Early effect): $J_C \approx (eD_B n_{B0}/x_B)e^{eV_{BE}/kT}$ rises, the output curves slope up and extrapolate to $-V_A$, the output resistance is V_A/I_C, and at the limit punch-through occurs (Example 12.8: x_B 0.648 $\rightarrow$ 0.597 μm, J_C 3.195 $\rightarrow$ 3.470 A/cm^2, $V_A \approx 90$ V).

(e) Both sides degenerately doped: E_F inside the bands, a 5 to 10 nm depletion layer, and electrons tunnel where filled states face empty ones. Forward current rises to a peak I_p, then falls (negative resistance) to a valley I_v as the bands uncross, then rises as normal diffusion current; reverse current is large. Curve: peak, valley, negative-resistance region, diffusion branch.

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5. (a) Describe diffusion process. Derive the expression for diffusion current density. Marks: 7
   (b) Draw the ideal low-frequency capacitance versus gate voltage curve of a MOS capacitor with p-type substrate and indicate different regions. Marks: 5
   (c) Show that diffusion length is the average length a carrier diffuses before recombination. Marks: 6
   (d) A silicon n-channel JFET at $T = 300K$ has a gate doping concentration of $N_a = 10^{18}cm^{-3}$ and a channel doping concentration of $N_d = 2 \times 10^{16}cm^{-3}$. Determine the metallurgical channel thickness, a, such that the pinchoff voltage is $V_p = -2.50\ V$. Marks: 7

5.(a) Describe diffusion process. Derive the expression for diffusion current density. Marks: 7

1. Theory

1. Random thermal motion (Core Idea)

Even with no field applied, every free carrier in a semiconductor moves fast and at random. At 300 K its **thermal velocity** is about 10^7 cm/s. It flies straight for a short **mean free path** l (a few tens of nm), collides with the lattice or an impurity, and sets off in a new random direction. With equal numbers of carriers everywhere this motion carries no net current: as many carriers cross any plane to the right as to the left.

2. A concentration gradient makes a net flow

If there are more carriers on the left than on the right, the random motion no longer balances. More carriers start out on the left, so more of them cross a given plane going right than come back going left. The result is a **net flow from high concentration to low concentration**. Nothing pushes the carriers; the flow comes from numbers alone, like a drop of ink spreading in water or a smell spreading in a room.

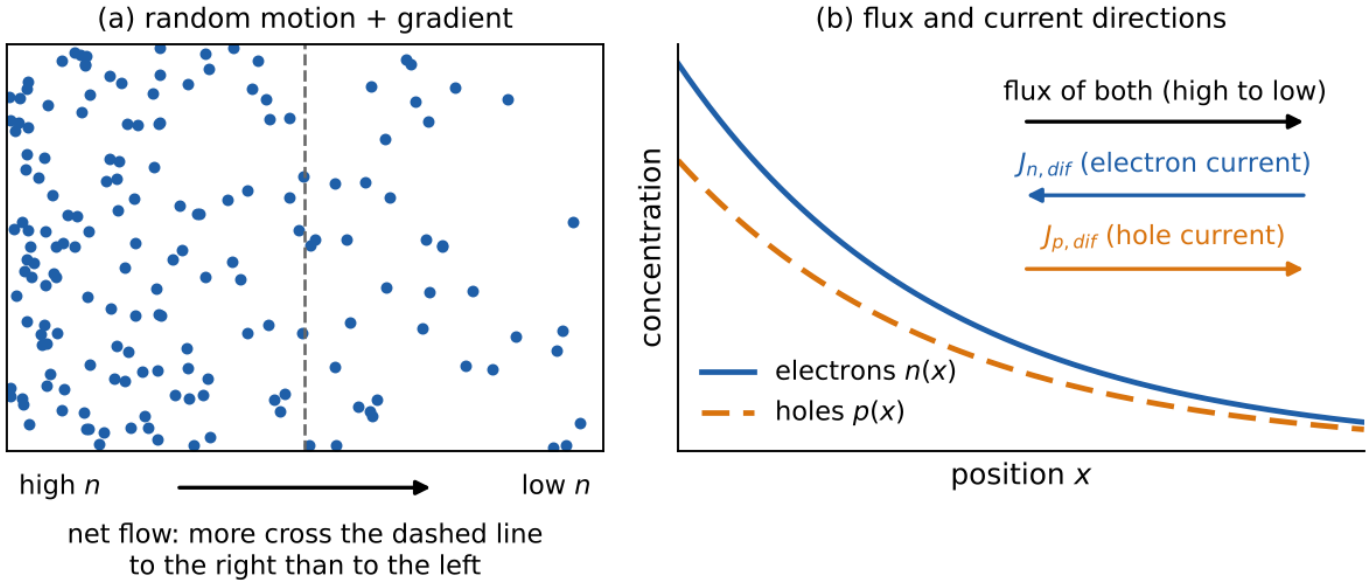


Figure 22: Diffusion: random motion of many carriers with a concentration gradient gives a net flow from high to low concentration

Reading the figure. In panel (a) the carriers (filled dots) are dense on the left and sparse on the right. Each moves at random, but more cross the dashed line to the right than to the left (black arrow). In panel (b) the solid blue curve is an electron profile $n(x)$ and the dashed orange curve a hole profile $p(x)$, both falling with x. The black arrow shows that both kinds of particles flow to $+x$ (high to low). The electron current (blue arrow) points to $-x$, because electrons are negative; the hole current (orange arrow) points to $+x$.

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3. Particle flux and current

The **flux** F is the number of carriers crossing a unit area per second ($\mathrm{cm}^{-2}\ \mathrm{s}^{-1}$). A current density is the flux times the charge of each carrier: $J = qF$, with $q = -e$ for electrons and $+e$ for holes.

2. Answer

Diffusion is the process by which particles flow from a region of high concentration to a region of low concentration because of their random thermal motion. The current it produces in a semiconductor is the **diffusion current**; it needs no electric field, only a gradient of the carrier concentration.

Derivation of the diffusion current density (Neamen, Sec. 5.2.1).

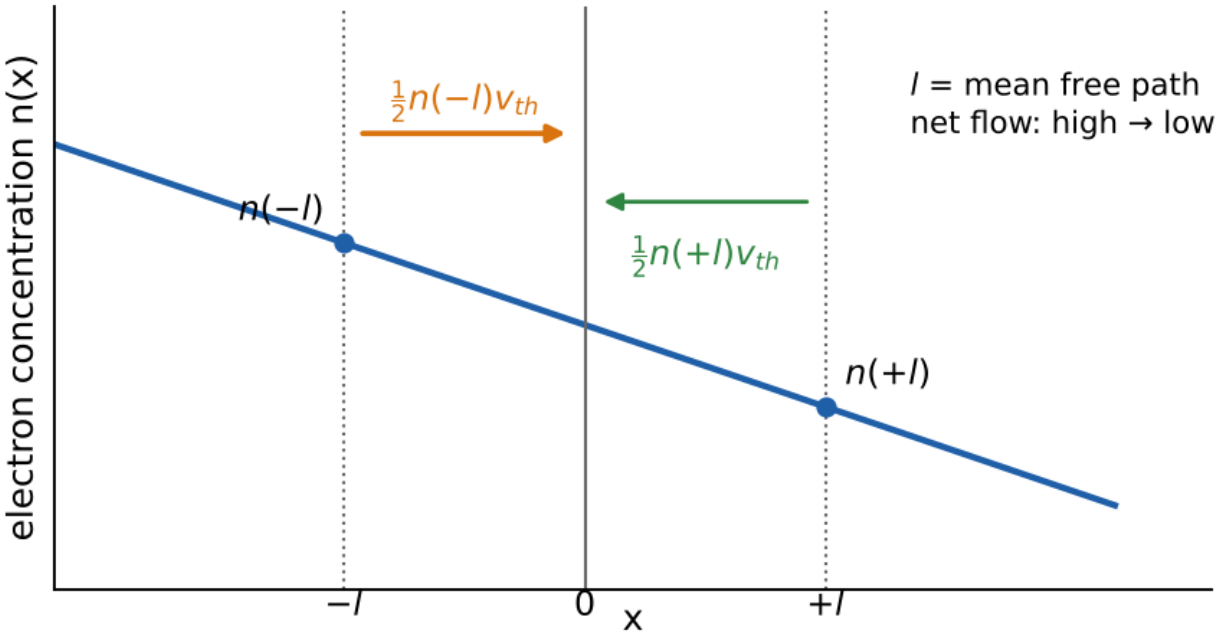


Figure 23: Electron concentration against distance with the two planes one mean free path from x = 0

Reading the figure. The solid blue line is an electron concentration $n(x)$ falling with x; the blue dots are its values at $x = -l$ and $x = +l$, one mean free path either side of the plane $x = 0$ (solid grey line). The orange arrow is the flow to the right, $\frac{1}{2}n(-l)v_{th}$; the green arrow the smaller flow to the left, $\frac{1}{2}n(+l)v_{th}$. Their difference is the net flow, from high to low concentration.

Step 1: The set-up

Take a one-dimensional electron concentration $n(x)$ that decreases with x (Neamen, Fig. 5.10). Electrons move with the mean thermal speed v_{th} and travel a mean free path l between collisions. Look at the plane $x = 0$.

Step 2: Which electrons cross the plane

In one mean free time, the electrons that can reach $x = 0$ are those within one mean free path of it, effectively those at $x = -l$ (moving right) and at $x = +l$ (moving left). At any instant *half* of the electrons at each place move towards the plane (in one dimension, half go right and half go left). So:

- flow to the right from $x = -l$: $\frac{1}{2}n(-l)v_{th}$
- flow to the left from $x = +l$: $\frac{1}{2}n(+l)v_{th}$

Step 3: The net flux

The net flux of electrons in the $+x$ direction is the difference (Neamen, Eq. 5.29):

$$F_n = \frac{1}{2}n(-l)v_{th} - \frac{1}{2}n(+l)v_{th} = \frac{1}{2}v_{th}\left[n(-l) - n(+l)\right]$$

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Step 4 (Basic): Taylor expansion

For a short distance l, a smooth function is approximated by its first two Taylor terms about $x = 0$:

$$n(\pm l) \approx n(0) \pm l\frac{dn}{dx}$$

So $n(-l) - n(+l) = \left[n(0) - l\frac{dn}{dx}\right] - \left[n(0) + l\frac{dn}{dx}\right] = -2l\frac{dn}{dx}$.

Step 5: The flux in terms of the gradient

Put Step 4 into Step 3 (Neamen, Eqs. 5.30, 5.31):

$$F_n = \frac{1}{2}v_{th}\left(-2l\frac{dn}{dx}\right) = -v_{th}l\frac{dn}{dx}$$

The minus sign says that the flux goes *down* the gradient: if n falls with x ($dn/dx < 0$), $F_n > 0$, to the right.

Step 6: From flux to current

Each electron carries charge $-e$, so (Neamen, Eq. 5.32)

$$J = -eF_n = +ev_{th}l\frac{dn}{dx}$$

Step 7: The diffusion coefficient

Define the **electron diffusion coefficient** $D_n = v_{th}l$ (cm^2/s). Then (Neamen, Eq. 5.33)

$$\boxed{J_{n,dif} = eD_n\frac{dn}{dx}}$$

Step 8: The same for holes

For holes the flux is the same, $F_p = -v_{th}l\,dp/dx = -D_p\,dp/dx$, but each hole carries $+e$, so $J = +eF_p$ (Neamen, Eq. 5.34):

$$\boxed{J_{p,dif} = -eD_p\frac{dp}{dx}}$$

Steps to remember

Half the carriers at $\pm l$ cross $x = 0 \rightarrow F = \frac{1}{2}v_{th}[n(-l) - n(+l)] \rightarrow$ Taylor: $n(\pm l) = n(0) \pm l\,dn/dx \rightarrow F = -v_{th}l\,dn/dx \rightarrow J = -eF = eD_n\,dn/dx$ with $D_n = v_{th}l \rightarrow$ holes: $J = -eD_p\,dp/dx$.

Directions: electrons and holes both flow from high to low concentration, but because their charges have opposite signs the electron diffusion current flows *up* the electron gradient and the hole diffusion current *down* the hole gradient.

Total current: with a field present the drift terms are added (Neamen, Eq. 5.35):

$$J = en\mu_n E_x + ep\mu_p E_x + eD_n\frac{dn}{dx} - eD_p\frac{dp}{dx}$$

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3. Keyword Explanation

1. Diffusion

Net flow of particles from high to low concentration by random thermal motion.

2. Mean free path

The average distance a carrier travels between collisions, $l = v_{th}\tau_c$.

3. Diffusion coefficient

$D = v_{th}l$ (cm^2/s); related to the mobility by $D/\mu = kT/e$ (2(c)).

4. Flux

The number of particles crossing unit area per unit time.

4. Notes and Remarks

- **Typical values (Si, 300 K):** $D_n \approx 35$ cm^2/s and $D_p \approx 12.4$ cm^2/s.
- **Where it matters:** diffusion is the main current of minority carriers in the neutral regions of diodes and in the base of a bipolar transistor (4(a), 4(d)); drift is negligible there because the field is very small.
- **Order-of-magnitude model:** the factor $\frac{1}{2}$ and the choice of $\pm l$ are simplifications; a full kinetic treatment gives the same form $J = eD\,dn/dx$ with a slightly different numerical relation between D, v_{th} and l.

5.(b) Draw the ideal low-frequency capacitance versus gate voltage curve of a MOS capacitor with p-type substrate and indicate different regions. Marks: 5

1. Theory

1. Capacitance of a MOS structure (Core Idea)

The MOS capacitor (3(a)) stores charge Q'_m on the gate and an equal and opposite charge in the semiconductor. Its **small-signal capacitance** per unit area is $C' = dQ'_m/dV_G$ (Neamen, Eq. 10.34): the change of charge for a small change of gate voltage. The value depends on *where* the changing charge sits in the semiconductor, which in turn depends on the bias region.

2. Two capacitors in series

In general a small change of gate charge is answered by a change of charge at some depth in the silicon. The oxide then acts as one capacitor, $C_{ox} = \epsilon_{ox}/t_{ox}$, and the silicon between the interface and that charge as a second capacitor, $C'_{SD} = \epsilon_s/x_d$, in series:

$$\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}}$$

3. "Low frequency"

At low frequency the electrons of the inversion layer have time to be generated and recombined with each change of the signal, so they respond to it. At high frequency they cannot follow.

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2. Answer

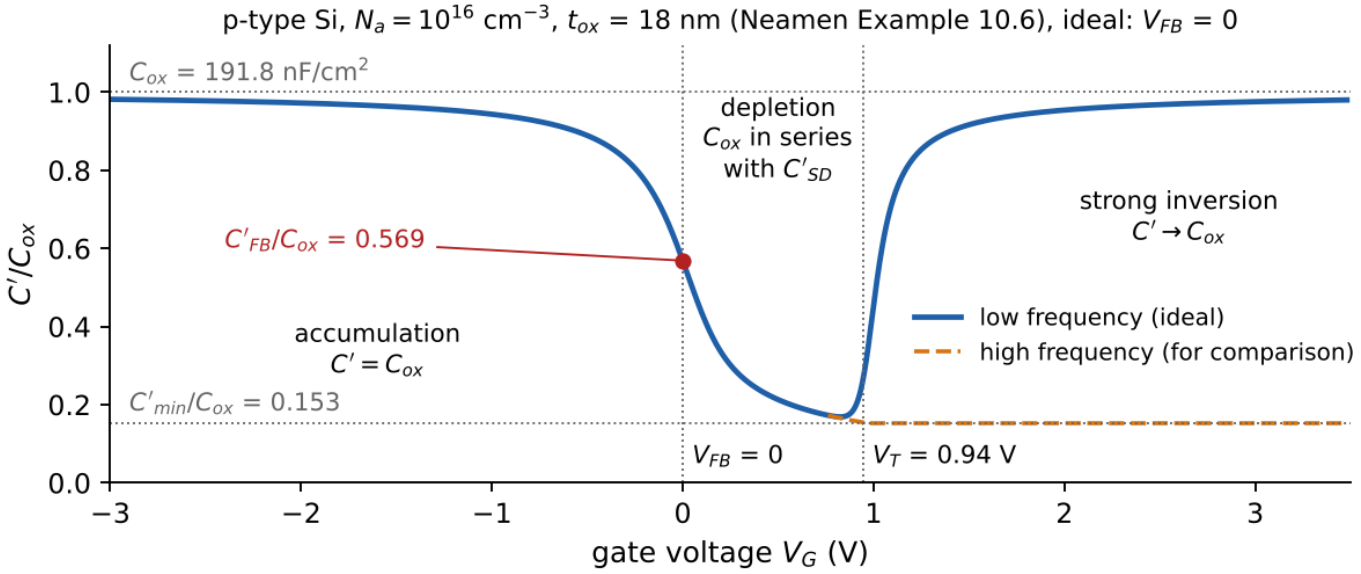


Figure 24: Ideal low-frequency C-V curve of a MOS capacitor with a p-type substrate

Reading the figure. Neamen's Example 10.6 capacitor (p-type Si, $N_a = 10^{16}$ cm^{-3}, $t_{ox} = 18$ nm), ideal ($V_{FB} = 0$). The solid blue curve is the low-frequency C'/C_{ox} against V_G. On the left (accumulation) it equals C_{ox}; it falls through the flat-band value 0.569 C_{ox} (red dot) at $V_G = 0$, through depletion to a minimum near threshold, $V_T = 0.94$ V, and rises back to C_{ox} in strong inversion. The dashed orange line is the high-frequency curve, which stays at $C'_{min} = 0.153\,C_{ox}$ (dotted grey line) in inversion; it is shown only for comparison.

Regions (p-type substrate):

1. **Accumulation ($V_G < V_{FB}$, negative):** holes gather at the oxide interface. A small change of V_G changes the charge right at the surface, so only the oxide acts (Neamen, Eq. 10.35):

$$C'(\text{acc}) = C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

2. **Flat band ($V_G = V_{FB}$):** no charge; the capacitance has a value set by the Debye length, $C'_{FB} = \epsilon_{ox}/[t_{ox} + (\epsilon_{ox}/\epsilon_s)\sqrt{(kT/e)(\epsilon_s/eN_a)}]$ (Eq. 10.40), a little below C_{ox}.
3. **Depletion ($V_{FB} < V_G < V_T$):** holes are pushed away; the charge changes at the edge of the depletion layer, at depth x_d. C_{ox} in series with $C'_{SD} = \epsilon_s/x_d$ (Eq. 10.37):

$$C'(\text{depl}) = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d}$$

As V_G rises, x_d grows and C' **falls**, reaching its minimum at threshold, where $x_d = x_{dT}$ (Eq. 10.38):

$$C'_{min} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_{dT}}$$

4. **Inversion ($V_G > V_T$):** the electron inversion layer at the surface responds to the low-frequency signal, so the charge again changes right at the surface and (Eq. 10.39)

$$C'(\text{inv}) = C_{ox}$$

The curve rises steeply back to C_{ox} just above V_T.

Numbers (Neamen, Example 10.6): $C_{ox} = (3.9)(8.85 \times 10^{-14})/(18 \times 10^{-7}) = 1.918 \times 10^{-7}$ F/cm^2; $\phi_{fp} = 0.3473$ V; $x_{dT} = 0.300$ μm; $C'_{min} = 2.93 \times 10^{-8}$ F/cm^2 $= 0.153\,C_{ox}$; $C'_{FB} = 1.09 \times 10^{-7}$ F/cm^2 $= 0.569\,C_{ox}$; ideal $V_T = |Q'_{SD}(\text{max})|/C_{ox} + 2\phi_{fp} = 0.94$ V.

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Shape to draw: high and flat at C_{ox} for negative V_G; falling through C'_{FB} at V_{FB}; a minimum C'_{min} at V_T; back up to C_{ox} for $V_G > V_T$. Regions labelled accumulation, depletion, inversion.

3. Keyword Explanation

1. Oxide capacitance

$C_{ox} = \epsilon_{ox}/t_{ox}$, the largest capacitance of the structure.

2. Flat-band capacitance

The capacitance at $V_G = V_{FB}$, set by the oxide in series with a Debye-length layer of the silicon.

3. Minimum capacitance

C'_{min}, at threshold, when the depletion layer has its maximum width x_{dT}.

4. Notes and Remarks

- **High frequency:** the inversion electrons cannot follow a fast signal, so the capacitance stays at C'_{min} in inversion (the dashed curve); a fast voltage sweep can even give deep depletion, below C'_{min} (Neamen, Sec. 10.2.2).
- **Real device:** a work function difference and oxide charge shift the whole curve along the voltage axis by $V_{FB} = \phi_{ms} - Q'_{ss}/C_{ox}$ without changing its shape (Sec. 10.2.3); interface traps also stretch it.
- **n-type substrate:** the curve is the mirror image: accumulation at positive V_G, inversion at negative V_G.
- **Book check:** Neamen's Example 10.6 gives $C_{ox} = 1.9175 \times 10^{-7}$, $C'_{min} = 2.925 \times 10^{-8}$ F/cm^2 ($C'_{min}/C_{ox} = 0.1525$), as above.

5.(c) Show that diffusion length is the average length a carrier diffuses before recombination. Marks: 6

1. Theory

1. Excess carriers and recombination (Core Idea)

Light or a forward-biased junction can add **excess** carriers to a region, more than its equilibrium number. In n-type material the excess holes δp are minority carriers. Each one survives, on average, for the **minority carrier lifetime** τ_{p0} before it recombines with an electron; the recombination rate is $\delta p/\tau_{p0}$.

2. Injection at one plane

Suppose excess holes are injected steadily at the plane $x = 0$ of a long n-type bar, with no electric field. They diffuse into the bar ($x > 0$) and recombine on the way. In steady state the hole continuity equation reduces to the **steady-state diffusion equation** (Neamen, Sec. 6.3, ambipolar equation at low injection):

$$D_p \frac{d^2(\delta p)}{dx^2} - \frac{\delta p}{\tau_{p0}} = 0$$

It says: what diffusion brings into a slice per second (the curvature term) equals what recombines there.

3. The diffusion length

Divide by D_p and write $L_p^2 = D_p \tau_{p0}$:

$$\frac{d^2(\delta p)}{dx^2} - \frac{\delta p}{L_p^2} = 0, \qquad L_p = \sqrt{D_p \tau_{p0}}$$

L_p has the unit of length: $D_p\tau_{p0}$ is in (cm^2/s)(s) = cm^2, so its square root is in cm. The question asks to show what this length means.

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2. Answer

Step 1: Solve the steady-state diffusion equation

The equation $d^2(\delta p)/dx^2 = \delta p/L_p^2$ has the general solution

$$\delta p(x) = Ae^{-x/L_p} + Be^{+x/L_p}$$

(check: each exponential, differentiated twice, gives itself divided by L_p^2).

Step 2: Apply the boundary conditions

- Far into a long bar the excess must vanish, $\delta p(\infty) = 0$, so $B = 0$.
- At the injection plane $\delta p(0) = \delta p_0$, so $A = \delta p_0$.

$$\delta p(x) = \delta p_0\, e^{-x/L_p}, \quad x \geq 0$$

Step 3: The probability that a hole reaches x

All holes start at $x = 0$. In steady state, the number of holes per unit volume at x is proportional to the number that have **survived** (not yet recombined) to get that far. So the fraction of injected holes that reach a distance x before recombining is

$$\frac{\delta p(x)}{\delta p_0} = e^{-x/L_p}$$

and the fraction that recombine between x and $x + dx$ is the drop of this survival fraction across dx:

$$P(x)\,dx = -\frac{d}{dx}\left(e^{-x/L_p}\right)dx = \frac{1}{L_p}e^{-x/L_p}\,dx$$

Check: $\int_0^\infty P(x)\,dx = 1$, so every hole recombines somewhere. (Equivalently, the recombination rate at x is $\delta p(x)/\tau_{p0} \propto e^{-x/L_p}$.)

Step 4: The average distance

The average (mean) distance travelled before recombination is

$$\langle x \rangle = \int_0^\infty x\,P(x)\,dx = \frac{1}{L_p}\int_0^\infty x\,e^{-x/L_p}\,dx$$

Step 5 (Basic): Integration by parts

With $u = x$, $dv = e^{-x/L_p}dx$, so $du = dx$, $v = -L_p e^{-x/L_p}$:

$$\int_0^\infty xe^{-x/L_p}dx = \left[-xL_pe^{-x/L_p}\right]_0^\infty + L_p\int_0^\infty e^{-x/L_p}dx = 0 + L_p\left[-L_pe^{-x/L_p}\right]_0^\infty = L_p^2$$

(the first bracket is 0 at both limits, since $xe^{-x/L_p} \to 0$ as $x \to \infty$).

Step 6: Result

$$\langle x \rangle = \frac{1}{L_p}\cdot L_p^2 = L_p$$

$$\boxed{\langle x \rangle = L_p = \sqrt{D_p\tau_{p0}}}$$

So the diffusion length is the **average distance a minority carrier diffuses before it recombines**. The same steps for excess electrons in p-type material give $\langle x \rangle = L_n = \sqrt{D_n\tau_{n0}}$.

Steps to remember

$D_p\,\delta p'' = \delta p/\tau_{p0} \to \delta p = \delta p_0 e^{-x/L_p} \to$ recombination probability $P(x) = (1/L_p)e^{-x/L_p} \to \langle x \rangle = \int xP\,dx \to$ by parts $= L_p$.

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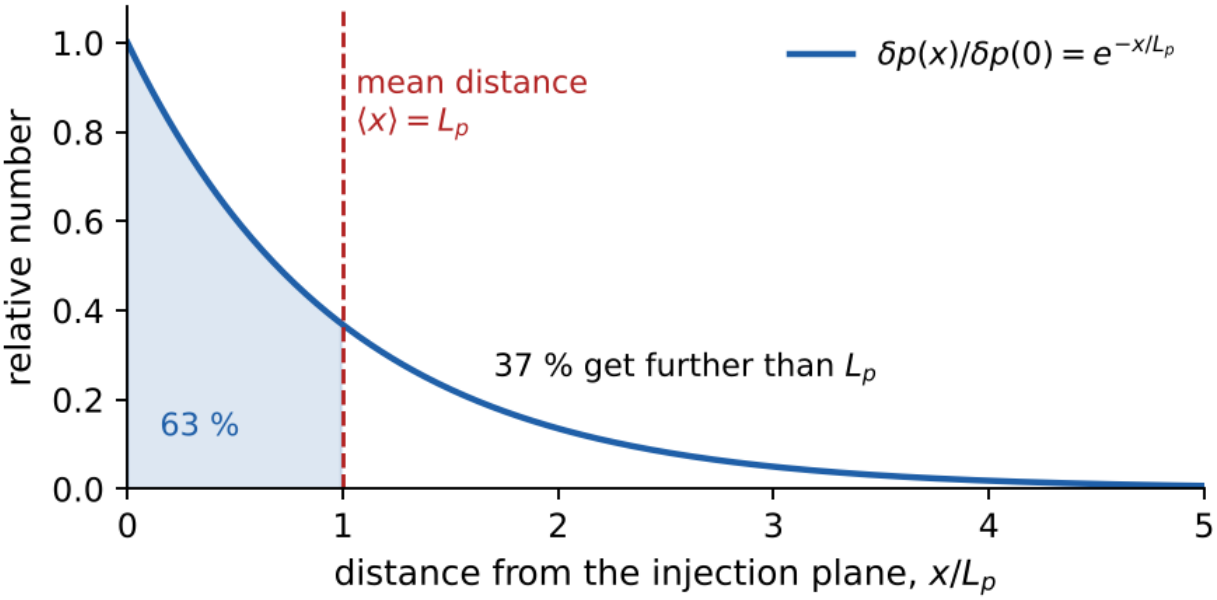


Figure 25: Excess carrier profile from an injection plane, with its mean distance

Reading the figure. The solid blue curve is $\delta p(x)/\delta p(0) = e^{-x/L_p}$. The dashed red line marks the mean distance $\langle x \rangle = L_p$. The shaded part shows that 63 % of the carriers recombine within one diffusion length; 37 % get further.

Example

$D_p = 10$ cm^2/s, $\tau_{p0} = 10^{-7}$ s: $L_p = \sqrt{10 \times 10^{-7}} = 10^{-3}$ cm $= 10$ μm; 95 % of the holes have recombined within $3L_p = 30$ μm.

3. Keyword Explanation

1. Minority carrier lifetime

τ_{p0}, the mean time an excess minority carrier exists before recombination.

2. Diffusion length

$L_p = \sqrt{D_p \tau_{p0}}$, the mean distance a minority carrier diffuses before recombination.

3. Low injection

Excess carriers much fewer than the majority carriers, so the lifetime is constant.

4. Notes and Remarks

- **Another way to see it:** a carrier diffusing for a time t spreads over a distance of order $\sqrt{Dt}$; putting $t = \tau$ gives $\sqrt{D\tau}$.
- **Why it matters:** it decides how far injected carriers reach: the neutral base of a transistor must be much thinner than L_B (4(d)), and a diode's current depends on L_n and L_p (4(a)).
- **Both directions:** if carriers are generated at $x = 0$ in an infinite bar, they diffuse both ways and the profile is $\delta p_0 e^{-|x|/L_p}$; the mean distance on each side is still L_p (Neamen, Example 6.4).

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5.(d) A silicon n-channel JFET at $T = 300K$ has a gate doping concentration of $N_a = 10^{18}cm^{-3}$ and a channel doping concentration of $N_d = 2 \times 10^{16}cm^{-3}$. Determine the metallurgical channel thickness, a, such that the pinchoff voltage is $V_p = -2.50\ V$. Marks: 7

1. Theory

1. The junction field-effect transistor (Core Idea)

An n-channel **pn JFET** is an n-type bar (the **channel**, doping N_d, metallurgical thickness a) with a heavily doped p^+ **gate** on it, and **source** and **drain** contacts at its two ends. Current flows along the channel from drain to source. The p^+n gate junction is reverse biased ($V_{GS} \leq 0$). Its depletion region, which in a one-sided junction lies almost all in the n channel, reaches a depth h into the channel and leaves only $a - h$ open for current.

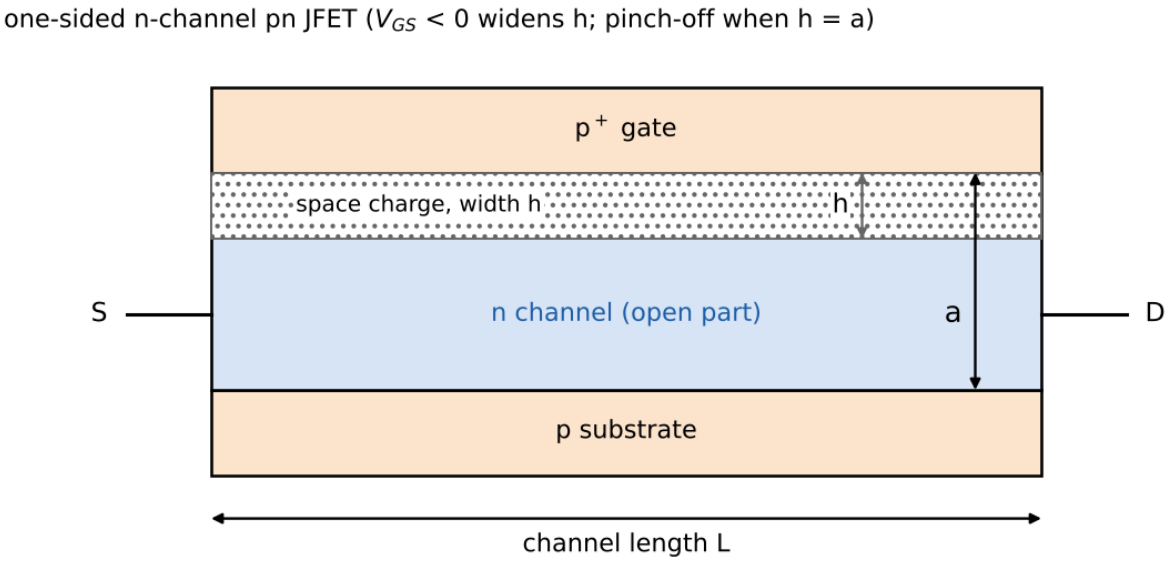


Figure 26: Cross section of a one-sided n-channel pn JFET

Reading the figure. The p^+ gate (top) and the p substrate (bottom) enclose the n channel of thickness a (black arrow). The dotted strip under the gate is the space charge region of width h (grey arrow); only the light blue part of the channel below it conducts between source S and drain D. A more negative V_{GS} widens h; when $h = a$ the channel is pinched off.

2. Depletion depth and pinch-off

For a one-sided p^+n junction with total reverse voltage $V_{bi} - V_{GS}$ (Neamen, Eq. 13.1):

$$h = \left[\frac{2\epsilon_s(V_{bi} - V_{GS})}{eN_d}\right]^{1/2}$$

Pinch-off is reached when the depletion region fills the channel, $h = a$. The total junction voltage needed is the **internal pinchoff voltage** V_{p0} (Eqs. 13.2, 13.3):

$$a = \left[\frac{2\epsilon_s V_{p0}}{eN_d}\right]^{1/2} \quad \Leftrightarrow \quad V_{p0} = \frac{ea^2N_d}{2\epsilon_s}$$

The gate-to-source voltage that does it is the **pinchoff voltage** V_p. Since $V_{bi} - V_p = V_{p0}$ (Eq. 13.4),

$$V_p = V_{bi} - V_{p0}$$

which is negative for an n-channel depletion-mode JFET.

3. Built-in potential of the gate junction

$$V_{bi} = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

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2. Answer

Given:

- Si, $T = 300$ K, n-channel JFET
- gate doping $N_a = 10^{18}$ cm^{-3}, channel doping $N_d = 2 \times 10^{16}$ cm^{-3}
- pinchoff voltage $V_p = -2.50$ V
- $n_i = 1.5 \times 10^{10}$ cm^{-3}, $\epsilon_s = 11.7 \times 8.85 \times 10^{-14}$ F/cm, $V_t = 0.0259$ V

Required: the metallurgical channel thickness a.

Step 1: Built-in potential of the p$^+$n gate junction

$$\frac{N_a N_d}{n_i^2} = \frac{(10^{18})(2 \times 10^{16})}{(1.5 \times 10^{10})^2} = 8.889 \times 10^{13}, \qquad \ln\!\left(8.889 \times 10^{13}\right) = 32.118$$

$$V_{bi} = (0.0259)(32.118) = 0.8319 \text{ V}$$

$$\boxed{V_{bi} = 0.832 \text{ V}}$$

Step 2: Internal pinchoff voltage

From $V_p = V_{bi} - V_{p0}$:

$$V_{p0} = V_{bi} - V_p = 0.8319 - (-2.50) = 3.332 \text{ V}$$

$$\boxed{V_{p0} = 3.33 \text{ V}}$$

This is the total reverse voltage across the gate junction at pinch-off: the built-in potential plus the 2.5 V applied.

Step 3: Channel thickness

$$a = \left[\frac{2\epsilon_s V_{p0}}{e N_d}\right]^{1/2} = \left[\frac{2(11.7)(8.85 \times 10^{-14})(3.332)}{(1.6 \times 10^{-19})(2 \times 10^{16})}\right]^{1/2}$$

$$= \left[\frac{6.900 \times 10^{-12}}{3.2 \times 10^{-3}}\right]^{1/2} = \left[2.156 \times 10^{-9}\right]^{1/2} = 4.644 \times 10^{-5} \text{ cm}$$

$$\boxed{a = 0.464 \ \mu\text{m}}$$

Check: $V_{p0} = e a^2 N_d / (2\epsilon_s) = (1.6 \times 10^{-19})(4.644 \times 10^{-5})^2 (2 \times 10^{16}) / [2(11.7)(8.85 \times 10^{-14})] = 3.33$ V ✔.

Final Answer: $V_{bi} = 0.832$ V, $V_{p0} = 3.33$ V, and the metallurgical channel thickness is $a = 0.464$ μm.

3. Keyword Explanation

1. Metallurgical channel thickness

a, the thickness of the doped channel between the gate junction and the substrate, before any depletion.

2. Internal pinchoff voltage

$V_{p0} = e a^2 N_d / 2\epsilon_s$, the total junction voltage that depletes the whole channel.

3. Pinchoff voltage

$V_p = V_{bi} - V_{p0}$, the gate-to-source voltage that turns the JFET off.

4. Notes and Remarks

- **Book check:** this is Neamen's Exercise Ex 13.1 word for word; the book's answer is $a = 0.464$ μm, as above.
- **One-sided assumption:** with $N_a = 50 N_d$ the depletion region on the gate side is only 2 % of h, so treating the junction as one-sided is accurate.

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- **Design view:** a thicker channel needs a more negative gate voltage to turn off ($V_{p0} \propto a^2$); a thinner one gives less current. Neamen's Example 13.1 ($N_d = 10^{16}$ cm^{-3}, $a = 0.75$ μm) gives $V_{p0} = 4.35$ V and $V_p = -3.54$ V.

5. Final Concise Answer

(a) Diffusion: net flow of carriers from high to low concentration by random thermal motion. Half the electrons at $x = \mp l$ cross $x = 0$: $F_n = \frac{1}{2} v_{th}[n(-l) - n(+l)]$; with $n(\pm l) = n(0) \pm l\, dn/dx$, $F_n = -v_{th} l\, dn/dx$; $J = -eF_n = eD_n\, dn/dx$ with $D_n = v_{th} l$; for holes $J_p = -eD_p\, dp/dx$.

(b) p-type, low frequency: $C' = C_{ox}$ in accumulation ($V_G < V_{FB}$); falls through C'_{FB} at flat band and through depletion as $\epsilon_{ox}/[t_{ox} + (\epsilon_{ox}/\epsilon_s)x_d]$ to C'_{min} at V_T; returns to C_{ox} in inversion (Example 10.6: $C'_{FB} = 0.569\, C_{ox}$, $C'_{min} = 0.153\, C_{ox}$, $V_T = 0.94$ V).

(c) Steady-state diffusion with recombination: $\delta p = \delta p_0 e^{-x/L_p}$, so a carrier recombines between x and $x + dx$ with probability $(1/L_p)e^{-x/L_p}dx$, and $\langle x \rangle = \int_0^\infty x(1/L_p)e^{-x/L_p}dx = L_p = \sqrt{D_p \tau_{p0}}$.

(d) $V_{bi} = 0.0259 \ln(8.889 \times 10^{13}) = 0.832$ V; $V_{p0} = V_{bi} - V_p = 3.33$ V; $a = [2\epsilon_s V_{p0}/(eN_d)]^{1/2} = 4.64 \times 10^{-5}$ cm $= 0.464$ μm.

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6. (a) The parameters of an n-channel silicon JFET are $\mu_n = 1000\ cm^2/V-s$, $N_d = 5\times10^{15} cm^{-3}$, $a = 0.50\ \mu m$, and $L = 2\ \mu m$. Determine the cutoff frequency. Marks: 6
   (b) Design the ratio of emitter doping to base doping in order to achieve an emitter injection efficiency factor of $\gamma = 0.9967$. Assume a *npn* bipolar transistor, and also consider $D_E = D_B$, $L_E = L_B$ and $x_E = X_B$. Marks: 6
   (c) What is current crowding? What is the impact of current crowding in high power devices? Marks: 7
   (d) With suitable energy band diagrams, explain Schottky contact. Marks: 6

6.(a) The parameters of an n-channel silicon JFET are $\mu_n = 1000\ cm^2/V-s$, $N_d = 5\times10^{15} cm^{-3}$, $a = 0.50\ \mu m$, and $L = 2\ \mu m$. Determine the cutoff frequency. Marks: 6

1. Theory

1. What limits the speed of a JFET (Core Idea)

A JFET is controlled by the voltage on its gate junction (5(d)). To change the channel current, the depletion region under the gate must widen or narrow, which means charge must flow into or out of the **gate capacitance** C_G. The input signal supplies this charging current. At high frequency the charging current grows ($\propto \omega C_G$), while the useful output current stays $g_m V_{gs}$. The **cutoff frequency** f_T is the frequency at which the input current equals the output current: above it the device no longer gives current gain (Neamen, Sec. 13.4.2).

2. The small-signal picture

In the small-signal model (Neamen, Fig. 13.22) the input sees $C_{gs} + C_{gd} = C_G$ and the output is a current source $g_m V_{gs}$. With the output shorted (Eq. 13.62):

$$|I_i| = 2\pi f C_G V_{gs}$$

Setting $|I_i| = g_m V_{gs}$ gives (Eqs. 13.63, 13.64)

$$f_T = \frac{g_m}{2\pi C_G}$$

2. Answer

Given:

- n-channel Si JFET: $\mu_n = 1000$ cm^2/V s, $N_d = 5\times10^{15}$ cm^{-3}
- channel thickness $a = 0.50\ \mu$m $= 0.50\times10^{-4}$ cm, channel length $L = 2\ \mu$m $= 2\times10^{-4}$ cm
- $\epsilon_s = 11.7\times8.85\times10^{-14}$ F/cm, $e = 1.6\times10^{-19}$ C

Required: the cutoff frequency f_T.

Step 1: Definition of the cutoff frequency

f_T is the frequency at which the input current through the gate capacitance equals the ideal output current $g_m V_{gs}$:

$$2\pi f_T C_G V_{gs} = g_m V_{gs} \quad \Rightarrow \quad f_T = \frac{g_m}{2\pi C_G}$$

Step 2: The largest transconductance

The maximum transconductance is the conductance of the fully open channel (Neamen, Eq. 13.65), for a channel width W:

$$g_{ms}(\max) = G_{01} = \frac{e\mu_n N_d W a}{L}$$

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Step 3: The smallest gate capacitance

The gate capacitance is smallest when the depletion region is widest, $h = a$: a parallel-plate capacitor of area WL and spacing a (Eq. 13.66):

$$C_G(\text{min}) = \frac{\epsilon_s W L}{a}$$

Step 4: The maximum cutoff frequency

Divide Step 2 by 2π times Step 3; the width W cancels (Neamen, Eq. 13.67):

$$f_T = \frac{e\mu_n N_d W a/L}{2\pi\epsilon_s W L/a} = \frac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$$

Step 5: Substitute the numbers

Numerator:

$$e\mu_n N_d a^2 = (1.6\times10^{-19})(1000)(5\times10^{15})(0.50\times10^{-4})^2 = 2.000\times10^{-9}$$

Denominator:

$$2\pi\epsilon_s L^2 = 2\pi(11.7)(8.85\times10^{-14})(2\times10^{-4})^2 = 2.602\times10^{-19}$$

$$f_T = \frac{2.000\times10^{-9}}{2.602\times10^{-19}} = 7.685\times10^{9}\ \text{Hz}$$

$$\boxed{f_T = 7.69\ \text{GHz}}$$

Meaning: the JFET can still give current gain up to about 7.7 GHz. The units check: (C)(cm^2/V s)(cm^{-3})(cm^2)/[(F/cm)(cm^2)] = C/(V s F) = 1/s ✔.

Final Answer: $f_T = e\mu_n N_d a^2/(2\pi\epsilon_s L^2) = 7.69$ GHz.

3. Keyword Explanation

1. Cutoff frequency

The frequency at which the small-signal current gain of the transistor falls to 1.

2. Transconductance

$g_m = \partial I_D/\partial V_{GS}$, the change of drain current per volt on the gate.

3. Gate capacitance

$C_G = C_{gs} + C_{gd}$, the capacitance of the gate depletion region.

4. Notes and Remarks

- **Book check:** this is Neamen's Exercise Ex 13.9 word for word; the book's answer is $f_T = 7.69$ GHz, as above.
- **Design trends:** $f_T \propto \mu_n N_d a^2/L^2$: a short channel and a high-mobility material raise it most; GaAs MESFETs, with μ_n several times larger, reach tens of GHz.
- **Upper estimate:** the formula uses the largest g_m and the smallest C_G, so it is a maximum; with very short channels the channel transit time can limit the speed instead (Neamen's comment).
- With $N_d = 5\times10^{15}$ cm^{-3} and $a = 0.5$ μm the internal pinchoff voltage is only $V_{p0} = ea^2N_d/2\epsilon_s = 0.966$ V.

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6.(b) Design the ratio of emitter doping to base doping in order to achieve an emitter injection efficiency factor of $\gamma = 0.9967$. Assume a *npn* bipolar transistor, and also consider $D_E = D_B$, $L_E = L_B$ and $x_E = X_B$. Marks: 6

1. Theory

1. Two currents across the emitter junction (Core Idea)

In an npn transistor in forward-active mode, the forward-biased B-E junction passes two currents:

- **electrons** injected from the emitter into the base, J_{nE}: these are the useful ones, which reach the collector;
- **holes** injected from the base into the emitter, J_{pE}: these are wasted, because they never reach the collector; they only add to the base current (4(b)).

2. Emitter injection efficiency

The **emitter injection efficiency** is the fraction of the emitter current that is useful (Neamen, Eq. 12.32):

$$\gamma = \frac{J_{nE}}{J_{nE} + J_{pE}} = \frac{1}{1 + J_{pE}/J_{nE}}$$

It should be as close to 1 as possible, since it multiplies into the current gain $\alpha = \gamma \alpha_T \delta$.

3. The two currents from the minority carrier distributions

Each is a minority diffusion current (Neamen, Eqs. 12.33, 12.34). Their ratio gives (Neamen, Eq. 12.35a)

$$\gamma = \frac{1}{1 + \dfrac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}}$$

where $p_{E0} = n_i^2/N_E$ is the equilibrium hole concentration in the emitter, $n_{B0} = n_i^2/N_B$ the equilibrium electron concentration in the base, D the diffusion coefficients, L the diffusion lengths and x_E, x_B the neutral widths. For short regions ($x \ll L$), $\tanh(x/L) \approx x/L$ and this becomes (Eq. 12.35b)

$$\gamma \approx \frac{1}{1 + \dfrac{N_B}{N_E} \cdot \dfrac{D_E}{D_B} \cdot \dfrac{x_B}{x_E}}$$

2. Answer

Given: npn transistor, $\gamma = 0.9967$, $D_E = D_B$, $L_E = L_B$, $x_E = x_B$.
Required: the ratio N_E/N_B (Neamen, Design Example 12.4).

Step 1: Start from the injection efficiency

$$\gamma = \frac{1}{1 + \dfrac{p_{E0} D_E L_B}{n_{B0} D_B L_E} \cdot \dfrac{\tanh(x_B/L_B)}{\tanh(x_E/L_E)}}$$

Step 2: Use the given equalities

With $D_E = D_B$, $L_E = L_B$ and $x_E = x_B$, the ratio $D_E/D_B = 1$, $L_B/L_E = 1$ and the two tanh factors are equal, so

$$\gamma = \frac{1}{1 + \dfrac{p_{E0}}{n_{B0}}}$$

Step 3: Write the minority concentrations in terms of the doping

$p_{E0} = n_i^2/N_E$ and $n_{B0} = n_i^2/N_B$ (mass-action law in each region), so

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$$\frac{p_{E0}}{n_{B0}} = \frac{n_i^2/N_E}{n_i^2/N_B} = \frac{N_B}{N_E} \quad \Rightarrow \quad \gamma = \frac{1}{1 + N_B/N_E}$$

Step 4: Solve for the doping ratio

$$1 + \frac{N_B}{N_E} = \frac{1}{\gamma} \quad \Rightarrow \quad \frac{N_B}{N_E} = \frac{1}{\gamma} - 1 = \frac{1}{0.9967} - 1 = 0.00331$$

$$\boxed{\frac{N_E}{N_B} = \frac{1}{0.00331} = 302}$$

Step 5: Check

$\gamma = 1/(1 + 1/302) = 0.99670$ ✔.

Meaning: the emitter must be doped about **300 times** more heavily than the base. For example, a base of $N_B = 5 \times 10^{16}$ cm^{-3} needs $N_E \approx 1.5 \times 10^{19}$ cm^{-3}. This is why bipolar transistors have an n$^+$ (or n^{++}) emitter.

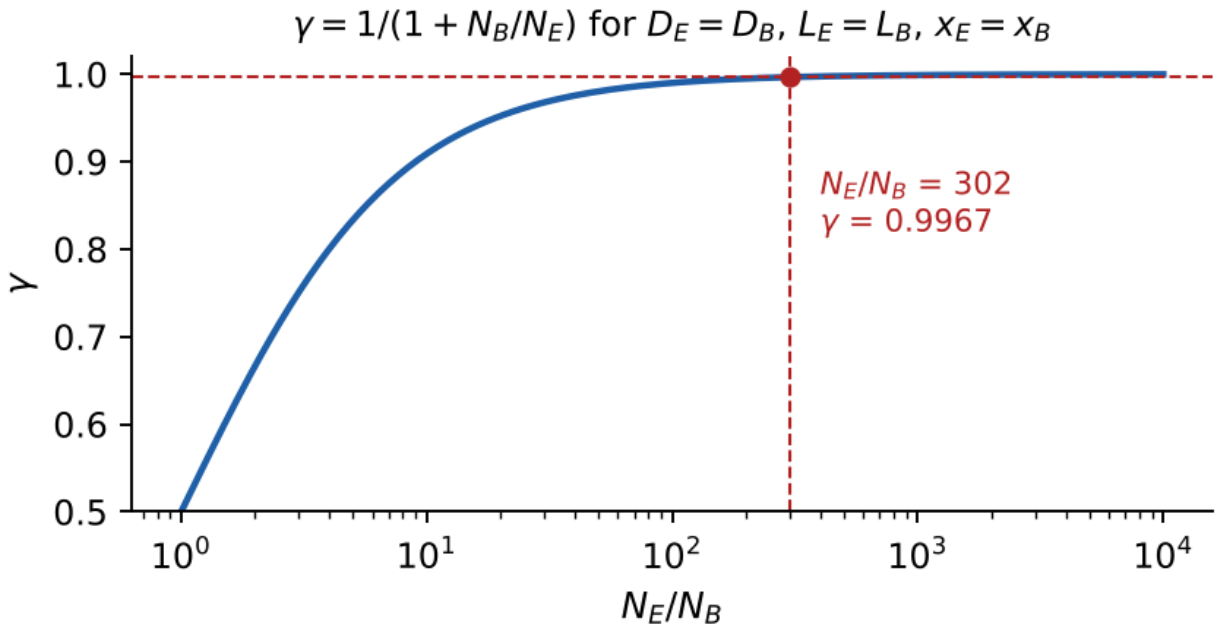


Figure 27: Emitter injection efficiency against the emitter-to-base doping ratio

Reading the figure. The solid blue curve is $\gamma = 1/(1 + N_B/N_E)$ against N_E/N_B on a log scale. It rises from 0.5 at equal doping, passes 0.909 at 10 and 0.990 at 100, and reaches the target 0.9967 at $N_E/N_B = 302$ (red dot, dashed red lines).

3. Keyword Explanation

1. Emitter injection efficiency

γ, the fraction of the emitter current carried by carriers injected into the base.

2. Common-base current gain

$\alpha = \gamma\alpha_T\delta$; $\beta = \alpha/(1 - \alpha)$.

4. Notes and Remarks

- **Book check:** this is Neamen's Design Example 12.4 word for word ("$x_E = X_B$" on the paper is $x_E = x_B$); the book's answer is $N_B/N_E = 0.00331$, $N_E/N_B = 302$.
- **Why 0.9967:** if $\gamma = \alpha_T = \delta = 0.9967$, then $\alpha = 0.9967^3 = 0.990$ and $\beta = 99 \approx 100$ (Neamen's reason for the number).

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- **Limit:** at very heavy emitter doping, band-gap narrowing raises p_{E0} above n_i^2/N_E and γ stops improving (Neamen, Sec. 12.4.3); heterojunction transistors avoid this.

6.(c) What is current crowding? What is the impact of current crowding in high power devices? Marks: 7

1. Theory

1. Base resistance (Core Idea)

The base of a bipolar transistor is a thin p layer (less than 1 μm thick) between the emitter and the collector. The base current enters through base contacts at the **sides** of the emitter and has to flow **sideways** (laterally) along this thin layer to reach the parts of the B-E junction under the emitter. A thin layer has a sizable resistance, so the lateral base current makes a voltage drop in the base.

2. Injection is exponential in the local voltage

The emitter is heavily doped and acts as one equipotential. The electron injection at each point of the B-E junction depends on the **local** B-E voltage as $\exp(eV_{BE}/kT)$. A difference of only $V_t \ln 2 = 18$ mV between two points doubles the injection at one of them.

2. Answer

Current crowding (emitter current crowding): the non-uniform distribution of emitter current across the emitter area of a bipolar transistor, in which the current is concentrated ("crowded") at the **edges** of the emitter, because the lateral base current makes a voltage drop in the base resistance under the emitter (Neamen, Sec. 12.4.4).

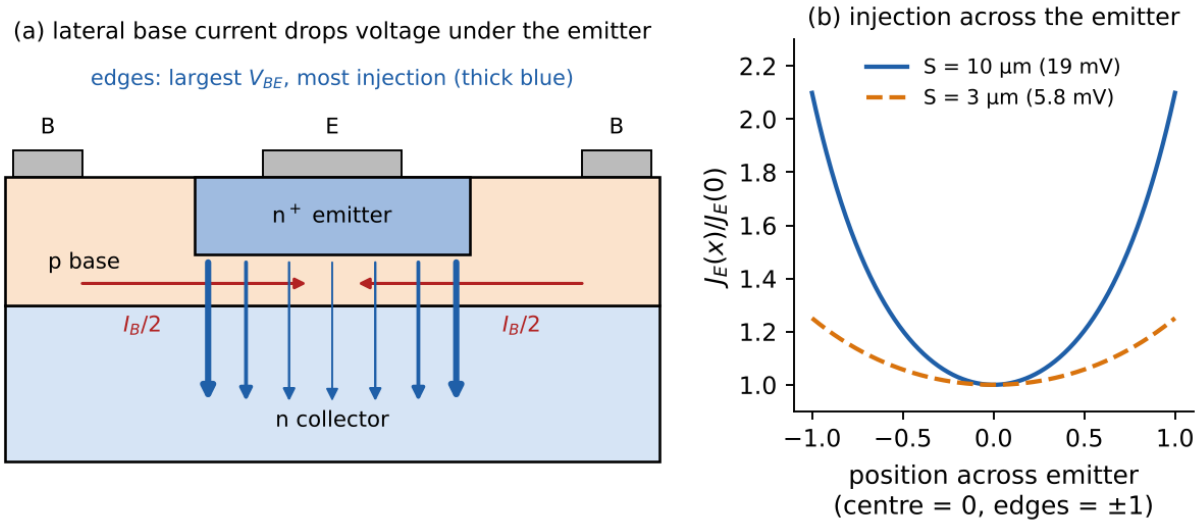


Figure 28: Emitter current crowding: lateral base current and the injection across the emitter

Reading the figure. Panel (a): cross section of an npn transistor with base contacts B on both sides of the emitter E. Half the base current, $I_B/2$ (red arrows), flows in from each side under the emitter. The blue arrows are the injected electron current; the thick ones at the emitter edges show more injection there than in the centre. Panel (b): the relative emitter current density across the emitter for Neamen's Problem 12.42 base ($N_B = 2\times10^{16}$ cm^{-3}, $x_B = 0.65$ μm, $\mu_p = 250$ cm^2/V s, $L = 25$ μm, $I_B/2 = 5$ μA): the solid blue curve, for a 10 μm wide emitter (19 mV drop), is 2.1 times higher at the edges than at the centre; the dashed orange curve, for a 3 μm emitter (5.8 mV), only 1.25 times.

Why it happens (npn):

1. The base current flows laterally from the base contacts towards the centre of the emitter through the base resistance r_b.

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2. The voltage drop $I_B r_b$ makes the base potential **lower at the emitter centre** than at its edges, so the forward bias V_{BE} is largest at the edges and smallest at the centre.
3. Injection depends exponentially on V_{BE}, so many more electrons are injected near the edges: the emitter current crowds to the edges.

Example (Neamen, Problem 12.42, upper estimate with the whole $I_B/2$ crossing half the emitter)

$\rho = 1/(e\mu_p N_B) = 1.25\ \Omega$ cm; for $S = 10\ \mu$m, $R = \rho(S/2)/(x_B L) = 3.85$ kΩ, $\Delta V = (5\ \mu\text{A})(3.85\ \text{k}\Omega) = 19.2$ mV, so injection at the edge is $e^{19.2/25.9} = 2.10$ times that at the centre. For $S = 3\ \mu$m: $\Delta V = 5.77$ mV and a ratio of 1.25.

Impact in high-power devices:

- **Local high current density** at the emitter edges: high-injection effects appear there first, lowering the current gain at high currents.
- **Local heating (hot spots):** the edges dissipate more power and get hotter; since injection grows with temperature, this can run away into **thermal runaway and second breakdown**, destroying the device.
- **Poor use of the emitter area:** the centre carries little current, so a wide emitter does not give proportionally more current; the effective area is smaller than the drawn area.
- **Larger effective base resistance and lower speed**, and a non-uniform base current.
- **Remedy:** power transistors use **many narrow emitter fingers** (interdigitated emitter and base, Neamen Fig. 12.29), so the lateral path and the base drop are small, and emitter ballast resistors to share the current evenly.

3. Keyword Explanation

1. Base spreading resistance

The resistance r_b of the thin base layer to the lateral base current.

2. Interdigitated structure

Emitter and base contacts arranged as interleaved narrow fingers.

3. Second breakdown

A destructive failure of a power transistor caused by current concentrating in a hot spot.

4. Notes and Remarks

- **Narrower is better:** the drop grows with the emitter width S (for a fixed current per unit area it grows as S^2), which is why the 3 μm emitter crowds much less than the 10 μm one.
- **The estimate is an upper bound:** in reality the base current leaves the lateral path gradually along the way, and a two-dimensional analysis would give a smaller drop (Neamen's comment).
- **Lateral current in MOSFETs:** a similar debiasing occurs along long gate fingers, which is why power MOSFETs are also built from many small cells.

6.(d) With suitable energy band diagrams, explain Schottky contact. Marks: 6

1. Theory

1. Work function and electron affinity (Core Idea)

- The **work function** of a metal, $e\phi_m$, is the energy needed to lift an electron from its Fermi level to the **vacuum level** (just outside the solid).
- In a semiconductor, E_F moves with doping, so we use the **electron affinity** $e\chi$: the energy from the conduction band edge E_c to the vacuum level (4.01 eV for Si). Its work function is $e\phi_s = e\chi + (E_c - E_F)$.

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2. Contact makes the Fermi levels equal

When two materials touch, electrons flow from the one with the higher Fermi level (smaller work function) into the other until the Fermi levels line up. For a metal with $\phi_m > \phi_s$ on n-type silicon, the semiconductor's Fermi level is higher, so electrons flow from the semiconductor into the metal:

- the metal gains a thin sheet of negative charge at its surface;
- the semiconductor near the interface loses its electrons and keeps the positive donor ions: a **depletion region** of width x_n;
- the bands of the semiconductor bend *up* near the interface.

2. Answer

Schottky contact: a rectifying metal-semiconductor contact, formed when a metal of large work function touches an n-type semiconductor ($\phi_m > \phi_s$; for p-type, $\phi_m < \phi_s$). A potential barrier forms at the interface, and the contact conducts easily in one direction and blocks in the other, like a pn diode (the **Schottky barrier diode**; Neamen, Sec. 9.1).

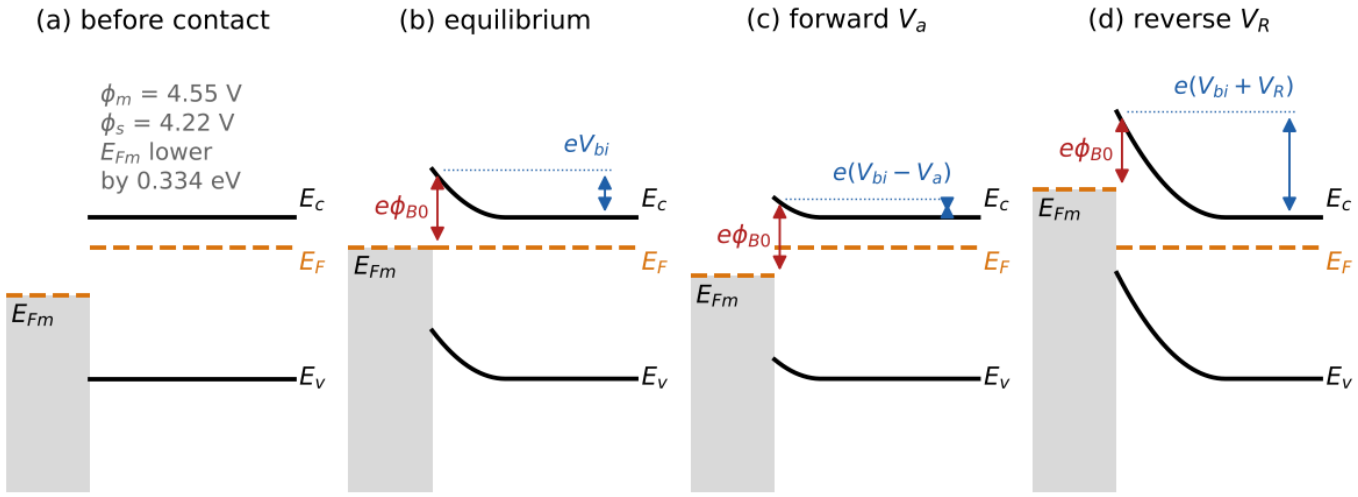


Figure 29: Energy band diagrams of a Schottky contact on n-type silicon: before contact, in equilibrium, forward and reverse bias

Reading the figure. Tungsten ($\phi_m = 4.55$ V) on n-Si with $N_d = 10^{16}$ cm^{-3} ($\phi_s = 4.22$ V). The grey block is the metal, filled up to E_{Fm}; the solid black lines are E_c and E_v of the semiconductor and the dashed orange lines the Fermi levels. (a) Before contact the metal Fermi level lies 0.334 eV below that of the semiconductor. (b) In equilibrium the Fermi levels line up, the bands bend up, and two barriers appear: $e\phi_{B0}$ (red) from the metal and eV_{bi} (blue) from the semiconductor. (c) Forward bias V_a (metal positive) lowers the semiconductor-side barrier to $e(V_{bi} - V_a)$; $e\phi_{B0}$ is unchanged. (d) Reverse bias V_R raises it to $e(V_{bi} + V_R)$ and widens the depletion region.

1. Before contact (panel a): the metal Fermi level is $e(\phi_m - \phi_s)$ below the semiconductor's.

2. Equilibrium (panel b): electrons move from the semiconductor into the metal until E_F is flat. The positive depletion layer bends the bands up. Two barriers result:

- the **Schottky barrier height** seen by electrons in the metal (Neamen, Eq. 9.1):

$$\phi_{B0} = \phi_m - \chi$$

- the **built-in potential** seen by electrons in the semiconductor (Eq. 9.2):

$$V_{bi} = \phi_{B0} - \phi_n, \qquad \phi_n = \frac{kT}{e} \ln\left(\frac{N_c}{N_d}\right)$$

The depletion width is that of a one-sided junction, $x_n = [2\epsilon_s(V_{bi} + V_R)/(eN_d)]^{1/2}$ (Eq. 9.7).

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3. Forward bias (panel c, metal positive): the semiconductor's Fermi level is raised by eV_a; the barrier from the semiconductor side falls to $V_{bi} - V_a$, and many electrons flow from the semiconductor over the barrier into the metal. The current rises exponentially. ϕ_{B0} does not change.
4. Reverse bias (panel d): the barrier from the semiconductor side rises to $V_{bi} + V_R$, the depletion layer widens, and only the small current of metal electrons that cross ϕ_{B0} flows.
Current-voltage relation: by thermionic emission over the barrier (Neamen, Eqs. 9.25, 9.26)

$$J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right], \qquad J_{sT} = A^*T^2 \exp\left(-\frac{e\phi_{B0}}{kT}\right)$$

with A^* the effective Richardson constant (about 120 A/cm^{2}K^2 for free electrons).

Example (Neamen, Example 9.1)

W on n-Si, $N_d = 10^{16}$ cm^{-3}: $\phi_{B0} = 4.55 - 4.01 = 0.54$ V; $\phi_n = 0.0259\ln(2.8 \times 10^{19}/10^{16}) = 0.206$ V; $V_{bi} = 0.334$ V; $x_n = 0.208$ μm and $|E_{max}| = 3.21 \times 10^4$ V/cm at zero bias.

Compared with a pn junction:

- current carried by **majority carriers** (thermionic emission), so there is no minority charge storage and the switching is very fast (no storage time, 2(a));
- J_{sT} is many orders of magnitude larger, so the turn-on voltage is lower (about 0.1 to 0.3 V against 0.6 V for Si; here 0.12 V against 0.60 V for 1 A/cm^2);
- used in fast rectifiers, switching power supplies, clamping diodes and MESFET gates.

3. Keyword Explanation

1. Schottky barrier height

$\phi_{B0} = \phi_m - \chi$ (ideal), the barrier electrons in the metal must cross.

2. Electron affinity

$e\chi$, the energy from E_c to the vacuum level.

3. Thermionic emission

Current carried by electrons with enough thermal energy to pass over a barrier.

4. Notes and Remarks

- **Real barrier heights:** interface states and image-force lowering (Neamen, Sec. 9.1.3) make the measured ϕ_{B0} differ from $\phi_m - \chi$ (for W on Si the measured value is about 0.67 V).
- **Numbers in the comparison:** J_{sT} is computed with $A^* = 120$ A/cm^{2}K^2 (9.5×10^{-3} A/cm^2) against an ideal pn junction with $N_a = N_d = 10^{16}$ cm^{-3}, $D_p = 10$, $D_n = 25$ cm^2/s, $\tau = 10^{-7}$ s (9.3×10^{-11} A/cm^2), illustrative values.
- **Book check:** Neamen's Example 9.1 gives $\phi_{B0} = 0.54$ V, $V_{bi} = 0.334$ V, $x_n = 0.208 \times 10^{-4}$ cm and $|E_{max}| = 3.21 \times 10^4$ V/cm, as above.
- **Ohmic contacts** are the other kind of metal-semiconductor contact (4(c)).

5. Final Concise Answer

(a) $f_T = g_m/(2\pi C_G)$ with $g_{ms}(\max) = e\mu_n N_d W a/L$ and $C_G(\min) = \epsilon_s WL/a$, so $f_T = e\mu_n N_d a^2/(2\pi\epsilon_s L^2) = 2.000 \times 10^{-9}/2.602 \times 10^{-19} = 7.69$ GHz.
(b) $\gamma = 1/[1 + (p_{E0}D_E L_B/n_{B0}D_B L_E)\tanh(x_B/L_B)/\tanh(x_E/L_E)]$ reduces with the given equalities to $\gamma = 1/(1 + N_B/N_E)$; $N_B/N_E = 1/0.9967 - 1 = 0.00331$, so $N_E/N_B = 302$.
(c) The lateral base current drops voltage in the base resistance, so V_{BE} is larger at the emitter edges and injection ($\propto e^{V_{BE}/V_t}$) crowds to the edges (Problem 12.42: 2.1 times more at the edges of a 10 μm

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emitter). In power devices it causes local high injection and gain fall-off, hot spots, thermal runaway and second breakdown, and wastes emitter area; cured by narrow interdigitated emitter fingers and ballast resistors.

(d) A rectifying metal-semiconductor contact ($\phi_m > \phi_s$ on n-type): electrons flow into the metal, a depletion layer forms, the bands bend up, giving a barrier $\phi_{B0} = \phi_m - \chi$ from the metal and $V_{bi} = \phi_{B0} - \phi_n$ from the semiconductor; forward bias lowers V_{bi} to $V_{bi} - V_a$, reverse bias raises it; $J = A^* T^2 e^{-e\phi_{B0}/kT}(e^{eV_a/kT} - 1)$; majority-carrier device, fast, low turn-on (W on Si: $\phi_{B0} = 0.54$ V, $V_{bi} = 0.334$ V).

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7. (a) Define the density of states and carrier concentration. Marks: 2

   (b) Explain with the help of equations and graphs- why intrinsic carrier concentration (n_i) for a typical semiconductor is high (~10^6-10^9/cc) at 300K even though the Fermi Dirac distribution is low at that temperature. Marks: 3

   (c) What is electron-effective mass? Two possible conduction bands in the E versus k diagram are given in Figure 1. State which band will result in the heavier electron effective mass;state why. Marks: 3

   (d) Using the necessary theorem explains how band structure is formed of a particular solid-state material. Marks: 3

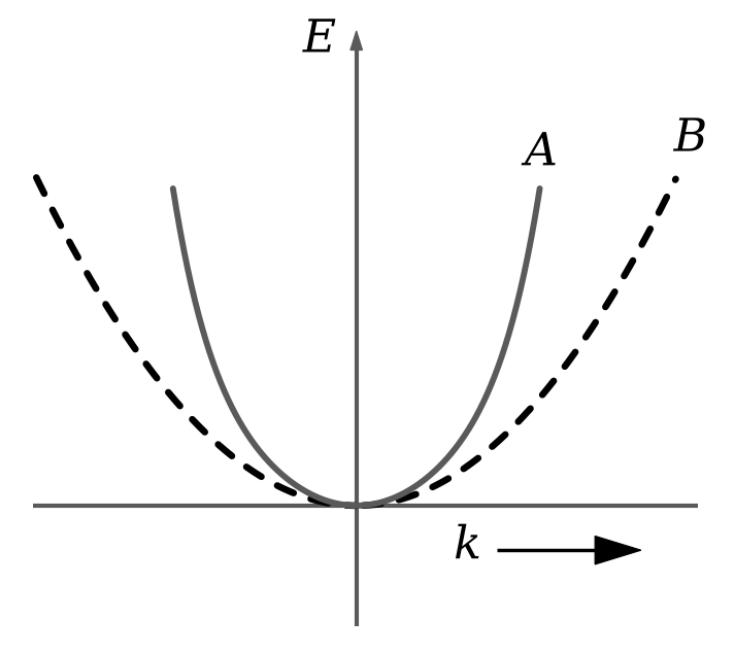


Figure 30: Figure 1: Two possible conduction bands A and B in the E versus k diagram (redrawn from the paper)

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7.(a) Define the density of states and carrier concentration. Marks: 2

1. Theory

1. Allowed states in a crystal (Core Idea)

Quantum mechanics allows an electron in a crystal only certain states, each with its own wave number k and energy E. In a band these states are packed very closely, so instead of counting them one by one we say how many lie in a small energy range. Each state can hold one electron (Pauli). Whether a state is actually filled is a separate question, answered by the Fermi-Dirac function $f_F(E)$ (2(b)).

2. Counting the states near a band edge

Near the bottom of the conduction band the energy is $E = E_c + \hbar^2 k^2/2m_n^*$ (Neamen, Eq. 3.70). Counting the k-states in a thin spherical shell in k-space and changing variable from k to E gives (Neamen, Eq. 3.72)

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}, \quad E \geq E_c$$

and for the valence band (Eq. 3.75)

$$g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E}, \quad E \leq E_v$$

So there are no states in the gap, and the number grows as the square root of the energy into each band.

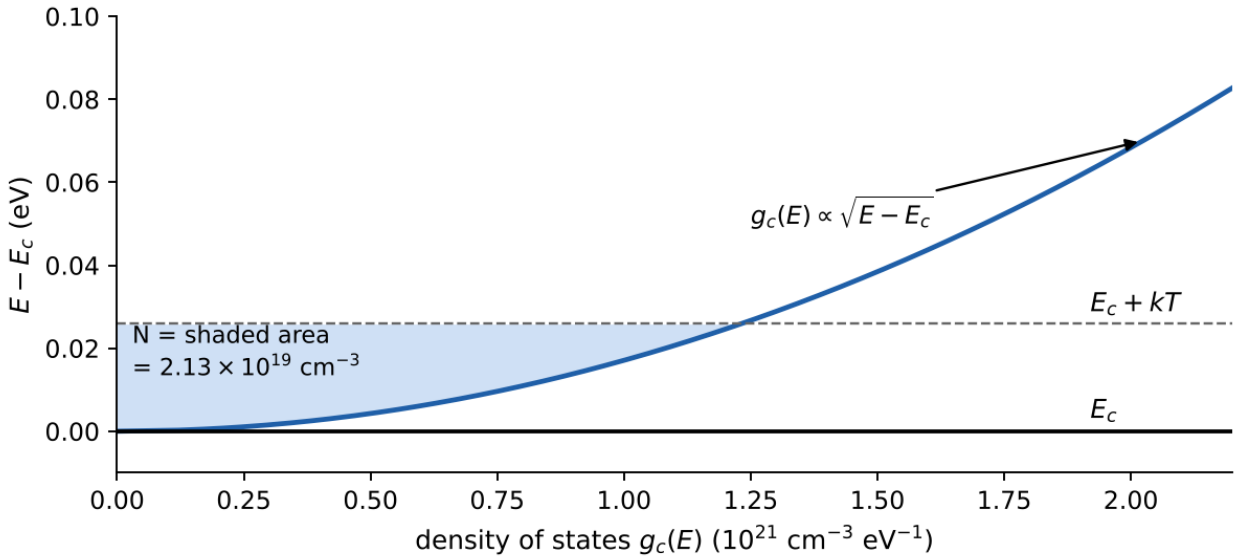


Figure 31: Density of states in the conduction band of Si; the shaded area between E_c and $E_c + kT$ is the number of states asked for

Reading the figure. The solid blue curve is $g_c(E)$ for Si, with energy upwards. It starts at zero at E_c (solid black line) and grows as $\sqrt{E - E_c}$. The shaded area between E_c and the dashed grey line $E_c + kT$ is the number of states between E_c and $E_c + kT$ (a 2020–21 question; here it only illustrates the shape).

2. Answer

Density of states: the density of states $g(E)$ is the number of allowed quantum states per unit energy per unit volume of the crystal at energy E (units cm^{-3} eV^{-1}). Near the conduction band edge $g_c(E) = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$ (Neamen, Eq. 3.72); it says how many states are **available**, not how many are filled.

Carrier concentration: the carrier concentration is the number of free charge carriers per unit volume: electrons in the conduction band (n_0) or holes in the valence band (p_0), in cm^{-3}. It is the number of

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available states times the probability that each is occupied, added over the band (Neamen, Eqs. 4.1 to 4.3; the result is Eq. 4.11):

$$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$$

and for holes $p_0 = \int g_v(E)[1 - f_F(E)]\, dE = N_v \exp[-(E_F - E_v)/kT]$ (Eq. 4.19).

3. Keyword Explanation

1. Quantum state

One allowed solution of the Schrödinger equation (a value of k and a spin); it holds at most one electron.

2. Effective density of states

$N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$, the result of the integral above; 2.8×10^{19} cm^{-3} for Si at 300 K.

4. Notes and Remarks

- **Marks on the paper:** the class test's header says Marks: 10, but its four questions carry 2 + 3 + 3 + 3 = 11. Both are kept as printed.
- **Two different quantities:** the density of states belongs to the band structure alone; the carrier concentration also depends on the temperature and the Fermi level (through f_F).
- **Units:** $g(E)$ is per unit energy and volume; integrating it over an energy range gives a number per unit volume, the unit of n_0.

7.(b) Explain with the help of equations and graphs- why intrinsic carrier concentration (n_i) for a typical semiconductor is high (~10^6-10^9/cc) at 300K even though the Fermi Dirac distribution is low at that temperature. Marks: 3

1. Theory

1. The intrinsic semiconductor (Core Idea)

In a pure semiconductor at 300 K a few electrons are thermally lifted across the gap; each leaves a hole, so $n_0 = p_0 = n_i$. The Fermi level of the intrinsic material, E_{Fi}, lies near the middle of the gap, about $E_g/2 = 0.56$ eV below E_c in Si.

2. Two factors make the carrier density

The electron concentration in the conduction band is (Neamen, Eq. 4.1)

$$n(E) = g_c(E) f_F(E)$$

summed over the band. It is the product of:

- the **density of states** $g_c(E)$: how many places there are for electrons, which is **enormous** (of the order of 10^{19} to 10^{20} per cm^3 within a few kT of the band edge);
- the **Fermi-Dirac probability** $f_F(E)$: the chance that each place is filled, which is **tiny** near E_c.

A tiny probability times a huge number of states can still be a sizeable number.

2. Answer

The paradox: with E_F near midgap in Si, the Fermi-Dirac probability of a state at E_c being filled is (Neamen, Eq. 3.79)

$$f_F(E_c) = \frac{1}{1 + \exp\left(\dfrac{0.56}{0.0259}\right)} \approx \exp\left(-\frac{0.56}{0.0259}\right) = 4.07 \times 10^{-10}$$

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only about 4 in 10^{10}. Yet $n_i = 1.5 \times 10^{10}$ cm^{-3} is a sizeable number.

The reason: the carrier concentration is the probability **times the number of states**, integrated over the band (Neamen, Eqs. 4.3 to 4.11):

$$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$$

The effective density of states $N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$ is **huge**: 2.8×10^{19} cm^{-3} for Si at 300 K. So

$$n_i \approx N_c\, f_F(E_c) = (2.8 \times 10^{19})(4.07 \times 10^{-10}) \approx 1.1 \times 10^{10} \text{ cm}^{-3}$$

With both bands (Neamen, Eq. 4.23):

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right) = \sqrt{(2.8 \times 10^{19})(1.04 \times 10^{19})}\, e^{-1.12/0.0518} = 6.9 \times 10^{9} \text{ cm}^{-3}$$

of the same order as the accepted 1.5×10^{10} cm^{-3}.

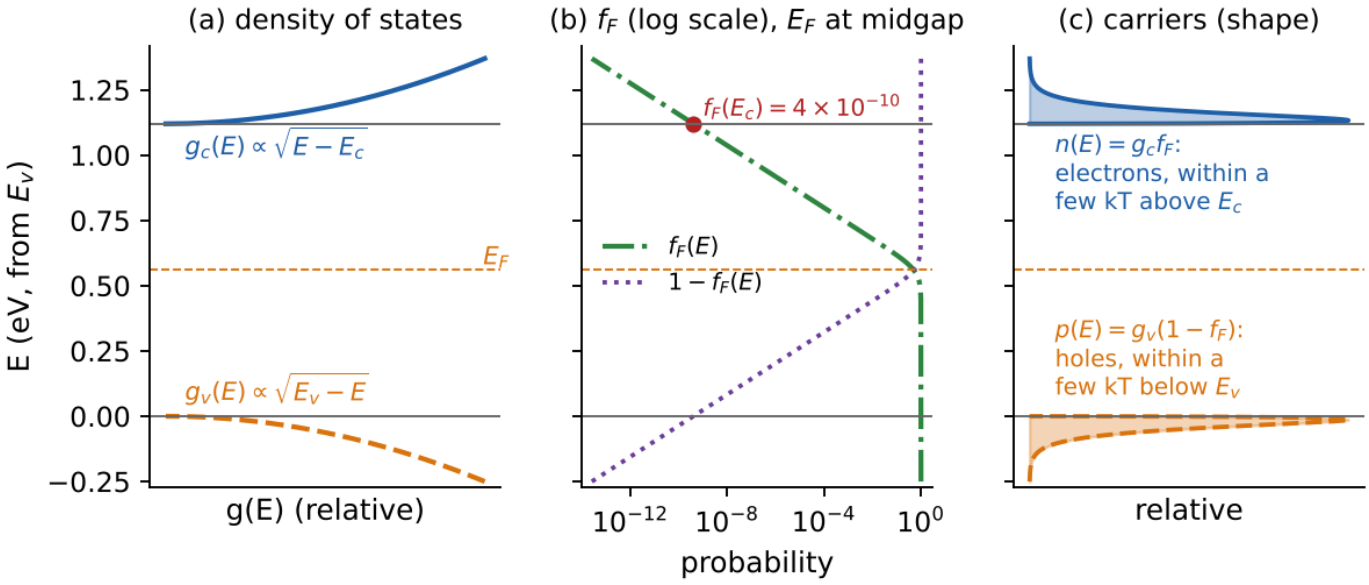


Figure 32: Density of states, Fermi-Dirac function and their product for intrinsic silicon

Reading the figure. Energy is upwards in all three panels, with E_c and E_v as grey lines and E_F (dashed orange) at midgap. Panel (a): the solid blue $g_c(E)$ and dashed orange $g_v(E)$ grow as square roots into the bands: many states. Panel (b): on a log scale, the dash-dot green $f_F(E)$ falls to 4×10^{-10} at E_c (red dot), and the dotted purple $1 - f_F$ falls just as fast below E_F. Panel (c): the products, electrons $n(E) = g_c f_F$ (blue, shaded) and holes $p(E) = g_v(1 - f_F)$ (orange dashed, shaded), are confined to a few kT next to each band edge; their areas are n_0 and p_0.

In words: each state has a tiny chance of holding an electron, but there are so many states ($\sim 10^{19}$ cm^{-3}) that the expected number of electrons is $\sim 10^{10}$ cm^{-3}. Compared with the 5×10^{22} Si atoms per cm^3 it is still very small (one free electron per 3×10^{12} atoms), but it is a large absolute number, and it is why a pure semiconductor conducts at all.

For other semiconductors the same formula gives n_i from about 10^6 to 10^{13} cm^{-3} at 300 K, set by E_g: GaAs ($E_g = 1.42$ eV) $\approx 2 \times 10^6$, Si $\approx 10^{10}$, Ge (0.66 eV) $\approx 2 \times 10^{13}$ cm^{-3}; the paper's range 10^6 to 10^9 corresponds to the wider-gap materials.

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3. Keyword Explanation

1. Intrinsic carrier concentration

n_i, the electron (and hole) concentration of a pure semiconductor; $n_i^2 = N_c N_v e^{-E_g/kT}$.

2. Effective density of states

N_c (or N_v), the number of states that, placed at the band edge, would hold the same number of carriers as the whole band.

4. Notes and Remarks

- **Why the simple estimate differs from 1.5×10^{10}:** $\sqrt{N_c N_v}e^{-E_g/2kT}$ with Neamen's N_c, N_v and E_g gives 6.9×10^9 cm^{-3}; Neamen notes that the accepted (measured) value 1.5×10^{10} cm^{-3} differs because of uncertainties in the effective masses and E_g (Sec. 4.1.3). The argument does not change.
- **Exponential sensitivity:** each extra 0.06 eV of half-gap divides n_i by 10, so E_g decides the order of magnitude.
- **Paper's notation:** "/cc" means per cubic centimetre (cm^{-3}).

7.(c) What is electron-effective mass? Two possible conduction bands in the E versus k diagram are given in Figure 1. State which band will result in the heavier electron effective mass;state why. Marks: 3

1. Theory

1. An electron in a crystal is not free (Core Idea)

A free electron has $E = \hbar^2 k^2/2m_0$, a parabola. In a crystal the electron also feels the periodic forces of the ions. Rather than add up all those forces, we keep the free-particle equations and give the electron a different mass, the **effective mass** m^*, which includes the effect of the lattice.

2. Mass from the curvature of E(k)

Differentiate the free-electron relation twice: $d^2E/dk^2 = \hbar^2/m$ (Neamen, Eq. 3.40), so

$$\frac{1}{m} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2} \qquad \text{(Neamen, Eq. 3.41)}$$

Near the bottom of a conduction band any $E(k)$ curve is close to a parabola, $E - E_c = C_1 k^2$ (Eq. 3.44), so $d^2E/dk^2 = 2C_1$ and (Eq. 3.47)

$$\frac{1}{m_n^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{2C_1}{\hbar^2}$$

With this mass the electron obeys Newton's law, $F = m_n^* a = -eE$ (Eq. 3.48).

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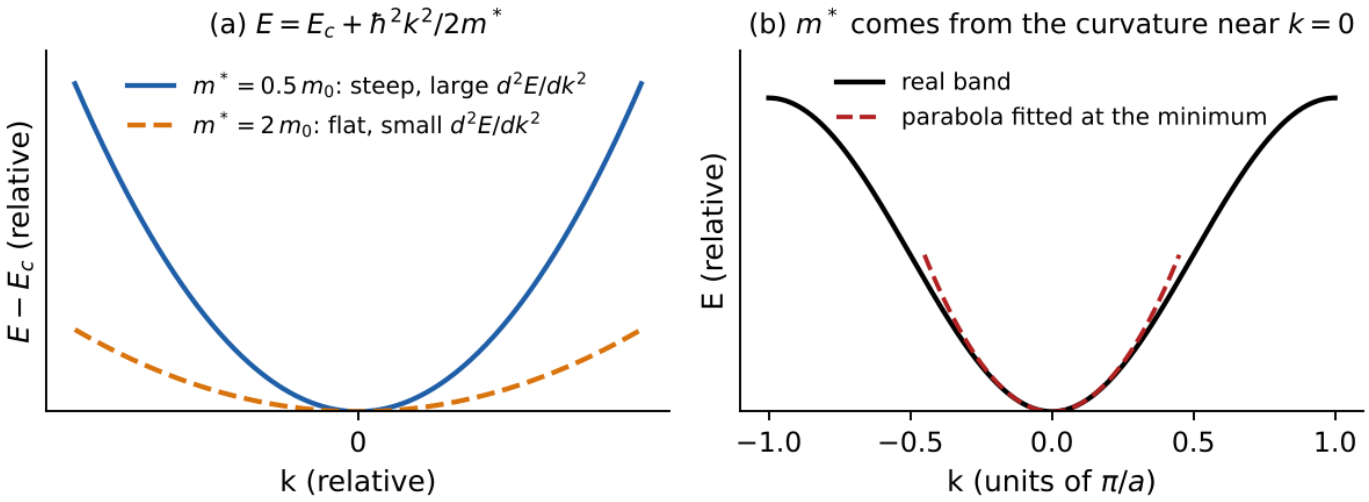


Figure 33: Effective mass and the curvature of the E-k curve

Reading the figure. Panel (a): two parabolas $E = E_c + \hbar^2k^2/2m^*$; the solid blue one ($m^* = 0.5\,m_0$) is narrow and steep, with a large d^2E/dk^2; the dashed orange one ($m^* = 2\,m_0$) is wide and flat, with a small d^2E/dk^2. Panel (b): a real band (solid black) is not a parabola, but near its minimum the dashed red parabola fits it; m^* comes from that curvature.

2. Answer

Electron effective mass: the mass m_n^* that an electron in the conduction band of a crystal appears to have when it is accelerated by an external force; it includes the effect of the internal periodic crystal forces. It is set by the curvature of the E-k curve (Neamen, Eq. 3.47):

$$m_n^* = \frac{\hbar^2}{d^2E/dk^2}$$

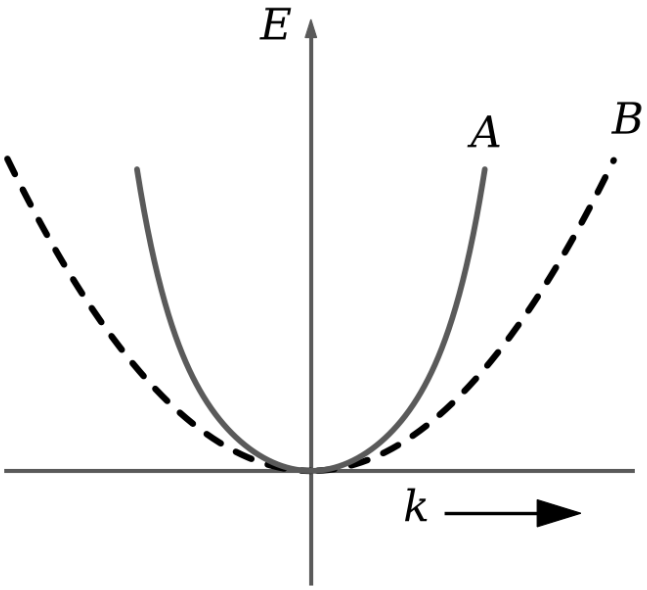


Figure 34: Repeated from the question: Figure 1: Two possible conduction bands A and B in the E versus k diagram (redrawn from the paper)

Which band is heavier: band B.
Why:

- Both bands have their minimum at $k = 0$. Band **A** (solid curve) is **narrow and steep**: its

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energy rises quickly with k, so its curvature d^2E/dk^2 is **large**.

- Band **B** (dashed curve) is **wide and flat**: its energy rises slowly with k, so its curvature d^2E/dk^2 is **small**.
- Since $m^* = \hbar^2/(d^2E/dk^2)$, the mass is inversely proportional to the curvature: the smaller curvature of B gives the **larger** effective mass.

$$\left(\frac{d^2E}{dk^2}\right)_B < \left(\frac{d^2E}{dk^2}\right)_A \quad \Rightarrow \quad m_B^* > m_A^*$$

So band B gives the heavier electron effective mass (from the drawing, its curvature is roughly half that of A, so its mass is roughly twice as large). A heavier electron accelerates less for the same force, so band B also gives a lower mobility.

3. Keyword Explanation

1. E-k diagram

The plot of electron energy against wave number (crystal momentum $\hbar k$) for a band.

2. Curvature

d^2E/dk^2, how sharply the E-k curve bends.

4. Notes and Remarks

- **Measuring the figure:** a parabola fitted to each curve of the scan gives a curvature ratio A/B of about 2.4 (between 1.8 and 2.7 depending on the height compared), so $m_B^* \approx 2$ to $2.5\,m_A^*$; the paper gives no scale, so only the comparison is asked.
- **Real materials:** GaAs has a steep conduction band minimum, $m_n^* = 0.067\,m_0$ (light, fast electrons); Si has $m_n^* = 1.08\,m_0$ (density-of-states mass), a flatter band.
- **Holes:** at a valence band maximum $d^2E/dk^2 < 0$, which gives a negative electron mass there, the reason holes are treated as positive carriers with positive mass (Neamen, Eq. 3.56 onward).

7.(d) Using the necessary theorem explains how band structure is formed of a particular solid-state material. Marks: 3

1. Theory

1. Energy levels of single atoms (Core Idea)

In an isolated atom, an electron can only have certain discrete energies (the 1s, 2s, 2p, 3s, 3p, … levels). Two identical atoms far apart have exactly the same set of levels.

2. What happens when atoms come close

When atoms are brought together to form a crystal, the outer electron wave functions of neighbouring atoms **overlap**, and the atoms interact. By the **Pauli exclusion principle** no two electrons in the interacting system may occupy the same quantum state. So each atomic level must **split** into as many distinct levels as there are atoms: two atoms give two levels, N atoms give N levels. In a crystal of about 5×10^{22} atoms per cm^3 these levels are so close together that they form a practically continuous **band** of allowed energies (Neamen, Sec. 3.1.1).

3. Bloch's theorem

The ions of a crystal are arranged periodically, with period a, so the potential energy is periodic, $V(x + a) = V(x)$. **Bloch's theorem** states that every solution of the Schrödinger equation in a periodic potential has the form (Neamen, Sec. 3.1.2)

$$\psi(x) = u(x)e^{jkx}$$

where $u(x)$ is periodic with the period of the lattice, $u(x + a) = u(x)$, and k is the wave number. The electron is a plane wave modulated by the lattice.

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2. Answer

Qualitative picture (Pauli principle): as the interatomic distance r shrinks to the crystal spacing r_0, each discrete atomic level splits into N closely spaced levels, which form an **allowed energy band**. The bands are separated by **forbidden bands (energy gaps)**, energies with no allowed states. In silicon the 3s and 3p levels split and then mix into two bands of $4N$ states each: the lower one (valence band) is full of the $4N$ valence electrons at 0 K and the upper one (conduction band) is empty, with the gap $E_g = 1.12$ eV between them.

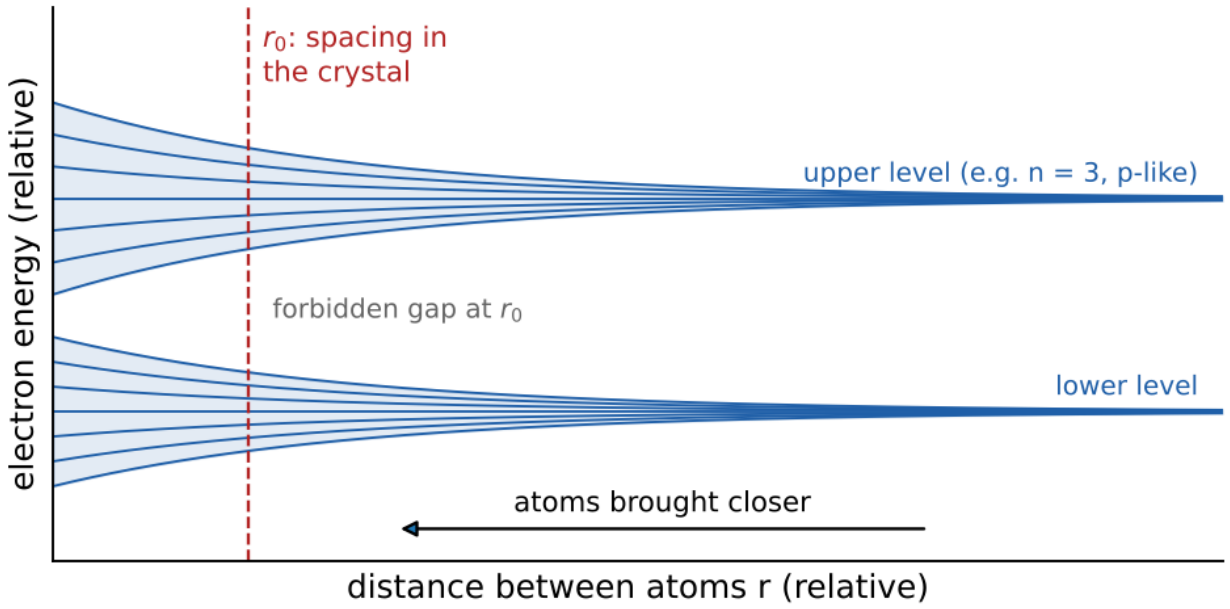


Figure 35: Splitting of two atomic levels into bands as the atoms are brought closer

Reading the figure. On the right, far apart, each atom has two sharp levels (blue lines). Moving left (atoms closer), each level splits into many levels (thin lines) that spread into a band (shaded). At the crystal spacing r_0 (dashed red line) the two bands are separated by a forbidden gap.

Quantitative picture (Bloch's theorem and the Kronig-Penney model, Neamen, Sec. 3.1.2):

1. Model the periodic potential of the ions as a row of square wells and barriers of period $a + b$.
2. By **Bloch's theorem**, $\psi(x) = u(x)e^{jkx}$ with u periodic. Solving the Schrödinger equation in the well and the barrier and matching ψ and $d\psi/dx$ at the boundaries (and using the periodicity of u) gives, for very thin high barriers, the condition (Neamen, Eqs. 3.24, 3.29, 3.30)

$$P'\frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka, \qquad \alpha^2 = \frac{2mE}{\hbar^2}$$

where P' measures the strength of the barriers.

3. The right-hand side, $\cos ka$, can only lie between -1 and $+1$. The left-hand side $f(\alpha a)$ goes outside this range for some energies. So:

   - energies for which $|f(\alpha a)| \leq 1$ are **allowed** (a real k exists): the **allowed bands**;
   - energies for which $|f(\alpha a)| > 1$ have no real k: the **forbidden bands** (gaps).

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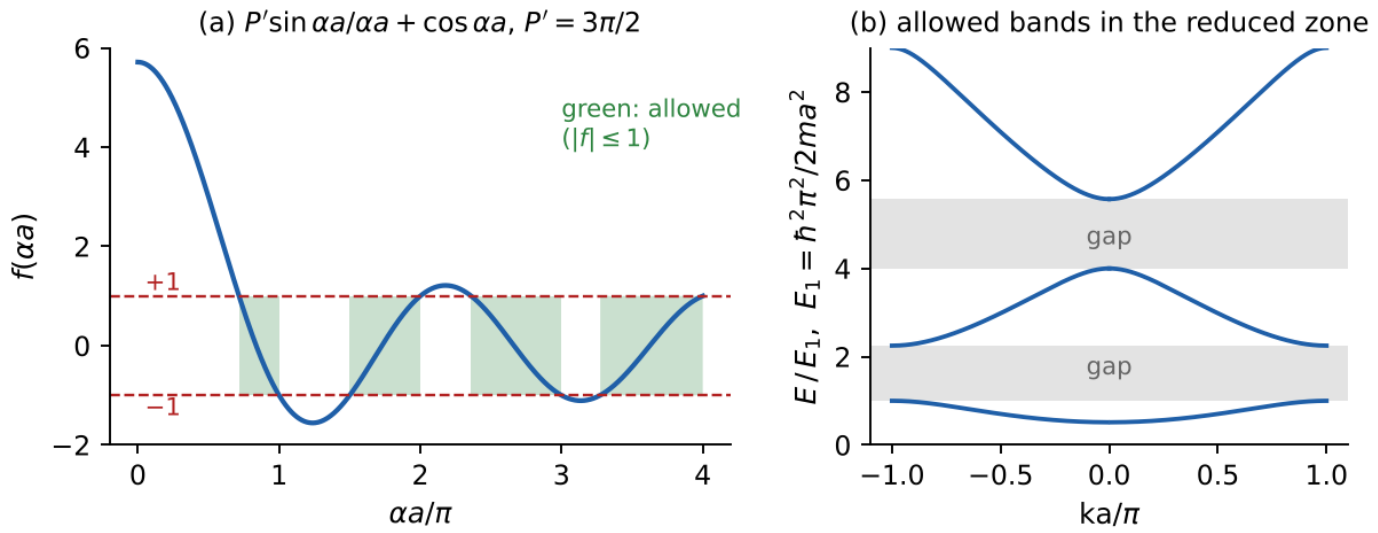


Figure 36: The Kronig-Penney function and the allowed bands it gives

Reading the figure. Panel (a): the solid blue curve is $f(\alpha a) = P'\sin\alpha a/\alpha a + \cos\alpha a$ with $P' = 3\pi/2$; the dashed red lines are ± 1, and the green shading marks the allowed ranges, $\alpha a/\pi$ from 0.72 to 1, 1.5 to 2, 2.36 to 3 and 3.27 to 4 (the plot stops at 4). Panel (b): the resulting E-k diagram in the reduced zone ($-\pi < ka < \pi$): three allowed bands (solid blue) separated by forbidden gaps (grey); each band edge falls at $ka = 0$ or $\pm\pi$, where the curves are flat.

Result: the periodic potential breaks the free-electron parabola into **allowed bands separated by forbidden gaps**; the gaps open at $ka = n\pi$. With $P' \to 0$ (no lattice) the gaps close and $E = \hbar^2k^2/2m$ again; with $P' \to \infty$ the bands shrink to the discrete levels of isolated atoms. Whether a material is a metal, semiconductor or insulator depends on how its electrons fill these bands and on the size of the gap.

3. Keyword Explanation

1. Bloch's theorem

In a periodic potential every electron wave function has the form $u(x)e^{jkx}$ with u having the lattice period.

2. Kronig-Penney model

A one-dimensional periodic array of square wells and barriers, solvable exactly, which shows the band structure.

3. Allowed and forbidden bands

Energy ranges with and without electron states.

4. Notes and Remarks

- **"The necessary theorem":** read here as Bloch's theorem (with the Pauli exclusion principle for the splitting); the Kronig-Penney model is how Neamen applies it. $P' = 3\pi/2$ is chosen for the figure only.
- **Band edges computed:** $|f| = 1$ at $\alpha a/\pi = 0.716,\ 1,\ 1.5,\ 2,\ 2.360,\ 3,\ 3.274,\ 4$ (calc script); panel (b) shows the first three bands.
- **Number of states:** each band holds $2N$ electrons (two spins per k) for N unit cells, which is why full and empty bands occur.

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5. Final Concise Answer

(a) Density of states: the number of allowed quantum states per unit energy per unit volume, $g_c(E) = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$. Carrier concentration: the number of free electrons (or holes) per unit volume, $n_0 = \int g_c(E) f_F(E)\, dE = N_c e^{-(E_c - E_F)/kT}$ (and $p_0 = N_v e^{-(E_F - E_v)/kT}$).

(b) $n_0 = \int g_c f_F\, dE$: $f_F(E_c)$ is tiny (4×10^{-10} for Si with E_F at midgap) but the number of states is huge ($N_c = 2.8 \times 10^{19}$ cm^{-3}), so their product is $\sim 10^{10}$ cm^{-3}; $n_i = \sqrt{N_c N_v} e^{-E_g/2kT}$ (from 10^6 for GaAs to 10^{13} cm^{-3} for Ge).

(c) $m^* = \hbar^2/(d^2E/dk^2)$, the mass that includes the crystal forces. Band **B** (wider, flatter, smaller curvature) gives the heavier effective mass (about twice that of A from the drawing).

(d) Atoms brought together: overlapping wave functions and the Pauli principle split each level into N levels, forming bands; Bloch's theorem $\psi = u(x)e^{jkx}$ in the periodic (Kronig-Penney) potential gives $P' \sin \alpha a/\alpha a + \cos \alpha a = \cos ka$, allowed only where the left side is between -1 and 1: allowed bands separated by forbidden gaps.

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8. (a) A silicon pn junction at T = 300 K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{20}$ [*sic*: should be 5×10^{15}] cm^{-3}. Determine x_n, x_p and W. Marks: 3

(b) Determine the position of the Fermi level with respect to the valence band energy in p type GaAs at T = 300 K. The doping concentrations are [unclear: N_a] $= 5 \times 10^{19}$ [*sic*: should be 5×10^{16}] cm^{-3} and $N_d = 10^{15}$ [*sic*: should be 4×10^{15}] cm^{-3}. Marks: 4

(c) Silicon at T = 300 K is doped with impurity concentrations of Nd = 5 × 1016 [*sic*: should be 10^{16}] /cm3 and Na = 2 × 1016 [*sic*: should be 10^{16}] /cm3. (a) What are the electron and hole mobilities? (b) Determine the conductivity and resistivity of the [unclear: material]. Marks: 3

8.(a) A silicon pn junction at T = 300 K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{20}$ [*sic*: should be 5×10^{15}] cm^{-3}. Determine x_n, x_p and W. Marks: 3

1. Theory

Note on the paper

The paper prints $N_a = 5 \times 10^{20}$ cm^{-3}. The question is Neamen's Exercise Ex 7.2 word for word, where $N_a = 5 \times 10^{15}$ cm^{-3}, so the solution uses 5×10^{15}. At 5×10^{20} cm^{-3} the p side would be degenerate (more acceptors than the 1.04×10^{19} cm^{-3} states of N_v), and the formulas below, which assume Boltzmann statistics, would not hold. The as-printed numbers are given in the Notes.

1. The PN junction in equilibrium (Core Idea)

A **pn junction** is one crystal that is p-type on one side (acceptors N_a, many holes) and n-type on the other (donors N_d, many electrons). The plane where the doping changes is the **metallurgical junction**, $x = 0$.

When the two sides meet, electrons diffuse from the n side (where they are many) into the p side, and holes diffuse the other way. Each electron that leaves the n side leaves behind a fixed positive donor ion; each hole that leaves the p side leaves a fixed negative acceptor ion. So a thin region around the junction loses its free carriers and keeps only the ion charges. This is the **space charge region** (or **depletion region**): it stretches a distance x_n into the n side and x_p into the p side, and its total width is

$$W = x_n + x_p$$

The ion charges make an electric field that points from the positive ions (n side) to the negative ions (p side). This field pushes electrons back to the n side and holes back to the p side. In equilibrium the push of the field exactly balances the diffusion, and no net current flows.

2. The built-in potential

In equilibrium the Fermi level is flat through the whole crystal. On the n side E_F sits above the intrinsic level E_{Fi}, and on the p side below it, so the bands must bend across the junction. The total bend, divided by e, is the **built-in potential** (Neamen, Eqs. 7.1 to 7.10):

$$V_{bi} = |\phi_{Fn}| + |\phi_{Fp}|, \qquad |\phi_{Fn}| = \frac{kT}{e} \ln \frac{N_d}{n_i}, \qquad \phi_{Fp} = \frac{kT}{e} \ln \frac{N_a}{n_i}$$

$$V_{bi} = \frac{kT}{e} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

where n_i is the intrinsic carrier concentration and the dopants are taken as completely ionized.

3. Charge, field and the charge balance

With the **depletion approximation** (no free carriers inside the space charge region, and sharp edges), the charge density is

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$$\rho(x) = -eN_a \quad (-x_p < x < 0), \qquad \rho(x) = +eN_d \quad (0 < x < x_n)$$

Poisson's equation $dE/dx = \rho(x)/\epsilon_s$ (Eq. 7.11) gives, after integrating with $E = 0$ at both edges (Eqs. 7.14 and 7.16),

$$E(x) = -\frac{eN_a}{\epsilon_s}(x + x_p) \quad (-x_p \le x \le 0), \qquad E(x) = -\frac{eN_d}{\epsilon_s}(x_n - x) \quad (0 \le x \le x_n)$$

The field is continuous at $x = 0$, so the two expressions agree there:

$$N_a x_p = N_d x_n \quad \text{(Eq. 7.17)}$$

The negative charge per unit area on the p side equals the positive charge per unit area on the n side. So the region reaches further into the **more lightly doped** side.

4. The widths

Integrating the field once more gives the potential; its total change across the region must be V_{bi} (Eq. 7.26):

$$V_{bi} = \frac{e}{2\epsilon_s}\left(N_d x_n^2 + N_a x_p^2\right)$$

Put $x_p = N_d x_n / N_a$ (from the charge balance) into it and solve for x_n; do the same for x_p (Eqs. 7.28, 7.29, 7.31):

$$x_n = \left[\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a}{N_d}\right)\left(\frac{1}{N_a + N_d}\right)\right]^{1/2}, \qquad x_p = \left[\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_d}{N_a}\right)\left(\frac{1}{N_a + N_d}\right)\right]^{1/2}$$

$$W = x_n + x_p = \left[\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$$

The field is largest at $x = 0$: $|E_{max}| = eN_d x_n/\epsilon_s = eN_a x_p/\epsilon_s$.

2. Answer

Given:

- Silicon, $T = 300$ K, zero bias
- $N_d = 5 \times 10^{16}$ cm^{-3}, $N_a = 5 \times 10^{15}$ cm^{-3} (the paper's 5×10^{20} read as in Neamen Ex 7.2)
- $n_i = 1.5 \times 10^{10}$ cm^{-3}, $kT/e = 0.0259$ V, $e = 1.6 \times 10^{-19}$ C
- $\epsilon_s = 11.7\epsilon_0 = 11.7 \times 8.85 \times 10^{-14} = 1.0355 \times 10^{-12}$ F/cm

Required: x_n, x_p and W.

Step 1: Built-in potential

$$\frac{N_a N_d}{n_i^2} = \frac{(5 \times 10^{15})(5 \times 10^{16})}{(1.5 \times 10^{10})^2} = \frac{2.5 \times 10^{32}}{2.25 \times 10^{20}} = 1.111 \times 10^{12}$$

$$V_{bi} = 0.0259 \ln\left(1.111 \times 10^{12}\right) = 0.0259 \times 27.736$$

$$\boxed{V_{bi} = 0.718 \text{ V}}$$

Step 2: The common factor

$$\frac{2\epsilon_s V_{bi}}{e} = \frac{2(1.0355 \times 10^{-12})(0.7184)}{1.6 \times 10^{-19}} = 9.298 \times 10^6 \text{ cm}^{-1}$$

and

$$N_a + N_d = 5 \times 10^{15} + 5 \times 10^{16} = 5.5 \times 10^{16} \text{ cm}^{-3}$$

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Step 3: Width on the n side

$$x_n = \left[(9.298\times10^{6})\left(\frac{5\times10^{15}}{5\times10^{16}}\right)\left(\frac{1}{5.5\times10^{16}}\right)\right]^{1/2} = \left[1.690\times10^{-11}\right]^{1/2}$$

$$\boxed{x_n = 4.11\times10^{-6}\ \text{cm} = 0.0411\ \mu\text{m}}$$

Step 4: Width on the p side

$$x_p = \left[(9.298\times10^{6})\left(\frac{5\times10^{16}}{5\times10^{15}}\right)\left(\frac{1}{5.5\times10^{16}}\right)\right]^{1/2} = \left[1.690\times10^{-9}\right]^{1/2}$$

$$\boxed{x_p = 4.11\times10^{-5}\ \text{cm} = 0.411\ \mu\text{m}}$$

Check with the charge balance: $N_a x_p = (5\times10^{15})(4.112\times10^{-5}) = 2.06\times10^{11}$ cm^{-2} and $N_d x_n = (5\times10^{16})(4.112\times10^{-6}) = 2.06\times10^{11}$ cm^{-2} ✔.

Step 5: Total width

$$W = x_n + x_p = 4.11\times10^{-6} + 4.11\times10^{-5}$$

$$\boxed{W = 4.52\times10^{-5}\ \text{cm} = 0.452\ \mu\text{m}}$$

Check with Eq. 7.31: $W = [(9.298\times10^{6})(5.5\times10^{16})/(2.5\times10^{32})]^{1/2} = [2.046\times10^{-9}]^{1/2} = 4.52\times10^{-5}$ cm ✔. The p side is ten times more lightly doped, so 91 % of the width lies in it.

For completeness, the peak field (Neamen's exercise also asks it): $|E_{max}| = eN_d x_n/\epsilon_s = (1.6\times10^{-19})(5\times10^{16})(4.112\times10^{-6})/(1.0355\times10^{-12}) = 3.18\times10^{4}$ V/cm.

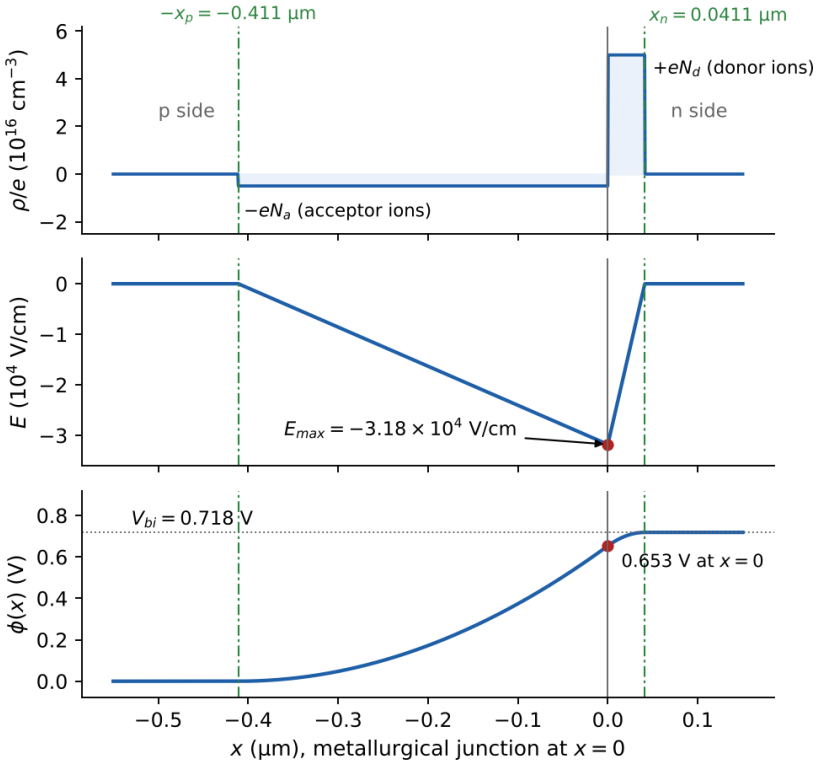


Figure 37: Space charge density, electric field and potential across this junction, drawn to scale with the computed values (the same junction as 2020–21 6(d))

Reading the figure. The two dash-dot vertical lines are the edges of the space charge region, $-x_p = -0.411$ μm on the left and $x_n = 0.0411$ μm on the right; the thin grey line is the junction. Top panel: the charge is a shallow negative block over the wide p part and a block ten times taller, but ten times narrower, over the n part; their areas are equal (the charge balance). Middle panel: the field falls in

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a straight line from zero at $-x_p$ to its peak -3.18×10^4 V/cm at the junction (marked with a dot), then returns steeply to zero at x_n. Bottom panel: the potential rises from 0 to $V_{bi} = 0.718$ V, most of it across the p side.

Steps to remember

$V_{bi} = (kT/e)\ln(N_a N_d/n_i^2) \to$ common factor $2\epsilon_s V_{bi}/e \to x_n = [\dots(N_a/N_d)/(N_a+N_d)]^{1/2}$, $x_p = [\dots(N_d/N_a)/(N_a+N_d)]^{1/2} \to W = x_n + x_p \to$ check $N_a x_p = N_d x_n$.

Final Answer: $V_{bi} = 0.718$ V; $x_n = 0.0411$ μm, $x_p = 0.411$ μm, $W = 0.452$ μm.

3. Keyword Explanation

1. Space charge region

The region around the junction emptied of free carriers, holding only the fixed ion charges; also called the depletion region.

2. Depletion approximation

The assumption that the space charge region has no free carriers and sharp edges, so its charge is exactly $-eN_a$ on the p side and $+eN_d$ on the n side.

3. Built-in potential

The potential difference between the neutral n and p regions in equilibrium, set by the doping.

4. Notes and Remarks

- **Book check:** Neamen's answer to Ex 7.2 is $x_n = 4.11 \times 10^{-6}$ cm, $x_p = 4.11 \times 10^{-5}$ cm, $W = 4.52 \times 10^{-5}$ cm, $|E_{max}| = 3.18 \times 10^4$ V/cm; the results above agree.
- **As printed ($N_a = 5 \times 10^{20}$):** the formulas would give $V_{bi} = 1.017$ V, $x_n = 0.162$ μm, $x_p = 1.6 \times 10^{-5}$ μm and $W = 0.162$ μm: an almost one-sided junction with all the width on the n side. They are not valid, because the p side is degenerate, so this reading is not used.
- **Repeat:** the same junction and numbers are asked in 2020–21 6(d).

8.(b) Determine the position of the Fermi level with respect to the valence band energy in p type GaAs at T = 300 K. The doping concentrations are [unclear: N_a] $= 5 \times 10^{19}$ [*sic:* should be 5×10^{16}] cm^{-3} and $N_d = 10^{15}$ [*sic:* should be 4×10^{15}] cm^{-3}. Marks: 4

1. Theory

Note on the paper

The paper prints [unclear: N_a] $= 5 \times 10^{19}$ and $N_d = 10^{15}$ cm^{-3} (the subscript of the first is faint). The question is Neamen's TYU 4.14 word for word, with $N_a = 5 \times 10^{16}$ and $N_d = 4 \times 10^{15}$ cm^{-3}, and these are used. As printed, $N_a = 5 \times 10^{19}$ is more than $N_v = 7.0 \times 10^{18}$ cm^{-3} of GaAs, so the material would be degenerate and E_F would lie inside the valence band (Notes).

1. The Fermi level and the carrier densities (Core Idea)

The **Fermi level** E_F is the energy at which a state would be filled with probability one half. Its position in the gap sets the carrier densities: the closer E_F is to E_v, the more holes. With the Boltzmann approximation (Neamen, Eq. 4.19)

$$p_0 = N_v \exp\left[\frac{-(E_F - E_v)}{kT}\right]$$

where N_v is the effective density of states in the valence band (7.0×10^{18} cm^{-3} for GaAs at 300 K, Table 4.1).

Taking the natural logarithm and solving for $E_F - E_v$ gives (Eq. 4.66)

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$$E_F - E_v = kT \ln\left(\frac{N_v}{p_0}\right)$$

2. Compensated p-type material

The paper's material has both acceptors and donors: it is **compensated**. The donors' electrons fill some of the acceptors, so only $N_a - N_d$ acceptors are left to make holes. Charge neutrality, $n_0 + N_a = p_0 + N_d$, with $n_0 = n_i^2/p_0$, gives a quadratic in p_0 whose positive root is (Eq. 4.62)

$$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$$

When $N_a - N_d \gg n_i$, the square root is just $(N_a - N_d)/2$, so $p_0 = N_a - N_d$.

3. Position relative to the intrinsic level

The same result can be written from the intrinsic Fermi level (Eq. 4.68):

$$E_{Fi} - E_F = kT \ln\left(\frac{p_0}{n_i}\right)$$

2. Answer

Given:

- p-type GaAs, $T = 300$ K, $kT = 0.0259$ eV
- $N_a = 5 \times 10^{16}$ cm^{-3}, $N_d = 4 \times 10^{15}$ cm^{-3} (the paper's values read as in Neamen TYU 4.14)
- GaAs: $N_v = 7.0 \times 10^{18}$ cm^{-3}, $n_i = 1.8 \times 10^{6}$ cm^{-3}, $E_g = 1.42$ eV (Neamen, Tables 4.1 and 4.2)

Required: $E_F - E_v$.

Step 1: Hole concentration

$$N_a - N_d = 5 \times 10^{16} - 4 \times 10^{15} = 4.6 \times 10^{16} \text{ cm}^{-3}$$

This is about 10^{10} times n_i, so n_i^2 in Eq. 4.62 is negligible:

$$p_0 = \frac{4.6 \times 10^{16}}{2} + \sqrt{\left(\frac{4.6 \times 10^{16}}{2}\right)^2 + (1.8 \times 10^6)^2} = 4.6 \times 10^{16} \text{ cm}^{-3}$$

Step 2: Fermi level above the valence band

$$\frac{N_v}{p_0} = \frac{7.0 \times 10^{18}}{4.6 \times 10^{16}} = 152.2$$

$$E_F - E_v = kT \ln\left(\frac{N_v}{p_0}\right) = 0.0259 \ln(152.2) = 0.0259 \times 5.025$$

$$\boxed{E_F - E_v = 0.130 \text{ eV}}$$

The Fermi level is 0.130 eV above the top of the valence band. This is more than $3kT = 0.078$ eV, so the Boltzmann approximation used in Eq. 4.19 is valid.

Step 3: Check against the intrinsic level

$$E_{Fi} - E_F = kT \ln\left(\frac{p_0}{n_i}\right) = 0.0259 \ln\left(\frac{4.6 \times 10^{16}}{1.8 \times 10^6}\right) = 0.0259 \times 23.96 = 0.621 \text{ eV}$$

The intrinsic level in GaAs is $E_{Fi} - E_v = E_g/2 + \frac{3}{4}kT \ln\left(m_p^*/m_n^*\right) = 0.71 + \frac{3}{4}(0.0259)\ln(0.48/0.067) = 0.748$ eV above E_v (Eq. 4.26a). Then $E_F - E_v = 0.748 - 0.621 = 0.127$ eV, which agrees with Step 2 to 3 meV; the small difference is because the book's n_i, N_v and N_c for GaAs are rounded separately.

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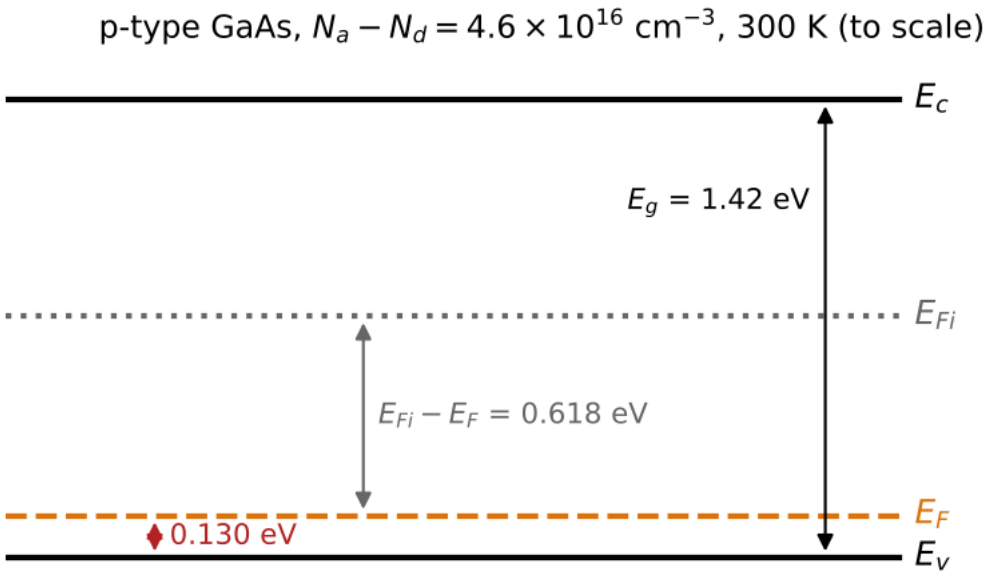


Figure 38: Band diagram of the p-type GaAs, energies to scale

Reading the figure. The solid black lines are E_c (top) and E_v (bottom), 1.42 eV apart. The dotted grey line is the intrinsic level E_{Fi}, slightly above midgap. The dashed line just above E_v is the Fermi level E_F: the short double arrow beside it is $E_F - E_v = 0.130$ eV, and the long double arrow between E_{Fi} and E_F is 0.618 eV (from the E_{Fi} of Eq. 4.26a). E_F near E_v means many holes: the material is strongly p-type.

Steps to remember

$p_0 = N_a - N_d$ (compensated, n_i negligible) $\rightarrow E_F - E_v = kT\ln(N_v/p_0) \rightarrow$ check $E_F - E_v > 3kT$ (Boltzmann valid).

Final Answer: $p_0 = 4.6 \times 10^{16}$ cm^{-3}, and the Fermi level is $E_F - E_v = 0.130$ eV above the valence band edge ($E_{Fi} - E_F \approx 0.62$ eV).

3. Keyword Explanation

1. Fermi level

The energy at which the occupation probability is one half; in equilibrium it is the same everywhere in the material.

2. Compensated semiconductor

A semiconductor with both donors and acceptors in the same region; the net doping is $|N_a - N_d|$.

3. Effective density of states

$N_v = 2(2\pi m_p^* kT/h^2)^{3/2}$: the number of valence-band states that, all placed at E_v, would give the same hole density.

4. Notes and Remarks

- **Book check:** Neamen's answer to TYU 4.14 is $E_F - E_v = 0.130$ eV, the same as above.
- **As printed:** with $N_a = 5 \times 10^{19}$ and $N_d = 10^{15}$ cm^{-3}, $kT\ln(N_v/p_0) = -0.051$ eV: the formula would put E_F below E_v, inside the valence band. The material would be degenerate and the Boltzmann formula would not apply, which is a further sign that the printed exponent is a misprint.

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8.(c) Silicon at T = 300 K is doped with impurity concentrations of Nd = 5 × 1016 [*sic*: should be 10^{16}] /cm3 and Na = 2 × 1016 [*sic*: should be 10^{16}] /cm3. (a) What are the electron and hole mobilities? (b) Determine the conductivity and resistivity of the [unclear: material]. Marks: 3

1. Theory

Note on the paper

The paper prints "5 × 1016" and "2 × 1016": the exponents lost their superscript. They are read as 5×10^{16} and 2×10^{16} cm^{-3}, as in Neamen's TYU 5.2, which this question copies word for word. The last word is cut at the edge of the scan and read as "material".

1. Drift and mobility (Core Idea)

In an electric field E, free carriers are accelerated but keep colliding with the vibrating lattice and with the ionized dopant atoms. On average they move at a steady **drift velocity** proportional to the field (Neamen, Eqs. 5.13, 5.14):

$$v_{dn} = -\mu_n E, \qquad v_{dp} = \mu_p E$$

The constant μ is the **mobility** (cm^2/V s). It is $e\tau/m^*$, where τ is the mean time between collisions: the fewer the collisions, the higher the mobility.

2. Mobility depends on the total impurity concentration

Two kinds of scattering limit the mobility. **Lattice (phonon) scattering** depends on the temperature only. **Ionized impurity scattering** depends on how many charged ions there are, and every ion counts, whether donor or acceptor, because a carrier is deflected by a positive or a negative ion alike. So the mobility is read from the **total ionized impurity concentration**

$$N_I = N_d^+ + N_a^-$$

not the net doping. Neamen's Figure 5.3 plots μ_n and μ_p for silicon at 300 K against N_I: both fall as N_I grows, and μ_p is always about a third of μ_n.

3. Conductivity and resistivity

The drift current density is (Eq. 5.19)

$$J_{drf} = e(\mu_n n + \mu_p p)E = \sigma E$$

so the **conductivity** and **resistivity** are (Eq. 5.20)

$$\sigma = e(\mu_n n + \mu_p p), \qquad \rho = \frac{1}{\sigma}$$

The carrier densities n and p come from the **net** doping, while the mobilities come from the **total** doping.

2. Answer

Given:

- Silicon, $T = 300$ K
- $N_d = 5 \times 10^{16}$ cm^{-3}, $N_a = 2 \times 10^{16}$ cm^{-3} (complete ionization)
- $n_i = 1.5 \times 10^{10}$ cm^{-3}, $e = 1.6 \times 10^{-19}$ C

Required: (a) μ_n and μ_p; (b) σ and ρ.

Step 1: Total ionized impurity concentration

$$N_I = N_d + N_a = 5 \times 10^{16} + 2 \times 10^{16} = 7 \times 10^{16} \text{ cm}^{-3}$$

Step 2: Read the mobilities (part (a))

From Neamen's Figure 5.3 at $N_I = 7 \times 10^{16}$ cm^{-3}:

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$$\boxed{\mu_n \approx 1000 \text{ cm}^2/\text{V s}, \qquad \mu_p \approx 350 \text{ cm}^2/\text{V s}}$$

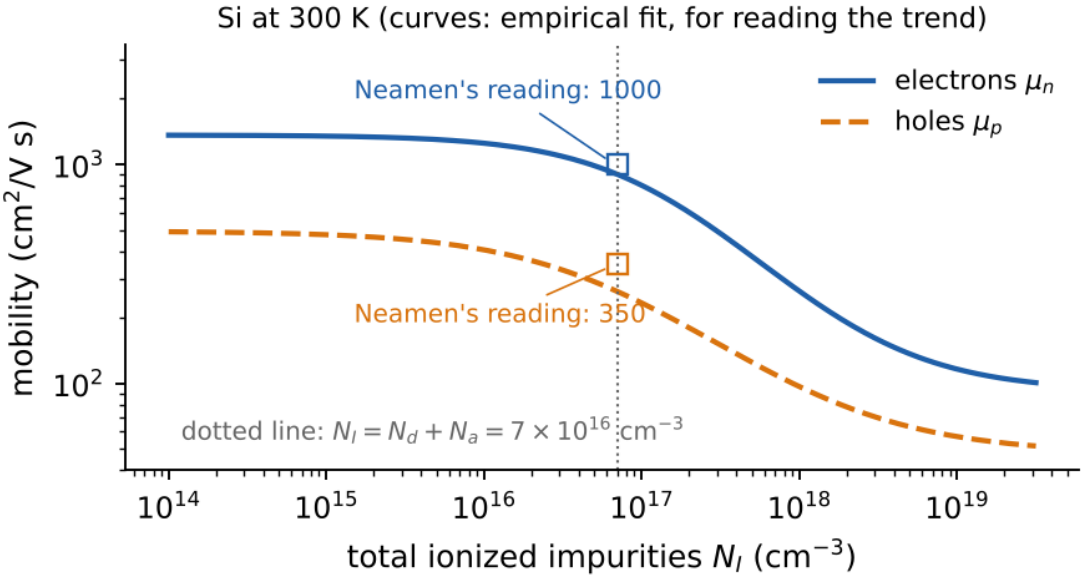


Figure 39: Electron and hole mobility in Si at 300 K against the total ionized impurity concentration

Reading the figure. The solid curve is the electron mobility and the dashed curve the hole mobility, both falling as the total impurity concentration grows. The dotted vertical line is $N_I = 7 \times 10^{16}$ cm^{-3}. The two squares mark Neamen's readings from his Figure 5.3, 1000 and 350 cm^2/V s. The curves are an empirical fit drawn only to show the trend; at this N_I they lie a little lower (900 and 262), which is why the book's readings are used in the answer.

Step 3: Carrier concentrations

$N_d > N_a$, so the material is n-type with net doping $N_d - N_a = 3 \times 10^{16}$ cm^{-3}, far above n_i:

$$n_0 = N_d - N_a = 3 \times 10^{16} \text{ cm}^{-3}, \qquad p_0 = \frac{n_i^2}{n_0} = \frac{(1.5 \times 10^{10})^2}{3 \times 10^{16}} = 7.5 \times 10^3 \text{ cm}^{-3}$$

Step 4: Conductivity (part (b))

$$\sigma = e(\mu_n n_0 + \mu_p p_0) = (1.6 \times 10^{-19}) \left[(1000)(3 \times 10^{16}) + (350)(7.5 \times 10^3)\right]$$

The hole term (2.6×10^6) is negligible beside the electron term (3×10^{19}):

$$\sigma = (1.6 \times 10^{-19})(3 \times 10^{19})$$

$$\boxed{\sigma = 4.8 \ (\Omega\text{-cm})^{-1}}$$

Step 5: Resistivity

$$\rho = \frac{1}{\sigma} = \frac{1}{4.8}$$

$$\boxed{\rho = 0.208 \ \Omega\text{-cm}}$$

Steps to remember

Mobility from the **total** $N_I = N_d + N_a$ (Fig. 5.3) $\rightarrow$ carriers from the **net** $N_d - N_a \rightarrow \sigma = e(\mu_n n + \mu_p p) \approx e\mu_n n_0 \rightarrow \rho = 1/\sigma$.

Final Answer: (a) $\mu_n \approx 1000$ cm^2/V s and $\mu_p \approx 350$ cm^2/V s; (b) $\sigma = 4.8$ $(\Omega\text{-cm})^{-1}$ and $\rho = 0.208$ Ω-

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cm.

3. Keyword Explanation

1. Mobility

The drift speed per unit electric field, $\mu = |v_d|/E$; it measures how freely a carrier moves through the crystal.

2. Ionized impurity scattering

The deflection of a moving carrier by the Coulomb field of a charged dopant ion; it grows with the number of ions and falls with temperature.

3. Conductivity

$\sigma = e(\mu_n n + \mu_p p)$, the ratio of the drift current density to the field.

4. Notes and Remarks

- **Book check:** Neamen's answer to TYU 5.2 is $\mu_n = 1000$, $\mu_p = 350$ cm^2/V s, $\sigma = 4.8$ $(\Omega\text{-cm})^{-1}$, $\rho = 0.208$ Ω-cm, the same as above.
- **A common mistake:** reading the mobility at the net doping 3×10^{16} cm^{-3} gives too high a mobility; the acceptors scatter electrons too. Reading it at N_d alone (5×10^{16}) would give about 985 cm^2/V s from the fit.
- **Reading a graph:** a value read from Figure 5.3 is good to about 5 %, so answers such as $\mu_n = 950$ to 1050 are equally right.

5. Final Concise Answer

(a) $V_{bi} = (kT/e)\ln(N_a N_d/n_i^2) = 0.718$ V; $x_n = [(2\epsilon_s V_{bi}/e)(N_a/N_d)/(N_a + N_d)]^{1/2} = 0.0411$ μm, $x_p = 0.411$ μm, $W = 0.452$ μm (with $N_a = 5 \times 10^{15}$ cm^{-3}, Neamen Ex 7.2).

(b) $p_0 = N_a - N_d = 4.6 \times 10^{16}$ cm^{-3}; $E_F - E_v = kT\ln(N_v/p_0) = 0.0259\ln(7.0 \times 10^{18}/4.6 \times 10^{16}) = 0.130$ eV (Neamen TYU 4.14 values).

(c) $N_I = N_d + N_a = 7 \times 10^{16}$ cm^{-3}: $\mu_n \approx 1000$, $\mu_p \approx 350$ cm^2/V s (Neamen Fig. 5.3); $n_0 = 3 \times 10^{16}$ cm^{-3}, $\sigma \approx e\mu_n n_0 = 4.8$ $(\Omega\text{-cm})^{-1}$, $\rho = 0.208$ Ω-cm.

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FORMULA SHEET

EEE421

Solid State Devices

Formula Sheet

Session 2018–19 paper *Every formula behind the 4th Year 1st Semester Final Examination, June 2022, with Class Test 01 and Term Test 2*

TOPICS

1 Bands, effective mass and the Fermi function — 4 formulas

2 Carriers in equilibrium — 10 formulas

3 Transport — 6 formulas

4 Excess carriers — 1 formula

5 PN junction — 7 formulas

6 Metal-semiconductor contacts — 2 formulas

7 MOS structure — 4 formulas

8 Bipolar transistor — 5 formulas

9 JFET — 2 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

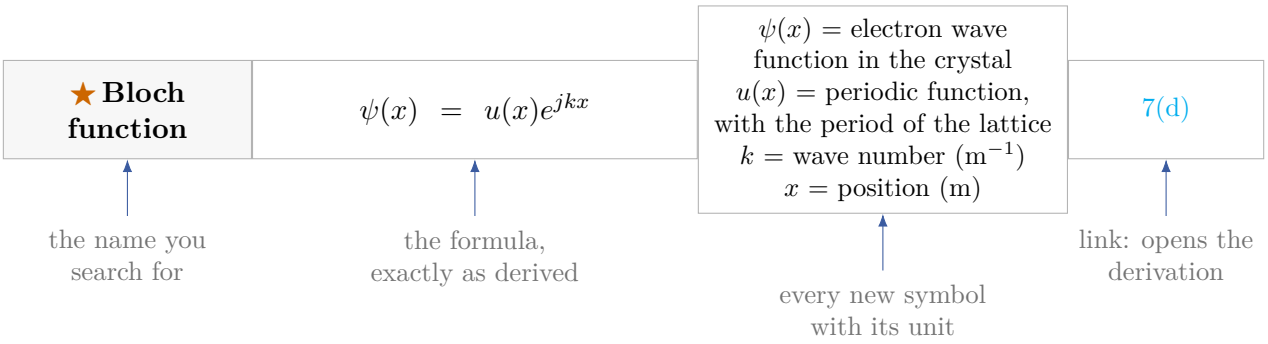
Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Bands, effective mass and the Fermi function 4

2 Carriers in equilibrium 4

3 Transport 6

4 Excess carriers 7

5 PN junction 7

6 Metal-semiconductor contacts 8

7 MOS structure 9

8 Bipolar transistor 10

9 JFET 11

10 Key formulas 12

11 Symbol index 14

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Conventions (Neamen): e is the magnitude of the electron charge, $kT = 0.0259$ eV and $V_t = kT/e = 0.0259$ V at 300 K; $\epsilon_s = \epsilon_r \epsilon_0$ with $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm. Si at 300 K: $N_c = 2.8 \times 10^{19}$ cm^{-3}, $N_v = 1.04 \times 10^{19}$ cm^{-3}, $n_i = 1.5 \times 10^{10}$ cm^{-3}, $E_g = 1.12$ eV, $\epsilon_r = 11.7$. References point to the solutions of the 2018–19 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Bands, effective mass and the Fermi function:

Title	Equation	Notation	Ref.
★ **Bloch function**	$\psi(x) = u(x)e^{jkx}$	$\psi(x)$ = electron wave function in the crystal $u(x)$ = periodic function, with the period of the lattice k = wave number (m^{-1}) x = position (m)	7(d)
★ **Kronig-Penney condition**	$P'\frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka,\ \alpha^2 = \frac{2mE}{\hbar^2}$	P' = strength of the potential barriers (dimensionless) a = width of the well (m) m = mass of the electron (kg) E = electron energy (J) $\hbar$ = reduced Planck constant, $h/2\pi$ (J s) allowed bands where the left side is between -1 and $+1$	7(d)
★ **Effective mass**	$\frac{1}{m_n^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$	m_n^* = electron effective mass (kg) smaller curvature (flatter band) → heavier mass; Figure 1: band B	7(c)
★ **Fermi-Dirac function**	$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$	$f_F(E)$ = probability that a state at energy E is occupied E_F = Fermi energy (eV) kT = thermal energy (0.0259 eV at 300 K) at 0 K: 1 below E_F, 0 above; highest filled level E_v	1(c), 2(b)

Table 1: Bands, effective mass and the Fermi function

2. Carriers in equilibrium:

Title	Equation	Notation	Ref.
★ **Density of states**	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$, $g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E}$	$g_c(E)$ = density of states in the conduction band (cm^{-3}eV^{-1}) $g_v(E)$ = density of states in the valence band (cm^{-3}eV^{-1}) m_p^* = hole effective mass (kg) h = Planck's constant (6.625×10^{-34} J s) E_c = conduction-band edge (eV) E_v = valence-band edge (eV)	7(a)
Thermal energy	$kT = 0.0259$ eV at 300 K	k = Boltzmann constant (8.62×10^{-5} eV/K) T = absolute temperature (K) $\approx 1/40$ eV	2(b)
★ **Electron concentration**	$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE =$ $N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	n_0 = thermal-equilibrium electron concentration (cm^{-3}) N_c = effective density of states in the conduction band (cm^{-3}) Si: $N_c = 2.8 \times 10^{19}$ cm^{-3}	7(a), 7(b)

Title	Equation	Notation	Ref.
Hole concentration	$p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	p_0 = thermal-equilibrium hole concentration (cm^{-3}) N_v = effective density of states in the valence band (cm^{-3}) Si: $N_v = 1.04 \times 10^{19}$, GaAs: 7.0×10^{18} cm^{-3}	7(a), 8(b)
★ **Intrinsic concentration**	$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$	n_i = intrinsic carrier concentration (cm^{-3}) E_g = band gap energy (eV) Si: 1.5×10^{10}, GaAs: 1.8×10^6 cm^{-3}	1(a), 7(b)
★ **Mass-action law**	$n_0 p_0 = n_i^2$	holds for any doping in thermal equilibrium	1(b), 8(c)
★ **Majority carriers**	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$ (and n_0 with $N_d - N_a$)	N_a = acceptor concentration (cm^{-3}) N_d = donor concentration (cm^{-3}) Ge, $N_a = 10^{15}$: $p_0 = 1.0006 \times 10^{15}$, $n_0 = 6.25 \times 10^{11}$ cm^{-3}	1(a), 1(b), 8(b)
Electrons left on donors	$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$	n_d = electrons still bound to donor atoms (cm^{-3}) E_d = donor energy level (eV) freeze-out below about 100 K	1(a)
★ **Fermi level from the band edge**	$E_F - E_v = kT \ln \frac{N_v}{p_0}$	GaAs, $p_0 = 4.6 \times 10^{16}$: 0.130 eV	8(b)
Fermi level from the intrinsic level	$E_{Fi} - E_F = kT \ln \frac{p_0}{n_i}$	E_{Fi} = intrinsic Fermi level (eV) 0.621 eV for 8(b)	8(b)

Table 2: Carriers in equilibrium

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3. Transport:

Title	Equation	Notation	Ref.
★ **Drift and conductivity**	$\sigma = e(\mu_n n + \mu_p p),\ \rho = \frac{1}{\sigma}$	σ = conductivity $(\Omega\text{-cm})^{-1}$ ρ = resistivity (Ω-cm) e = magnitude of the electron charge (1.6×10^{-19} C) μ_n = electron mobility (cm^2/V s) μ_p = hole mobility (cm^2/V s) n = electron concentration (cm^{-3}) p = hole concentration (cm^{-3}) N_I = total ionized impurity concentration, $N_d + N_a$ (cm^{-3}) μ read at N_I; 8(c): 4.8 $(\Omega\text{-cm})^{-1}$, 0.208 Ω-cm	8(c)
★ **Diffusion current**	$J_{n,dif} = eD_n\frac{dn}{dx},\ J_{p,dif} = -eD_p\frac{dp}{dx}$	$J_{n,dif}$ = electron diffusion current density (A/cm^2) $J_{p,dif}$ = hole diffusion current density (A/cm^2) D_n = electron diffusion coefficient (cm^2/s) D_p = hole diffusion coefficient (cm^2/s) F = particle flux ($cm^{-2}s^{-1}$) v_{th} = thermal velocity (cm/s) l = mean free path (cm) from the flux $F = -v_{th} l\, dn/dx$	5(a)
Total current	$J = en\mu_n E + ep\mu_p E + eD_n\frac{dn}{dx} - eD_p\frac{dp}{dx}$	J = total current density (A/cm^2) E = electric field (V/cm) drift plus diffusion	5(a)
★ **Einstein relation**	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$	kT/e = thermal voltage (0.0259 V at 300 K)	2(c)
★ **Hall voltage**	$V_H = \frac{I_x B_z}{epd};\ p = \frac{I_x B_z}{edV_H},\ n = -\frac{I_x B_z}{edV_H}$	V_H = Hall voltage (V) I_x = current along x (A) B_z = magnetic flux density along z (T) d = sample thickness, along B_z (cm) sign of V_H gives the type	1(d)
Hall mobility	$\mu_p = \frac{I_x L}{epV_x W d}$	L = sample length (cm) V_x = applied voltage along x (V) W = sample width (cm)	1(d)

Table 3: Transport

4. Excess carriers:

Title	Equation	Notation	Ref.
★ **Diffusion length**	$L_p = \sqrt{D_p \tau_{p0}}$, $\delta p(x) = \delta p_0 e^{-x/L_p}$	L_p = hole diffusion length (cm) τ_{p0} = minority-carrier (hole) lifetime (s) $\delta p(x)$ = excess hole concentration at x (cm^{-3}) δp_0 = excess hole concentration at $x = 0$ (cm^{-3}) mean distance before recombination $\langle x \rangle = L_p$	5(c)

Table 4: Excess carriers

5. PN junction:

Title	Equation	Notation	Ref.
★ **Built-in potential**	$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$	V_{bi} = built-in potential barrier (V) 8(a): 0.718 V	2(d), 8(a)
★ **Charge balance**	$N_a x_p = N_d x_n$	x_n = space-charge width on the n side (cm) x_p = space-charge width on the p side (cm) the region reaches further into the lighter side	8(a)
★ **Space charge widths**	$x_n = \left[\frac{2\epsilon_s V_{bi}}{e} \frac{N_a}{N_d} \frac{1}{N_a + N_d}\right]^{1/2}$, $x_p = \left[\frac{2\epsilon_s V_{bi}}{e} \frac{N_d}{N_a} \frac{1}{N_a + N_d}\right]^{1/2}$	ϵ_s = permittivity of the semiconductor, $11.7\epsilon_0$ for Si (F/cm) 8(a): $x_n = 0.0411$, $x_p = 0.411$ μm	8(a)
★ **Total width**	$W = \left\{\frac{2\epsilon_s(V_{bi} + V_R)}{e} \left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}$	W = total space-charge width (cm) V_R = reverse-bias voltage (V) 8(a): 0.452 μm	2(d), 8(a)
★ **Junction capacitance**	$C' = \frac{\epsilon_s}{W} = \left\{\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right\}^{1/2}$	C' = junction capacitance per unit area (F/cm^2) $C' \propto 1/\sqrt{V_{bi} + V_R}$; $(1/C')^2$ linear in V_R	2(d)
★ **Ideal diode**	$J = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$	p_{n0} = thermal-equilibrium hole (minority) concentration in the n region (cm^{-3}) n_{p0} = thermal-equilibrium electron (minority) concentration in the p region (cm^{-3}) L_n = electron diffusion length (cm) V_a = applied forward voltage (V) minority current decays as $e^{-(x-x_n)/L_p}$	4(a)

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Title	Equation	Notation	Ref.
★ **Storage time**	$\text{erf}\sqrt{\frac{t_s}{\tau_{p0}}} = \frac{I_F}{I_F + I_R}$	t_s = storage time (s) I_F = forward current before switching (A) I_R = reverse current while the stored charge is removed (A) Q_p = stored minority-carrier charge (C) stored charge $Q_p = I_F \tau_{p0}$	2(a)

Table 5: PN junction

6. Metal-semiconductor contacts:

Title	Equation	Notation	Ref.
★ **Schottky barrier**	$\phi_{B0} = \phi_m - \chi,\ V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e} \ln \frac{N_c}{N_d}$	ϕ_{B0} = ideal Schottky barrier height (V) ϕ_m = metal work function (V) χ = electron affinity (V) ϕ_n = depth of the Fermi level below E_c, $(E_c - E_F)/e$ (V) 6(d): 0.54 V, 0.334 V	6(d)
Thermionic emission	$J = A^* T^2 \exp\left(-\frac{e\phi_{B0}}{kT}\right)\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$	A^* = effective Richardson constant (A/cm^{2}K^2) majority carriers; ohmic: low barrier or tunnelling	6(d), 4(c)

Table 6: Metal-semiconductor contacts

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7. MOS structure:

Title	Equation	Notation	Ref.
★ **Fermi potential**	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	ϕ_{fp} = Fermi potential of the p-type substrate, $(E_{Fi} - E_F)/e$ (V) V_t = thermal voltage, kT/e (V) ϕ_s = surface potential (V) inversion at $\phi_s = 2\phi_{fp}$	3(a), 3(b)
★ **Threshold voltage**	$V_{TN} = \frac{\lvert Q'_{SD}(\max)\rvert - Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp}$, $x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$	V_{TN} = threshold voltage of the n-channel MOSFET (V) $Q'_{SD}(\max)$ = maximum space-charge (depletion) charge per unit area (C/cm^2) Q'_{ss} = fixed oxide charge per unit area (C/cm^2) C_{ox} = oxide capacitance per unit area, ϵ_{ox}/t_{ox} (F/cm^2) ϕ_{ms} = metal-semiconductor work function difference (V) x_{dT} = maximum space-charge width (cm) 3(b): -0.262 V	3(b)
★ **Body effect**	$\Delta V_T = \gamma\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right]$, $\gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}$	ΔV_T = shift of the threshold voltage (V) γ = body-effect coefficient (V$^{1/2}$) V_{SB} = source-to-body reverse bias (V) 3(c): $\gamma = 0.578$ V$^{1/2}$, $\Delta V_T = 0.264$ V	3(c)
★ **MOS capacitance**	$\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}}$, $C'_{min} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_{dT}}$	C'_{SD} = space-charge capacitance per unit area (F/cm^2) C'_{min} = minimum capacitance per unit area (F/cm^2) ϵ_{ox} = oxide permittivity, $3.9\epsilon_0$ (F/cm) t_{ox} = oxide thickness (cm) accumulation and low-frequency inversion: C_{ox}; 5(b): $C_{min}/C_{ox} = 0.153$	5(b)

Table 7: MOS structure

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8. Bipolar transistor:

Title	Equation	Notation	Ref.
★ **Collector current**	$i_C = I_S \exp\left(\frac{v_{BE}}{V_t}\right),\ A_v = -g_m R_C = -\frac{I_C R_C}{V_t}$	i_C = collector current (A) I_S = saturation current (A) v_{BE} = base-emitter voltage (V) A_v = voltage gain g_m = transconductance, I_C/V_t (A/V) R_C = collector load resistance (Ω) I_C = dc collector current (A) amplification	3(d)
Base current	$I_B \approx I_{pE} + I_{RB} + I_R - I_{CBO}$	I_B = base current (A) I_{pE} = hole current injected from the base into the emitter (A) I_{RB} = recombination current in the base (A) I_R = recombination current in the B-E space-charge region (A) I_{CBO} = B-C reverse-bias current (A)	4(b)
★ **Early effect**	$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE} + V_A},\ J_C \approx \frac{eD_B n_{B0}}{x_B} e^{eV_{BE}/kT}$	V_{CE} = collector-emitter voltage (V) V_A = Early voltage (V) J_C = collector current density (A/cm^2) D_B = minority-carrier diffusion coefficient in the base (cm^2/s) n_{B0} = thermal-equilibrium minority-carrier concentration in the base (cm^{-3}) x_B = neutral base width (cm) V_{BE} = base-emitter voltage (V) 4(d): $V_A = 90.3$ V	4(d)
★ **Injection efficiency**	$\gamma \approx \frac{1}{1 + \frac{N_B}{N_E}\frac{D_E}{D_B}\frac{x_B}{x_E}}$	γ = emitter injection efficiency N_B = base doping (cm^{-3}) N_E = emitter doping (cm^{-3}) D_E = minority-carrier diffusion coefficient in the emitter (cm^2/s) x_E = neutral emitter width (cm) equal D, L, x: $N_E/N_B = 302$ for $\gamma = 0.9967$	6(b)
Current crowding	$\Delta V = \frac{I_B}{2} R$, injection $\propto e^{eV_{BE}/kT}$	ΔV = drop of V_{BE} from the emitter edge to its centre (V) R = lateral base resistance under half the emitter (Ω) S = emitter stripe width (cm) 18 mV doubles the injection; 6(c): 19.2 mV ($S = 10$ μm)	6(c)

Table 8: Bipolar transistor

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9. JFET:

Title	Equation	Notation	Ref.
★ **Pinch-off voltage**	$V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}$, $V_p = V_{bi} - V_{p0}$	V_{p0} = internal pinch-off voltage (V) V_p = pinch-off voltage, gate to source (V) a = metallurgical channel thickness (cm) 5(d): $V_{p0} = 3.33$ V, $a = 0.464$ μm	5(d)
★ **Cutoff frequency**	$f_T = \frac{g_m}{2\pi C_G} = \frac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$	f_T = cutoff frequency (Hz) C_G = gate capacitance, $C_{gs} + C_{gd}$ (F) L = channel length (cm) 6(a): 7.69 GHz	6(a)

Table 9: JFET

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10. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Bloch function	$\psi(x) = u(x)e^{jkx}$
Kronig-Penney condition	$P'\frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka,\ \alpha^2 = \frac{2mE}{\hbar^2}$
Effective mass	$\frac{1}{m_n^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$
Fermi-Dirac function	$f_F(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{kT}\right)}$
Density of states	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E-E_c},\ g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v-E}$
Electron concentration	$n_0 = \int_{E_c}^{\infty} g_c(E)f_F(E)\,dE = N_c \exp\left[-\frac{E_c-E_F}{kT}\right]$
Intrinsic concentration	$n_i^2 = N_cN_v \exp\left(-\frac{E_g}{kT}\right)$
Mass-action law	$n_0p_0 = n_i^2$
Majority carriers	$p_0 = \frac{N_a-N_d}{2} + \sqrt{\left(\frac{N_a-N_d}{2}\right)^2 + n_i^2}$ (and n_0 with $N_d - N_a$)
Fermi level from the band edge	$E_F - E_v = kT\ln\frac{N_v}{p_0}$
Drift and conductivity	$\sigma = e(\mu_n n + \mu_p p),\ \rho = \frac{1}{\sigma}$
Diffusion current	$J_{n,dif} = eD_n\frac{dn}{dx},\ J_{p,dif} = -eD_p\frac{dp}{dx}$
Einstein relation	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$
Hall voltage	$V_H = \frac{I_xB_z}{epd};\ p = \frac{I_xB_z}{edV_H},\ n = -\frac{I_xB_z}{edV_H}$
Diffusion length	$L_p = \sqrt{D_p\tau_{p0}},\ \delta p(x) = \delta p_0 e^{-x/L_p}$
Built-in potential	$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_aN_d}{n_i^2}\right)$
Charge balance	$N_ax_p = N_dx_n$
Space charge widths	$x_n = \left[\frac{2\epsilon_sV_{bi}}{e}\frac{N_a}{N_d}\frac{1}{N_a+N_d}\right]^{1/2},\ x_p = \left[\frac{2\epsilon_sV_{bi}}{e}\frac{N_d}{N_a}\frac{1}{N_a+N_d}\right]^{1/2}$
Total width	$W = \left\{\frac{2\epsilon_s(V_{bi}+V_R)}{e}\left[\frac{N_a+N_d}{N_aN_d}\right]\right\}^{1/2}$
Junction capacitance	$C' = \frac{\epsilon_s}{W} = \left\{\frac{e\epsilon_sN_aN_d}{2(V_{bi}+V_R)(N_a+N_d)}\right\}^{1/2}$

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Title	Equation
Ideal diode	$J = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$
Storage time	$\text{erf}\sqrt{\frac{t_s}{\tau_{p0}}} = \frac{I_F}{I_F + I_R}$
Schottky barrier	$\phi_{B0} = \phi_m - \chi,\ V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d}$
Fermi potential	$\phi_{fp} = V_t \ln\frac{N_a}{n_i}$
Threshold voltage	$V_{TN} = \frac{\lvert Q'_{SD}(\max)\rvert - Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp},\ x_{dT} = \left(\frac{4\epsilon_s\phi_{fp}}{eN_a}\right)^{1/2}$
Body effect	$\Delta V_T = \gamma\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right],\ \gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}$
MOS capacitance	$\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}},\ C'_{min} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s)x_{dT}}$
Collector current	$i_C = I_S \exp\left(\frac{v_{BE}}{V_t}\right),\ A_v = -g_m R_C = -\frac{I_C R_C}{V_t}$
Early effect	$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE} + V_A},\ J_C \approx \frac{eD_B n_{B0}}{x_B} e^{eV_{BE}/kT}$
Injection efficiency	$\gamma \approx \frac{1}{1 + \frac{N_B}{N_E}\frac{D_E}{D_B}\frac{x_B}{x_E}}$
Pinch-off voltage	$V_{p0} = \frac{ea^2 N_d}{2\epsilon_s},\ V_p = V_{bi} - V_{p0}$
Cutoff frequency	$f_T = \frac{g_m}{2\pi C_G} = \frac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$

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11. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
$\psi(x)$	electron wave function in the crystal	Bloch function
$u(x)$	periodic function, with the period of the lattice	Bloch function
k	wave number (m^{-1})	Bloch function
x	position (m)	Bloch function
P'	strength of the potential barriers (dimensionless)	Kronig-Penney condition
a	width of the well (m)	Kronig-Penney condition
m	mass of the electron (kg)	Kronig-Penney condition
E	electron energy (J)	Kronig-Penney condition
$\hbar$	reduced Planck constant, $h/2\pi$ (J s)	Kronig-Penney condition
m_n^*	electron effective mass (kg)	Effective mass
$f_F(E)$	probability that a state at energy E is occupied	Fermi-Dirac function
E_F	Fermi energy (eV)	Fermi-Dirac function
kT	thermal energy (0.0259 eV at 300 K)	Fermi-Dirac function
$g_c(E)$	density of states in the conduction band ($\mathrm{cm}^{-3}\mathrm{eV}^{-1}$)	Density of states
$g_v(E)$	density of states in the valence band ($\mathrm{cm}^{-3}\mathrm{eV}^{-1}$)	Density of states
m_p^*	hole effective mass (kg)	Density of states
h	Planck's constant (6.625×10^{-34} J s)	Density of states
E_c	conduction-band edge (eV)	Density of states
E_v	valence-band edge (eV)	Density of states
k	Boltzmann constant (8.62×10^{-5} eV/K)	Thermal energy
T	absolute temperature (K)	Thermal energy
n_0	thermal-equilibrium electron concentration (cm^{-3})	Electron concentration
N_c	effective density of states in the conduction band (cm^{-3})	Electron concentration
p_0	thermal-equilibrium hole concentration (cm^{-3})	Hole concentration
N_v	effective density of states in the valence band (cm^{-3})	Hole concentration
n_i	intrinsic carrier concentration (cm^{-3})	Intrinsic concentration
E_g	band gap energy (eV)	Intrinsic concentration
N_a	acceptor concentration (cm^{-3})	Majority carriers
N_d	donor concentration (cm^{-3})	Majority carriers
n_d	electrons still bound to donor atoms (cm^{-3})	Electrons left on donors
E_d	donor energy level (eV)	Electrons left on donors
E_{Fi}	intrinsic Fermi level (eV)	Fermi level from the intrinsic level
σ	conductivity $(\Omega\text{-cm})^{-1}$	Drift and conductivity
ρ	resistivity (Ω-cm)	Drift and conductivity
e	magnitude of the electron charge (1.6×10^{-19} C)	Drift and conductivity
μ_n	electron mobility ($\mathrm{cm}^2/\mathrm{V\ s}$)	Drift and conductivity

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Symbol	Meaning	First used in
μ_p	hole mobility ($\text{cm}^2/\text{V s}$)	Drift and conductivity
n	electron concentration (cm^{-3})	Drift and conductivity
p	hole concentration (cm^{-3})	Drift and conductivity
N_I	total ionized impurity concentration, $N_d + N_a$ (cm^{-3})	Drift and conductivity
$J_{n,dif}$	electron diffusion current density (A/cm^2)	Diffusion current
$J_{p,dif}$	hole diffusion current density (A/cm^2)	Diffusion current
D_n	electron diffusion coefficient (cm^2/s)	Diffusion current
D_p	hole diffusion coefficient (cm^2/s)	Diffusion current
F	particle flux ($\text{cm}^{-2}\text{s}^{-1}$)	Diffusion current
v_{th}	thermal velocity (cm/s)	Diffusion current
l	mean free path (cm)	Diffusion current
J	total current density (A/cm^2)	Total current
E	electric field (V/cm)	Total current
kT/e	thermal voltage (0.0259 V at 300 K)	Einstein relation
V_H	Hall voltage (V)	Hall voltage
I_x	current along x (A)	Hall voltage
B_z	magnetic flux density along z (T)	Hall voltage
d	sample thickness, along B_z (cm)	Hall voltage
L	sample length (cm)	Hall mobility
V_x	applied voltage along x (V)	Hall mobility
W	sample width (cm)	Hall mobility
L_p	hole diffusion length (cm)	Diffusion length
τ_{p0}	minority-carrier (hole) lifetime (s)	Diffusion length
$\delta p(x)$	excess hole concentration at x (cm^{-3})	Diffusion length
δp_0	excess hole concentration at $x = 0$ (cm^{-3})	Diffusion length
V_{bi}	built-in potential barrier (V)	Built-in potential
x_n	space-charge width on the n side (cm)	Charge balance
x_p	space-charge width on the p side (cm)	Charge balance
ϵ_s	permittivity of the semiconductor, $11.7\epsilon_0$ for Si (F/cm)	Space charge widths
W	total space-charge width (cm)	Total width
V_R	reverse-bias voltage (V)	Total width
C'	junction capacitance per unit area (F/cm^2)	Junction capacitance
p_{n0}	thermal-equilibrium hole (minority) concentration in the n region (cm^{-3})	Ideal diode
n_{p0}	thermal-equilibrium electron (minority) concentration in the p region (cm^{-3})	Ideal diode
L_n	electron diffusion length (cm)	Ideal diode
V_a	applied forward voltage (V)	Ideal diode
t_s	storage time (s)	Storage time

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Symbol	Meaning	First used in
I_F	forward current before switching (A)	Storage time
I_R	reverse current while the stored charge is removed (A)	Storage time
Q_p	stored minority-carrier charge (C)	Storage time
ϕ_{B0}	ideal Schottky barrier height (V)	Schottky barrier
ϕ_m	metal work function (V)	Schottky barrier
χ	electron affinity (V)	Schottky barrier
ϕ_n	depth of the Fermi level below E_c, $(E_c - E_F)/e$ (V)	Schottky barrier
A^*	effective Richardson constant (A/cm^2K^2)	Thermionic emission
ϕ_{fp}	Fermi potential of the p-type substrate, $(E_{Fi} - E_F)/e$ (V)	Fermi potential
V_t	thermal voltage, kT/e (V)	Fermi potential
ϕ_s	surface potential (V)	Fermi potential
V_{TN}	threshold voltage of the n-channel MOSFET (V)	Threshold voltage
$Q'_{SD}(\max)$	maximum space-charge (depletion) charge per unit area (C/cm^2)	Threshold voltage
Q'_{ss}	fixed oxide charge per unit area (C/cm^2)	Threshold voltage
C_{ox}	oxide capacitance per unit area, ϵ_{ox}/t_{ox} (F/cm^2)	Threshold voltage
ϕ_{ms}	metal-semiconductor work function difference (V)	Threshold voltage
x_{dT}	maximum space-charge width (cm)	Threshold voltage
ΔV_T	shift of the threshold voltage (V)	Body effect
γ	body-effect coefficient ($V^{1/2}$)	Body effect
V_{SB}	source-to-body reverse bias (V)	Body effect
C'_{SD}	space-charge capacitance per unit area (F/cm^2)	MOS capacitance
C'_{min}	minimum capacitance per unit area (F/cm^2)	MOS capacitance
ϵ_{ox}	oxide permittivity, $3.9\epsilon_0$ (F/cm)	MOS capacitance
t_{ox}	oxide thickness (cm)	MOS capacitance
i_C	collector current (A)	Collector current
I_S	saturation current (A)	Collector current
v_{BE}	base-emitter voltage (V)	Collector current
A_v	voltage gain	Collector current
g_m	transconductance, I_C/V_t (A/V)	Collector current
R_C	collector load resistance (Ω)	Collector current
I_C	dc collector current (A)	Collector current
I_B	base current (A)	Base current
I_{pE}	hole current injected from the base into the emitter (A)	Base current
I_{RB}	recombination current in the base (A)	Base current
I_R	recombination current in the B-E space-charge region (A)	Base current
I_{CBO}	B-C reverse-bias current (A)	Base current
V_{CE}	collector-emitter voltage (V)	Early effect
V_A	Early voltage (V)	Early effect

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Symbol	Meaning	First used in
J_C	collector current density (A/cm^2)	Early effect
D_B	minority-carrier diffusion coefficient in the base (cm^2/s)	Early effect
n_{B0}	thermal-equilibrium minority-carrier concentration in the base (cm^{-3})	Early effect
x_B	neutral base width (cm)	Early effect
V_{BE}	base-emitter voltage (V)	Early effect
γ	emitter injection efficiency	Injection efficiency
N_B	base doping (cm^{-3})	Injection efficiency
N_E	emitter doping (cm^{-3})	Injection efficiency
D_E	minority-carrier diffusion coefficient in the emitter (cm^2/s)	Injection efficiency
x_E	neutral emitter width (cm)	Injection efficiency
ΔV	drop of V_{BE} from the emitter edge to its centre (V)	Current crowding
R	lateral base resistance under half the emitter (Ω)	Current crowding
S	emitter stripe width (cm)	Current crowding
V_{p0}	internal pinch-off voltage (V)	Pinch-off voltage
V_p	pinch-off voltage, gate to source (V)	Pinch-off voltage
a	metallurgical channel thickness (cm)	Pinch-off voltage
f_T	cutoff frequency (Hz)	Cutoff frequency
C_G	gate capacitance, $C_{gs} + C_{gd}$ (F)	Cutoff frequency
L	channel length (cm)	Cutoff frequency

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